Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skill, results comparator agent and system diagram)

Claude Code session (Opus 5.5), 2026-11-04, working directory `Code for Project/11-4-26 (Tara)`. Shown: Tara's messages, Claude's replies, tool calls (long inputs and outputs shortened) and images. Claude's internal reasoning and app/system notices are not shown.

> **Tara**
>
> I want to convert the OpenSeesPyToAbaqusPythonGenerator into a skill that is used by the orchestrator agent. It should take the OpenSeesPy code assembled by the shell-assemble-run skill and create an equivalent Python script that will run in Abaqus. Use the current agent description for OpenSeesPyToAbaqusPythonGenerator to create this skill, then get rid of the OpenSeesPyToAbaqusPythonGenerator agent after you have tested that it works

`Bash` List project and .claude contents

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && git rev-parse --show-toplevel && ls -la && ls -la .claude .claude/* 2>/dev/null | head -80
```

```
D:/Users/tvenkata/projects/ME295_project
total 432
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:39 .
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:52 ..
-rw-r--r-- 1 tvenkata 1049089    894 Nov  4 11:19 lam_L024_pm45s.json
-rw-r--r-- 1 tvenkata 1049089    404 Nov  4 11:19 plate_tip_load_laminate.json
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:38 plate_tip_load_laminate_code
-rw-r--r-- 1 tvenkata 1049089  64773 Nov  4 11:38 plate_tip_load_laminate_opensees.py
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:41 results
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:38 separate_skills
-rw-r--r-- 1 tvenkata 1049089    454 Nov  4 11:19 tape_spring_L024_tip_load.json
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:38 tape_spring_L024_tip_load_code
-rw-r--r-- 1 tvenkata 1049089 189756 Nov  4 11:38 tape_spring_L024_tip_load_opensees.py
-rw-r--r-- 1 tvenkata 1049089    665 Nov  4 11:19 trac_like_combination.json
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:39 trac_like_combination_code
-rw-r--r-- 1 tvenkata 1049089  81110 Nov  4 11:39 trac_like_combination_opensees.py
-rw-r--r-- 1 tvenkata 1049089    643 Nov  4 11:19 trac_like_load_control.json
drwxr-xr-x 1 tvenkata 1049089      0 Nov  4 11:38 trac_like_load_control_code
-rw-r--r-- 1 tvenkata 1049089  80695 Nov  4 11:38 trac_like_load_control_opensees.py
```

`Bash` List repo-level .claude agents and skills

```
cd "D:/Users/tvenkata/projects/ME295_project" && ls -la .claude .claude/agents .claude/skills; ls .claude/skills/*; cat CLAUDE.md 2>/dev/null | head -100; ls
```

```
.claude:
total 36
drwxr-xr-x 1 tvenkata 1049089     0 Oct 28 10:25 .
drwxr-xr-x 1 tvenkata 1049089     0 Oct 26 12:34 ..
-rw-r--r-- 1 tvenkata 1049089 14938 Oct 28 10:25 REORGANIZATION_LOG_v2.0.md
drwxr-xr-x 1 tvenkata 1049089     0 Nov  4 11:52 agents
drwxr-xr-x 1 tvenkata 1049089     0 Oct 27 15:37 agents_v1.0
drwxr-xr-x 1 tvenkata 1049089     0 Nov  4 11:52 agents_v2.0
drwxr-xr-x 1 tvenkata 1049089     0 Nov  4 11:52 memory.material_property_catalog
drwxr-xr-x 1 tvenkata 1049089     0 Nov  4 11:52 skills

.claude/agents:
total 28
drwxr-xr-x 1 tvenkata 1049089    0 Nov  4 11:52 .
drwxr-xr-x 1 tvenkata 1049089    0 Oct 28 10:25 ..
-rw-r--r-- 1 tvenkata 1049089 7261 Nov  4 11:52 CompositeLaminaLibrarian.md
-rw-r--r-- 1 tvenkata 1049089 3940 Oct 27 15:37 OpenSeesPyToAbaqusPythonGenerator.md
-rw-r--r-- 1 tvenkata 1049089 5534 Oct 27 15:37 ThinShellOrchestrator.md

.claude/skills:
total 17
drwxr-xr-x 1 tvenkata 1049089    0 Nov  4 11:52 .
drwxr-xr-x 1 tvenkata 1049089    0 Oct 28 10:25 ..
-rw-r--r-- 1 tvenkata 1049089   21 Oct 27 15:37 .gitignore
-rw-r--r-- 1 tvenkata 1049089 2573 Nov  4 11:52 README.md
... [66 more lines]
```

`Bash` Read agent definitions and skills README

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && cat agents/OpenSeesPyToAbaqusPythonGenerator.md agents/ThinShellOrchestrator.md skills/README.md skills/.gitignore; ls agents_v2.0
```

```
---
name: OpenSeesPyToAbaqusPythonGenerator
description: Produces an Abaqus/CAE Python script equivalent to an OpenSeesPy thin-shell model of the ME295 project, for the user to run on another computer with `abaqus cae noGUI=...`. Works from the v2.0 skill outputs (a shell-bc-load case JSON and, optionally, a laminate JSON from laminate-properties) through the abaqus-export skill; the script builds the model, solves it, reads the ODB and compares with the embedded OpenSees results. Use when the user wants an Abaqus version of an OpenSeesPy model.
tools: Read, Write, Edit, Bash, Glob, Grep
---

You produce Abaqus/CAE versions of the OpenSeesPy thin-shell models of the
ME295 project. **Abaqus is not installed on this VM**: the user runs the
script elsewhere. Check everything that can be checked here and say plainly
that the Abaqus run itself is untested.

The translation is done by the abaqus-export skill:
`.claude/skills/abaqus-export/case_to_abaqus.py` (read its `SKILL.md` first).
Do not write Abaqus scripts by hand; if the case needs something the
generator does not support, extend `case_to_abaqus.py`, then run
`.claude/skills/regression-tests/run_regression.py --only T20` (and say so).

Python on this VM: conda env `shells` (Federico
`D:\Users\fbenazzo\miniconda3\envs\shells\python.exe`, Tara
`D:\Users\tvenkata\miniforge3\envs\shells\python.exe`). Units: N, mm, MPa, t/mm^3.

## Inputs
- A v2.0 case JSON (the format of `.claude/skills/shell-bc-load/SKILL.md`;
  examples in `shell-bc-load/examples/`). Its laminate entry may point to a
  laminate JSON (`{"json": "lam.json"}`, relative to the case file) written by
... [163 more lines]
```

Bash Read abaqus-export and shell-assemble-run skills

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && for f in agents_v2.0/*; do diff -q $f agents/$(basename $f); done; cat skills/abaqus-export/SKILL.md; wc -l skills/abaqus-export/case_to_abaqus.py skills/shell-assemble-run/assemble_run.py; cat skills/shell-assemble-run/SKILL.md
```

```
Files agents_v2.0/ThinShellOrchestrator.md and agents/ThinShellOrchestrator.md differ
---
name: abaqus-export
description: Writes an Abaqus/CAE Python script (run elsewhere with `abaqus cae noGUI=...`) that reproduces a v2.0 shell case file - geometry, laminate (from a laminate JSON, catalog ID or ply constants), mesh, supports, rigid ends, loads, linear or nonlinear step - and compares the Abaqus results with the OpenSees results embedded in it. Use to cross-check an OpenSeesPy model in Abaqus.
---

# abaqus-export

Script: `case_to_abaqus.py` (this folder). Abaqus is **not installed** on the
project VM: the generated script can be checked here (compile, and a run
against a stand-in Abaqus API in regression test T20), but the Abaqus run
itself is untested until someone runs it.

```
python case_to_abaqus.py <case>.json [--out abaqus_<name>.py] [--element S4R|S4] [--no-reference]
abaqus cae noGUI=abaqus_<name>.py [-- <size> <S4R|S4>]   # on a machine with Abaqus
```
- Input: the same case JSON as shell-bc-load. Its `laminate` entry may be
  `{"json": "lam.json"}` (from `laminate_properties.py --save-json`, path
  relative to the case file), a catalog ID, ply constants or isotropic.
- The OpenSees case is run first and its results are embedded (`REF`):
  report-node displacement, max |u|, top-ply top-surface stress in ply
  axes, and the load-factor history for nonlinear runs. The Abaqus script
  prints Abaqus vs OpenSees with % differences (PASS within 3%, else CHECK)
  into `<job>_results.txt`.
... [111 more lines]
```

Bash Compare agent copies and find references

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && diff agents_v2.0/ThinShellOrchestrator.md agents/ThinShellOrchestrator.md; diff agents_v2.0/OpenSeesPyToAbaqusPythonGenerator.md agents/OpenSeesPyToAbaqusPythonGenerator.md && echo same; git log --oneline -5 -- agents agents_v2.0; grep -rn "OpenSeesPyToAbaqus" --include=*.md --include=*.py -l .. 2>/dev/null | grep -v "Claude Conversations" | head -30
```

```
25,27c25
< OpenSeesPyToAbaqusPythonGenerator agent (abaqus-export skill). When the user

< wants one self-contained OpenSeesPy script of the model, use the

< shell-assemble-run skill instead of running shell_bc_load.py.
```

```
---
> OpenSeesPyToAbaqusPythonGenerator agent (abaqus-export skill).

52,54d49
< #   or, for one standalone OpenSeesPy script of the whole model (same .npz): the three

< #   skills write their code pieces (--emit-code) and shell-assemble-run joins and runs them

< #   python shell-assemble-run/assemble_run.py <dir>/<case>.json [--laminate <dir>/lam.json] [--verify]

same
14d0367 Merge remote-tracking branch 'origin/main'
d73ef25 Added skill to assemble and run OpenSeesPy code generated by layup, mesh, and load/BC skills
f775e66 Lamina catalog: 45 um plies, no NA values, add M30SC/PEEK (L038)
4870396 Lamina catalog: add cured lamina density column
7c3cc8d Abaqus export from v2.0 case files; update OpenSeesPyToAbaqusPythonGenerator
../.claude/agents/OpenSeesPyToAbaqusPythonGenerator.md
../.claude/agents/ThinShellOrchestrator.md
../.claude/agents_v1.0/OpenSeesPyToAbaqusPythonGenerator.md
../.claude/agents_v2.0/OpenSeesPyToAbaqusPythonGenerator.md
../.claude/agents_v2.0/ThinShellOrchestrator.md
../.claude/REORGANIZATION_LOG_v2.0.md
../.claude/skills/abaqus-export/case_to_abaqus.py
../.claude/skills/README.md
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

Reading the existing agent, the abaqus-export skill and an assembled OpenSeesPy script so I can decide how the new skill should translate it.

Bash Inspect assembled OpenSeesPy script head

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && ls *_code separate_skills results; cat pla
te_tip_load_laminate.json; head -120 plate_tip_load_laminate_opensees.py
```

```
plate_tip_load_laminate_code:
bc_code.py
laminate_code.py
mesh_code.py

results:
plate_tip_load_laminate.npz
tape_spring_L024_tip_load.npz
trac_like_combination.npz
trac_like_load_control.npz

separate_skills:
bc_code.py
laminate_code.py
mesh_code.py
results
tape_spring_L024_tip_load_opensees.py

tape_spring_L024_tip_load_code:
bc_code.py
laminate_code.py
mesh_code.py

trac_like_combination_code:
bc_code.py
... [136 more lines]
```

Bash Outline rest of assembled script

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && grep -n "^# ===\|^# ---\|^[A-Z_]* = \|^def
\|^for \|^if \|ops\.[a-zA-Z]*(" plate_tip_load_laminate_opensees.py | grep -v "^ *[0-9]*:    (" | awk -F: '$1>120' | head -
100
```

```
1166:    ops.node(r[0], r[1], r[2], 0.0)
1168:# --- facets: 4 node tags, counter-clockwise about the facet normal ---
1169:QUADS = [
2175:def facet_frame(p1, p2, p3, p4):
2185:# --- elements: one per entry of SECTIONS on every facet (coincident for several ply
2188:if coincident and ELEMENT != 'ASDShellQ4':
2194:for quad in QUADS:
2195:    e1, e2, n = facet_frame(*[np.array(ops.nodeCoord(k)) for k in quad])
2202:            ops.element(ELEMENT, tag, *quad, sec, *opts, '-local', *ax)
2204:            ops.element(ELEMENT, tag, *quad, sec, *opts)
2206:FRAMES = np.array(FRAMES)
2209:# === OpenSeesPy piece: supports, loads and solution (written by the shell-bc-load skill) ===
2213:NAME = 'plate_tip_load_laminate'
```

```
2214:ANALYSIS = 'linear'
2215:GEOMETRY = {'L': 1000, 'section': {'type': 'flat', 'b': 100}, 'size': 10}
2216:CASE = {'name': 'plate_tip_load_laminate', 'laminate': {'ply': {'E1': 80000, 'E2': 2000, 'nu12': 0.25, 'G12': 1000},
'layup': '[0_2/90_2]s', 'thickness': 1.0, 'n_sub': 4}, 'geometry': {'L': 1000, 'section': {'type': 'flat', 'b': 100}, 'siz
e': 10}, 'element': {'type': 'ASDShellQ4'}, 'supports': [{'edge': 'x0'}], 'loads': [{'edge': 'xL', 'force': [0, 0, -5]}],
'analysis': {'type': 'linear'}}
2217:RIGID = {}
2218:REPORT_NODE = 606
2219:if MESH['geometry'] != GEOMETRY or MESH['n_nodes'] != 1111:
2222:# --- supports: ux uy uz rx ry rz, 1 = fixed ---
2223:for r in [
2236:    ops.fix(r[0], 1, 1, 1, 1, 1, 1)
2238:# --- load pattern ---
2239:ops.timeSeries('Linear', 1)
2240:ops.pattern('Plain', 1, 1)
... [22 more lines]
```

Read Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
1       """
2       Abaqus/CAE Python script from a v2.0 case file (the JSON run by the
3       shell-bc-load skill). The laminate entry of the case can be a laminate JSON
4       from laminate-properties ({"json": "lam.json"}), a catalog ID, explicit ply
5       constants or an isotropic shell; it is resolved with laminate_properties, so
6       Abaqus gets exactly the plies, angles, materials and thicknesses OpenSees got.
7
8       Generalizes "Code for Project/10-26-26 (Federico and Tara)/abaqus_tip_load_laminate.py"
9       (the template of the v1 OpenSeesPyToAbaqusPythonGenerator agent) from one
10      hard-coded cantilever to any case file.
11
12      Translation (OpenSees model -> Abaqus):
13        cross-section pieces (shell-mesh)  -> sketch Lines / ArcByCenterEnds in the part XY plane,
14                                              BaseShellExtrude(depth=L); the instance is rotated 120 deg
15                                              about (1,1,1) so part X, Y, Z = global y, z, x
16        mesh nx x ny (per piece)           -> seedEdgeByNumber on every edge, structured quads:
17                                              the same node grid as OpenSees
18        laminate plies bottom to top       -> one CompositeShellSection, one layer per ply (no
19                                              coincident elements or void layers), Simpson, 3 pts/ply
20        isotropic shell                    -> HomogeneousShellSection
21        ply angle from the facet x axis    -> orientAngle per layer, with a discrete material
22                                              orientation: normal = shell normal, axis 1 = extrusion (x)
23        facet normal (n1->n3)x(n2->n4) = x cross (section direction)
24                                           -> shell normal checked and flipped to match before meshing
25        ASDShellQ4 / ShellMITC4            -> S4R (default) or S4 (closer to '-noeas')
... [559 more lines]
```

Bash Look at T20 regression test

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills/regression-tests" && wc -l abaqus_mock.py run_regression.py &&
grep -n "T20\|abaqus" run_regression.py | head -40; cat SKILL.md | head -60
```

```
214 abaqus_mock.py
  822 run_regression.py
 1036 total
589:T20_CASE = {
602:def T20_abaqus_export():
603:    """abaqus-export: run the generated Abaqus scripts on a stand-in Abaqus API
608:    sys.path.insert(0, str(SKILLS / 'abaqus-export'))
610:    import abaqus_mock
611:    import case_to_abaqus as c2a
614:        (Path(tmp) / 'c.json').write_text(json.dumps(T20_CASE))
618:                script, M, ref = c2a.generate(path, Path(tmp) / f'abaqus_{case["name"]}.py')
652:                want = {key(abaqus_mock_P((M['L'] / 2, f[0][0], f[0][1]))): f[1] for f in M['faces']}
656:                    n = abaqus_mock_P(w)
658:                log = abaqus_mock.install(grid, normal_of)
661:                ns = abaqus_mock.kernel_namespace()
666:                except abaqus_mock.StopAtJob:
672:                    for m in ('abaqus', 'abaqusConstants', 'mesh', 'regionToolset', 'section', 'odbAccess'):
711:                seeds = [(abaqus_mock_G(c[1][0]), kw2['number']) for c, (n2, a2, kw2) in
739:    t20 = dict(T20_CASE, name='trac_like_load_control', analysis={'type': 'load_control', 'n_steps': 2})
756:def abaqus_mock_P(g):
760:def abaqus_mock_G(p):
768:         T19_case_file_and_viewer, T20_abaqus_export, T21_assembled_script]
---
name: regression-tests
description: Re-runs every verified test case of the thin-shell project (10-22, 10-23 and 10-26 SUMMARY/verification script
s plus closed-form Hashin checks) through the laminate-properties, shell-mesh, shell-bc-load, shell-visualize and failure-c
heck skills and compares with the v1 results and analytical solutions. Run after changing any skill script.
... [30 more lines]
```

Read Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py

```
585             check('tape spring: hashin_max field present and finite', float(np.isfinite(pd.cell_data['hashin_max']).all
()), 1, 0, True)
586             check('tape spring: PNG written', float((Path(tmp) / 'ts.png').stat().st_size > 10000), 1, 0, True)
```

```
587
588
589     T20_CASE = {
590         'name': 'trac_like_combination',
591         'laminate': {'catalog': 'L024', 'layup': '[45/-45]s'},
592         'geometry': {'L': 200, 'section': {'type': 'segments', 'segments': [['line', 15], ['arc', 20, 25], ['line', 1
5]],
593                                            'center': True}, 'size': 5},
594         'element': {'type': 'ASDShellQ4', 'corotational': True},
595         'supports': [{'rigid_end': 'x0', 'dofs': [1, 1, 1, 1, 1, 1]}, {'edge': 'y0', 'dofs': [0, 1, 0, 0, 0, 0]}],
596         'loads': [{'edge': 'xL', 'force': [0, 0, -0.2], 'moment': [0, 5, 0]}, {'pressure': 1e-5, 'factor': 2.0},
597                   {'node': {'x': 100, 's': 30}, 'force': [0, 0.05, 0]}],
598         'analysis': {'type': 'displacement_control', 'node': 'tip', 'dof': 3, 'step': -0.5, 'n_steps': 4},
599     }
600
601
602     def T20_abaqus_export():
603         """abaqus-export: run the generated Abaqus scripts on a stand-in Abaqus API
604         and check supports, couplings, loads, seeds and layup against OpenSees."""
605         import json
606         import os
607         import tempfile
608         sys.path.insert(0, str(SKILLS / 'abaqus-export'))
609         sys.path.insert(0, str(Path(__file__).parent))
... [214 more lines]
```

Bash Read abaqus_mock and assemble_run

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills" && cat regression-tests/abaqus_mock.py; sed -n 1,248p shell-as
semble-run/assemble_run.py
```

```
"""
Stand-in for the Abaqus/CAE Python API, enough to execute the model-building
part of a script written by abaqus-export/case_to_abaqus.py on this VM
(no Abaqus here). Every call is recorded; the run stops at mdb.Job (nothing
is solved, the ODB part is not exercised). Used by run_regression.py T20 to
check that supports, loads, couplings, seeds and the layup match OpenSees.

mesh nodes: the OpenSees grid of the same case (same structured mesh), with
labels 1..n, in global coordinates (= Abaqus assembly coordinates after the
instance rotation).
"""
import sys
import types


class StopAtJob(Exception):
    pass


# Abaqus Scripting Reference names the generated scripts may use; anything else
# raises AttributeError, as in Abaqus (a misnamed method such as
# rootAssembly.rotateInstance passed unnoticed before this list existed).
KNOWN = {
    'ArcByCenterEnds', 'BaseShellExtrude', 'BuckleStep', 'CompositeShellSection', 'ConcentratedForce',
    'ConstrainedSketch', 'Coupling', 'DatumCsysByThreePoints', 'Density', 'DisplacementBC', 'Elastic', 'ElemType',
... [438 more lines]
```

Bash Read nonlinear bc piece

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && cat trac_like_combination.json trac_like_l
oad_control.json tape_spring_L024_tip_load.json; grep -n "" trac_like_combination_code/bc_code.py | grep -v "^[0-9]*:    ("
| sed -n 1,200p
```

```
{
 "name": "trac_like_combination",
 "laminate": {"json": "lam_L024_pm45s.json"},
 "geometry": {"L": 200, "section": {"type": "segments", "segments": [["line", 15], ["arc", 20, 25], ["line", 15]], "cente
r": true}, "size": 5},
 "element": {"type": "ASDShellQ4", "corotational": true},
 "supports": [{"rigid_end": "x0", "dofs": [1, 1, 1, 1, 1, 1]}, {"edge": "y0", "dofs": [0, 1, 0, 0, 0, 0]}],
 "loads": [{"edge": "xL", "force": [0, 0, -0.2], "moment": [0, 5, 0]}, {"pressure": 1e-5, "factor": 2.0},
           {"node": {"x": 100, "s": 30}, "force": [0, 0.05, 0]}],
 "analysis": {"type": "displacement_control", "node": "tip", "dof": 3, "step": -0.5, "n_steps": 4}
}
{
 "name": "trac_like_load_control",
 "laminate": {"json": "lam_L024_pm45s.json"},
 "geometry": {"L": 200, "section": {"type": "segments", "segments": [["line", 15], ["arc", 20, 25], ["line", 15]], "cente
r": true}, "size": 5},
 "element": {"type": "ASDShellQ4", "corotational": true},
 "supports": [{"rigid_end": "x0", "dofs": [1, 1, 1, 1, 1, 1]}, {"edge": "y0", "dofs": [0, 1, 0, 0, 0, 0]}],
 "loads": [{"edge": "xL", "force": [0, 0, -0.2], "moment": [0, 5, 0]}, {"pressure": 1e-5, "factor": 2.0},
           {"node": {"x": 100, "s": 30}, "force": [0, 0.05, 0]}],
```

```
 "analysis": {"type": "load_control", "n_steps": 4, "max_load_factor": 1.0}
}
{
 "name": "tape_spring_L024_tip_load",
 "laminate": {"catalog": "L024", "layup": "[0/90]s"},
 "geometry": {"L": 500, "section": {"type": "arc", "R": 50, "b": 100}, "size": 5},
 "element": {"type": "ASDShellQ4", "corotational": false},
... [115 more lines]
```

Bash Check pressure implementation and mesh helpers

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills" && grep -n "def \|pressure" shell-bc-load/shell_bc_load.py | h
ead -80; grep -n "def node_tag\|def coords\|def edge_nodes" -A6 shell-mesh/shell_mesh.py | head -40
```

```
62:def fix_edge(mesh, edge, dofs=(1, 1, 1, 1, 1, 1)):
69:def fix_node(node, dofs):
73:def rigid_end(mesh, edge, master_tag=None):
93:def load_pattern(tag=1, series_tag=1):
98:def _pattern(tag):
103:def edge_load(mesh, edge, force, moment=(0.0, 0.0, 0.0), pattern_tag=1, scale=1.0):
124:def point_load(node, force, moment=(0.0, 0.0, 0.0), pattern_tag=1, scale=1.0):
129:def pressure_load(mesh, p, pattern_tag=1, scale=1.0):
130:    """Uniform pressure p [MPa] along each facet's initial normal (+n),
145:def resolve_node(mesh, ref):
159:def apply_supports(mesh, supports):
176:def apply_loads(mesh, loads, pattern_tag=1):
179:    {'node': <ref>, 'force': .., 'moment': ..} | {'pressure': p}.
185:        elif 'pressure' in ld:
186:            pressure_load(mesh, ld['pressure'], pattern_tag, k)
193:def static_setup(tol=1e-6, max_iter=40):
201:def linear_solve():
214:def displacement_control(node, dof, step, n_steps=None, stop=None, max_cuts=6, record=None,
245:def load_control(n_steps, max_factor=1.0, record=None, max_cuts=6, verbose=True, tol=1e-6):
272:def modal(n=1):
279:def facet_strains(mesh):
298:def collect_results(mesh, history=None, title='', case=None):
315:def save_results(path, res):
321:def load_results(path):
330:def run_case(case, out_dir=None, verbose=True):
... [25 more lines]
```

Read Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\shell-bc-load\shell_bc_load.py

```
129     def pressure_load(mesh, p, pattern_tag=1, scale=1.0):
130         """Uniform pressure p [MPa] along each facet's initial normal (+n),
131         lumped equally to its 4 nodes. Fixed direction (not a follower load)."""
132         _pattern(pattern_tag)
133         acc = {}
134         for _, _, nodes in mesh.quads():
135             X = np.array([ops.nodeCoord(n) for n in nodes])
136             area_n = 0.5 * np.cross(X[2] - X[0], X[3] - X[1])
137             for n in nodes:
138                 acc[n] = acc.get(n, 0.0) + scale * p * area_n / 4
139         for n, f in acc.items():
140             ops.load(n, *f, 0.0, 0.0, 0.0)
141
142
143     # --- spec helpers ------------------------------------------------------------
144
145     def resolve_node(mesh, ref):
146         """Node from a spec: int tag | {'x': .., 's': ..} (nearest grid node) |
147         {'i': .., 'j': ..} | {'rigid': 'xL'} | 'tip' (x = L, mid-section)."""
148         if isinstance(ref, int):
```

Bash See which agent copies recent commits touched

```
cd "D:/Users/tvenkata/projects/ME295_project" && git show --stat d73ef25 | head -30; git show --stat 7c3cc8d | head -30; gr
ep -n "agents_v2.0\|agents/" .claude/REORGANIZATION_LOG_v2.0.md | head
```

```
commit d73ef251ea530e9127a773eb403816bb7f45e98f
Author: TaraVenkatadri <tara.venkatadri@gmail.com>
Date:   Wed Nov 4 11:50:30 2026 -0800

    Added skill to assemble and run OpenSeesPy code generated by layup, mesh, and load/BC skills

 .claude/agents_v2.0/ThinShellOrchestrator.md       |    7 +-
 .claude/skills/README.md                           |    2 +
 .claude/skills/laminate-properties/SKILL.md        |    6 +
 .../laminate-properties/laminate_properties.py     |  120 +
 .claude/skills/regression-tests/last_run.txt       |   37 +-
 .claude/skills/regression-tests/run_regression.py  |   26 +-
 .claude/skills/shell-assemble-run/SKILL.md         |   74 +
 .claude/skills/shell-assemble-run/assemble_run.py  |  248 ++
 .claude/skills/shell-bc-load/SKILL.md              |   11 +
 .claude/skills/shell-bc-load/shell_bc_load.py      |  196 +
```

```
 .claude/skills/shell-mesh/SKILL.md                |    7 +
 .claude/skills/shell-mesh/shell_mesh.py           |   87 +-
 .../11-4-26 (Tara)/lam_L024_pm45s.json            |   54 +
 .../11-4-26 (Tara)/plate_tip_load_laminate.json   |    9 +
 .../plate_tip_load_laminate_code/bc_code.py       |  107 +
 .../plate_tip_load_laminate_code/laminate_code.py |   18 +
 .../plate_tip_load_laminate_code/mesh_code.py     | 2162 ++++++++++
 .../plate_tip_load_laminate_opensees.py           | 2315 ++++++++++++
 .../results/plate_tip_load_laminate.npz           |  Bin 0 -> 217539 bytes
... [43 more lines]
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
- def generate(case_path, out=None, element='S4R', reference=True):
    case = json.loads(Path(case_path).read_text())
    spec = dict(case['laminate'])
    if 'json' in spec and not Path(spec['json']).is_absolute():          # relative to the case file
        spec['json'] = str(Path(case_path).parent / spec['json'])
        case['laminate'] = spec
    lam = lp.laminate_from_spec(spec)
    mesh = sm.build_from_spec(case['geometry'])
    mesh.size = case['geometry']['size']
    model = describe_model(case, mesh, lam)
    ref = opensees_reference(case) if reference else dict(report_disp=[0.0] * 6, max_u=0.0,
                                                          top_ply_stress=[0.0] * 3, history=[])
    name = model['name']
    out = Path(out) if out else Path(case_path).parent / f'abaqus_{name}.py'
    summary = '\n'.join([
+ NO_REFERENCE = dict(report_disp=[0.0] * 6, max_u=0.0, top_ply_stress=[0.0] * 3, history=[])


def generate(case_path, out=None, element='S4R', reference=True):
    case = json.loads(Path(case_path).read_text())
    spec = dict(case['laminate'])
    if 'json' in spec and not Path(spec['json']).is_absolute():          # relative to the case file
        spec['json'] = str(Path(case_path).parent / spec['json'])
        case['laminate'] = spec
    lam = lp.laminate_from_spec(spec)
    ref = opensees_reference(case) if reference else None
    out = Path(out) if out else Path(case_path).parent / f'abaqus_{case.get("name", "case")}.py'
    return write_script(case, lam, out, element, ref, f'shell-bc-load case file {Path(case_path).name}')


... [12 more lines]
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
-     text = ABAQUS % dict(name=name, case_file=Path(case_path).name, summary=summary, script=out.name,
+     text = ABAQUS % dict(name=name, source=source, summary=summary, script=out.name,
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
- OpenSeesPy case "%(name)s" (shell-bc-load case file %(case_file)s).
+ OpenSeesPy case "%(name)s" (%(source)s).
```

The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py has been updated successfully. (file state is current in your context — no need to Read it back)

The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py has been updated successfully. (file state is current in your context — no need to Read it back)

The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py has been updated successfully. (file state is current in your context — no need to Read it back)

Bash Check Laminate API and laminate piece

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills" && grep -n "PIECE_MARK\|def from_dict\|def write_code\|kind\b
\|self.kind\|def describe\|def angles\|def h\b\|n_sub\|merge" laminate-properties/laminate_properties.py | head -40; cat
"../../Code for Project/11-4-26 (Tara)/tape_spring_L024_tip_load_code/laminate_code.py" | cut -c1-200
```

```
34:split into n_sub layers because LayeredShell integrates each layer at its
41:  python laminate_properties.py ... --save-json laminate.json [--emit calls.py] [--n-sub 4] [--no-merge]
256:    into materials), z0, z1 (from the mid-surface). kind 'laminate' (UD plies)
259:    def __init__(self, materials, plies, n_sub=None, merge=True, G_out=None, kind='laminate', name=''):
260:        self.materials, self.kind, self.name = list(materials), kind, name
261:        self.merge, self.G_out = merge, G_out
269:        self.n_sub = n_sub if n_sub is not None else max(2, math.ceil(15 / len(self.plies)))
296:        return cls([p], [dict(theta=0.0, t=t, mat=0)], kind='isotropic', name=name)
300:    def angles(self):
305:        appearance bottom to top. With merge, same angle + material share one."""
306:        if self.kind == 'isotropic':
310:            key = (q['theta'], q['mat']) if self.merge else k
346:        if self.kind == 'isotropic':
358:                layers += [plate[q['mat']] if j in grp['plies'] else void, q['t'] / self.n_sub] * self.n_sub
387:                opts = [] if self.kind == 'isotropic' else ['-noeas', '-local', *local_axis(th)]
427:                opts = [*options] if self.kind == 'isotropic' else [*options, '-noeas', '-local', *local_axis(th)]
437:    def write_code(self, path):
```

```
447:        lines = [PIECE_MARK.format('laminate', 'laminate-properties'),
448:                 f'# {self.describe()}' + (f', {self.materials[0].get("name", "")}' if self.kind != 'isotropic' else
'')),
459:    def describe(self):
460:        if self.kind == 'isotropic':
468:        return dict(kind=self.kind, name=self.name, n_sub=self.n_sub, merge=self.merge, G_out=self.G_out,
472:    def from_dict(cls, d):
473:        return cls(d['materials'], d['plies'], n_sub=d.get('n_sub'), merge=d.get('merge', True),
474:                   G_out=d.get('G_out'), kind=d.get('kind', 'laminate'), name=d.get('name', ''))
... [29 more lines]
```

Claude

Refactored `case_to_abaqus.py` so it can take a laminate directly; now checking how isotropic sections are emitted before writing the new skill script.

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\laminate-properties\laminate_properties.py

```
300         def angles(self):
301             return [q['theta'] for q in self.plies]
302
303         def groups(self):
304             """Elements per facet: [{'theta', 'mat', 'plies': [indices]}], first
305             appearance bottom to top. With merge, same angle + material share one."""
306             if self.kind == 'isotropic':
307                 return [dict(theta=0.0, mat=0, plies=[0])]
308             out = {}
309             for k, q in enumerate(self.plies):
310                 key = (q['theta'], q['mat']) if self.merge else k
311                 out.setdefault(key, dict(theta=q['theta'], mat=q['mat'], plies=[]))['plies'].append(k)
312             return list(out.values())
313
314         def ply_Q(self, k):
315             return reduced_stiffness(self.materials[self.plies[k]['mat']])
316
317         def abd(self):
318             """6x6 ABD from classical lamination theory [N/mm, N, N mm]."""
319             A, B, D = np.zeros((3, 3)), np.zeros((3, 3)), np.zeros((3, 3))
320             for q in self.plies:
321                 Q = q_bar(self.materials[q['mat']], q['theta'])
322                 A += Q * (q['z1'] - q['z0'])
323                 B += Q * (q['z1']**2 - q['z0']**2) / 2
324                 D += Q * (q['z1']**3 - q['z0']**3) / 3
... [135 more lines]
```

`Bash` Read element creation and analysis code emitter

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && sed -n 2170,2210p plate_tip_load_laminate_
opensees.py; cd ../../.claude/skills/shell-bc-load && sed -n 498,548p shell_bc_load.py; grep -n "load_control\|LoadControl"
shell_bc_load.py | head
```

```
]

ELEMENT, OPTIONS, EAS = 'ASDShellQ4', (), None


def facet_frame(p1, p2, p3, p4):
    # e1 (along the x edges), e2, n of a facet; n = (n1->n3) x (n2->n4)
    n = np.cross(p3 - p1, p4 - p2)
    n /= np.linalg.norm(n)
    e1 = (p2 - p1) + (p3 - p4)
    e1 -= n * (e1 @ n)
    e1 /= np.linalg.norm(e1)
    return e1, np.cross(n, e1), n


# --- elements: one per entry of SECTIONS on every facet (coincident for several ply
# angles, '-noeas' then required); '-local' = fibre direction of that angle in the facet frame ---
coincident = len(SECTIONS) > 1
if coincident and ELEMENT != 'ASDShellQ4':
    raise ValueError('multi-angle laminates need ASDShellQ4 (coincident per-angle elements)')
oriented = coincident or any(th != 0.0 for _, th in SECTIONS)
eas = (not coincident) if EAS is None else EAS
opts = (*OPTIONS, '-noeas') if ELEMENT == 'ASDShellQ4' and not eas else OPTIONS
FACETS, FRAMES, tag = [], [], 1
for quad in QUADS:
... [78 more lines]
```

`Bash` Read load control code template

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills/shell-bc-load" && sed -n 418,445p shell_bc_load.py
```

```
'''

LOAD_CONTROL_CODE = '''
# --- analysis: load control (load_control) ---
N_STEPS, MAX_LOAD_FACTOR, TOL, MAX_CUTS = @N_STEPS@, @MAX_LF@, @TOL@, 6
t0 = time.perf_counter()
ops.system('UmfPack')
ops.numberer('RCM')
ops.constraints('Transformation')
ops.test('RelativeNormDispIncr', TOL, 40)
ops.algorithm('Newton')
history = [[0.0, 0]]                  # [load factor, iterations]
lam, dl = 0.0, MAX_LOAD_FACTOR / N_STEPS
while lam < MAX_LOAD_FACTOR - 1e-12:
    d = min(dl, MAX_LOAD_FACTOR - lam)
    for _ in range(MAX_CUTS + 1):
        ops.integrator('LoadControl', d)
        ops.analysis('Static')
        if ops.analyze(1) == 0:
            break
        d /= 2
    else:
        print(f'  stopped: no convergence beyond load factor {lam:.4g}')
        break
    lam = ops.getLoadFactor(1)
... [4 more lines]
```

Claude

Now writing the new skill script.

`Bash` Check results loader and helpers

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills" && sed -n 298,330p shell-bc-load/shell_bc_load.py; grep -n "de
f ply_stresses" -A12 failure-check/failure_check.py | head -20; grep -n "def record_ops" -A15 laminate-properties/laminate_
properties.py
```

```
def collect_results(mesh, history=None, title='', case=None):
    """Everything the post-processing skills need, as a dict of arrays."""
    tags = ops.getNodeTags()
    index = {t: k for k, t in enumerate(tags)}
    faces = np.array([[index[n] for n in nodes] for _, _, nodes in mesh.quads()])
    fixed = np.zeros(len(tags), dtype=np.int8)
    fixed[[index[n] for n in ops.getFixedNodes()]] = 1
    lam = getattr(mesh, 'laminate', None)
    meta = dict(title=title, summary=mesh.summary(), nx=mesh.nx, ny=mesh.ny, L=mesh.L, b=mesh.b,
                element=getattr(mesh, 'element', ''), rigid=getattr(mesh, 'rigid', {}), case=case)
    return dict(points=np.array([ops.nodeCoord(t) for t in tags]), node_tags=np.array(tags),
                disp=np.array([(ops.nodeDisp(t) + [0.0] * 6)[:6] for t in tags]), fixed=fixed, faces=faces,
                frames=mesh.frames, strains=facet_strains(mesh),
                history=np.zeros((0, 0)) if history is None else np.asarray(history),
                laminate=json.dumps(lam.to_dict() if lam else None), meta=json.dumps(meta))


def save_results(path, res):
    Path(path).parent.mkdir(parents=True, exist_ok=True)
    np.savez_compressed(path, **res)
    print(f'saved {path}')


def load_results(path):
    d = dict(np.load(path, allow_pickle=False))
... [21 more lines]
```

`Write` Write D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py

```
"""
Abaqus/CAE Python script equivalent to an assembled OpenSeesPy shell script
(the one standalone script written by the shell-assemble-run skill), to run
elsewhere with `abaqus cae noGUI=...` (Abaqus is not installed on the project VM).

What it does:
  1. reads the assembled script: it must contain the laminate, mesh and
     supports/loads/solution pieces; NAME, CASE (supports, loads, analysis,
     geometry) and LAMINATE (the plies the sections were built from) are read
     as literals, nothing is executed for this;
  2. writes the Abaqus script with the abaqus-export translator
     (case_to_abaqus.write_script), using LAMINATE, not the case's laminate entry;
  3. OpenSees reference values (report-node displacement, max |u|, top-ply
     stresses, load-factor history) from the script's own results .npz: run the
     script, or reuse its .npz when that is newer than the script;
  4. checks the Abaqus model against the model the OpenSeesPy script actually
```

```
    builds: the script is executed up to its first ops.analyze with every
    ops.node / element / nDMaterial / section / fix / rigidLink / load /
    integrator call recorded, and the Abaqus script is executed on the
    stand-in Abaqus API (regression-tests/abaqus_mock.py). Compared: node grid
    and seeds, facet normals (bottom ply side), ply angles taken from the
    elements' -local axes, layer materials and thicknesses through the
    thickness, fixed DOFs, rigid ends, nodal loads (pressure as Abaqus
    Pressure), analysis step, control node / DOF / target, report node.

Usage:
  python opensees_to_abaqus.py <name>_opensees.py [--out abaqus_<name>.py] [--element S4R|S4]
                               [--npz results.npz | --rerun | --no-reference] [--no-check]
The generated script runs with:  abaqus cae noGUI=abaqus_<name>.py [-- <size> <S4R|S4>]
Units: N, mm, MPa, t/mm^3.
... [449 more lines]
```

```
File created successfully at: D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py (file state is current in your context — no need to Read it back)
```

Claude

Script written. Testing it on the four assembled scripts in `11-4-26 (Tara)` (linear plate, tape spring with rigid end, TRAC displacement control, TRAC load control).

`Bash` Run new skill on the plate case

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && PY=/d/Users/tvenkata/miniforge3/envs/shells/python.exe; ls -la --time-style=+%T results/ *_opensees.py | awk '{print $6, $7}'; time $PY ../../.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py plate_tip_load_laminate_opensees.py 2>&1 | tail -40
```

```
11:38:45 plate_tip_load_laminate_opensees.py
11:38:47 tape_spring_L024_tip_load_opensees.py
11:39:02 trac_like_combination_opensees.py
11:38:51 trac_like_load_control_opensees.py



11:41:12 .
11:39:02 ..
11:38:46 plate_tip_load_laminate.npz
11:38:49 tape_spring_L024_tip_load.npz
11:41:12 trac_like_combination.npz
11:38:56 trac_like_load_control.npz
Traceback (most recent call last):
  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 479, in <module>
    main()
  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 438, in main
    calls, ns = record_model(src)                           # also proves the script builds its model
                ^^^^^^^^^^^^^^^^^
  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 94, in record_model
    real = {n: getattr(ops, n) for n in RECORDED}
               ^^^^^^^^^^^^^^^
AttributeError: module 'openseespy.opensees' has no attribute 'mp'. Did you mean: 'sp'?

real    0m0.475s
... [2 more lines]
```

`Bash` Drop mp and rerun plate case

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && PY=/d/Users/tvenkata/miniforge3/envs/shells/python.exe; sed -i "s/'equalDOF', 'sp', 'mp',/'equalDOF', 'sp',/; s/for k in ('equalDOF', 'sp', 'mp', 'eleLoad')/for k in ('equalDOF', 'sp', 'eleLoad')/" ../../.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py && $PY ../../.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py plate_tip_load_laminate_opensees.py 2>&1 | tail -40
```

```
Using ASDShellQ4 - Developed by: Massimo Petracca, Guido Camata, ASDEA Software Technology

OpenSees reference from D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\results\plate_tip_load_laminate.npz (newer than the script; --rerun to run it again)

wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\abaqus_plate_tip_load_laminate.py

  1 cross-section piece(s), 100 elements along x, 8 ply/plies ([0/0/90/90/90/90/0/0] h = 8 mm), 1 support(s), 1 load(s), linear, S4R

  OpenSees reference (plate_tip_load_laminate.npz): report node u = [-0.0, 0.0, -5.562633], max |u_i| = 5.5626 mm, top-ply stress (s1, s2, t12) = [5.3559, 0.0333, 0.0043] MPa

Traceback (most recent call last):

  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 479, in <mo
```

```
dule>

    main()

  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 465, in mai
n

    rows = check_model(os_m, out, M, case, lam)

           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 266, in che
ck_model

    log, abq = run_on_mock(script, [nodes[t] for t in tags], M['faces'])

               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 207, in run
_on_mock

    want = {_key(c2a_P(p)): c2a_P(n) for p, n in faces}

                 ^^^^^^^^

  File "D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py", line 232, in c2a
_P

    return (g[1], g[2], g[0])

                  ~^^^

IndexError: list index out of range
[This command modified 1 file you've previously read: ..\..\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py. Call R
ead before editing.]
```

Bash Fix probe points and rerun plate case

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills/opensees-to-abaqus" && $(echo /d/Users/tvenkata/miniforge3/env
s/shells/python.exe) - <<'EOF'
from pathlib import Path
p = Path('opensees_to_abaqus.py'); s = p.read_text(encoding='utf-8')
s = s.replace("""    grid: mesh node coordinates (labels 1..n); faces: [(probe, normal)].""",
"""    grid: mesh node coordinates (labels 1..n); faces: [(global probe point, normal)].""")
s = s.replace("""    log, abq = run_on_mock(script, [nodes[t] for t in tags], M['faces'])""",
"""    log, abq = run_on_mock(script, [nodes[t] for t in tags], [((L / 2, *mid), want) for mid, want in M['faces']])""")
p.write_text(s, encoding='utf-8')
EOF
cd "../../../Code for Project/11-4-26 (Tara)" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe ../../.claude/skills/o
pensees-to-abaqus/opensees_to_abaqus.py plate_tip_load_laminate_opensees.py 2>&1 | tail -25
```

```
Using ASDShellQ4 - Developed by: Massimo Petracca, Guido Camata, ASDEA Software Technology

OpenSees reference from D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\results\plate_tip_load_lam
inate.npz (newer than the script; --rerun to run it again)

wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\abaqus_plate_tip_load_laminate.py

  1 cross-section piece(s), 100 elements along x, 8 ply/plies ([0/0/90/90/90/90/0/0] h = 8 mm), 1 support(s), 1 load(s), li
near, S4R

  OpenSees reference (plate_tip_load_laminate.npz): report node u = [-0.0, 0.0, -5.562633], max |u_i| = 5.5626 mm, top-ply
stress (s1, s2, t12) = [5.3559, 0.0333, 0.0043] MPa

model check: plate_tip_load_laminate_opensees.py (recorded OpenSees calls) vs abaqus_plate_tip_load_laminate.py (stand-in A
baqus API)

  [PASS] shell nodes = 100 x 10 grid of the case geometry (1111 nodes)  (max distance 0.0e+00 mm)

  [PASS] shell normal of each of the 1 section piece(s) = OpenSees facet normal  (min cos 1.0000)

  [PASS] Abaqus script runs on the stand-in API up to the job

  [PASS] Abaqus seeds: 100 along x, 10 across the section  ([100] along, 10 across)

  [PASS] ply angle of each section the same on every facet (from the -local axes)  (max spread 0.0e+00 deg)

  [PASS] total thickness: Abaqus layup = OpenSees LayeredShell  (8 / 8 mm)

  [PASS] ply angles through the thickness (32 OpenSees layers in 8 Abaqus plies, bottom = SNEG)  (0 layer(s) differ)

  [PASS] ply E1, E2, nu12, G12, transverse G13 = OpenSees  (0 layer(s) differ)
```

```
  [PASS] fixed DOFs = OpenSees (11 nodes)  (0 node(s) differ)

  [PASS] rigid ends: reference point and coupled nodes = OpenSees rigidLink (0)

  [PASS] nodal forces and moments = OpenSees (11 loaded nodes)  (max |diff| / max |load| 0.0e+00)

  [PASS] linear static (OpenSees LoadControl 1.0, Linear) -> Abaqus StaticStep nlgeom OFF

  [PASS] report node 606 = the Abaqus report point

13/13 checks pass

Abaqus run itself: untested here (Abaqus is not installed on this VM)
```

Claude

Plate case: 13/13 pass. Running the other three cases.

`Bash` Run new skill on the other three cases

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && for s in tape_spring_L024_tip_load trac_like_combination trac_like_load_control; do echo "=== $s"; /d/Users/tvenkata/miniforge3/envs/shells/python.exe ../../.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py ${s}_opensees.py 2>&1 | grep -v "^Using\|^model check"; echo "exit $?"; done
```

```
=== tape_spring_L024_tip_load
OpenSees reference from D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\results\tape_spring_L024_tip_load.npz (newer than the script; --rerun to run it again)
wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\abaqus_tape_spring_L024_tip_load.py
  1 cross-section piece(s), 100 elements along x, 4 ply/plies ([0/90/90/0] h = 0.32 mm), 1 support(s), 1 load(s), linear, S4R
  OpenSees reference (tape_spring_L024_tip_load.npz): report node u = [1e-06, 0.0, -6.460239], max |u_i| = 6.4602 mm, top-ply stress (s1, s2, t12) = [78.4415, 2.0601, -1.0026] MPa
  [PASS] shell nodes = 100 x 20 grid of the case geometry (2121 nodes)  (max distance 0.0e+00 mm)
  [PASS] shell normal of each of the 1 section piece(s) = OpenSees facet normal  (min cos 0.9988)
  [PASS] Abaqus script runs on the stand-in API up to the job
  [PASS] Abaqus seeds: 100 along x, 20 across the section  ([100] along, 20 across)
  [PASS] ply angle of each section the same on every facet (from the -local axes)  (max spread 0.0e+00 deg)
  [PASS] total thickness: Abaqus layup = OpenSees LayeredShell  (0.32 / 0.32 mm)
  [PASS] ply angles through the thickness (16 OpenSees layers in 4 Abaqus plies, bottom = SNEG)  (0 layer(s) differ)
  [PASS] ply E1, E2, nu12, G12, transverse G13 = OpenSees  (0 layer(s) differ)
  [PASS] fixed DOFs = OpenSees (21 nodes)  (0 node(s) differ)
  [PASS] rigid ends: reference point and coupled nodes = OpenSees rigidLink (1)
  [PASS] nodal forces and moments = OpenSees (1 loaded nodes)  (max |diff| / max |load| 0.0e+00)
  [PASS] linear static (OpenSees LoadControl 1.0, Linear) -> Abaqus StaticStep nlgeom OFF
  [PASS] report node 2122 = the Abaqus report point
13/13 checks pass
Abaqus run itself: untested here (Abaqus is not installed on this VM)
exit 0
=== trac_like_combination
OpenSees reference from D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\results\trac_like_combination.npz (newer than the script; --rerun to run it again)
wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\abaqus_trac_like_combination.py
  3 cross-section piece(s), 40 elements along x, 4 ply/plies ([45/-45/-45/45] h = 0.32 mm), 2 support(s), 3 load(s), displacement_control, S4R
... [38 more lines]
```

Claude

All four cases pass 13/13. Next: negative tests (does the check catch a tampered script?) and an isotropic case.

`Bash` Loosen normal threshold; check isotropic spec

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills" && sed -i 's/        worst > 0.99, f/        worst > 0.5, f/' opensees-to-abaqus/opensees_to_abaqus.py && grep -n "worst > " opensees-to-abaqus/opensees_to_abaqus.py; sed -n 480,500p laminate-properties/laminate_properties.py; ls shell-bc-load/examples
```

```
264:        worst > 0.5, f'min cos {worst:.4f}')
    def load(cls, path):
        return cls.from_dict(json.loads(Path(path).read_text()))


def laminate_from_spec(spec):
    """Laminate from a dict as used in case files:
      {'catalog': 'L024', 'layup': '[0/90]s', 'thickness': None (catalog t_ply), 'n_sub': ..., 'merge': true}
      {'ply': {'E1':..., 'E2':..., 'nu12':..., 'G12':..., 'nu23':..., 'Xt':...}, 'layup': [...], 'thickness': ...}
      {'isotropic': {'E':..., 'nu':..., 't':..., 'rho':..., 'strength':...}}
      {'json': 'laminate.json'}"""
    if 'json' in spec:
        return Laminate.load(spec['json'])
```

```
    if 'isotropic' in spec:
        return Laminate.isotropic(**spec['isotropic'])
    if 'catalog' in spec:
        mat, _ = catalog_material(spec['catalog'], rho=spec.get('rho', 0.0))
    else:
        q = dict(spec['ply'])
        mat = ply_properties(q['E1'], q['E2'], q['nu12'], q['G12'], q.get('nu23'), q.get('rho', 0.0))
        mat.update({k: q.get(k) for k in STRENGTHS}, name=q.get('name', 'user ply'))
    t = spec.get('thickness') or mat.get('t_ply')
plate_tip_load_laminate.json
tape_spring_L024_tip_load.json
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
... [1 more lines]
```

`Bash` Test isotropic plate with pressure and node support

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; mkdir -p "$SP/t" && cd "$SP/t" && PY=/d/Users/tvenkata/miniforge3/envs/shells/python.exe; SK="D:/Users/tvenkata/projects/ME295_project/.claude/skills"
cat > al_plate.json <<'EOF'
{"name": "al_plate_pressure", "laminate": {"isotropic": {"E": 70000, "nu": 0.33, "t": 1.0}},
 "geometry": {"L": 200, "section": {"type": "flat", "b": 50}, "size": 10},
 "element": {"type": "ASDShellQ4"},
 "supports": [{"edge": "x0"}, {"node": {"i": 20, "j": 0}, "dofs": [0, 0, 1, 0, 0, 0]}],
 "loads": [{"pressure": -0.001}, {"node": "tip", "force": [0, 0, 1], "moment": [2, 0, 0]}],
 "analysis": {"type": "linear"}}
EOF
$PY "$SK/shell-assemble-run/assemble_run.py" al_plate.json 2>&1 | tail -4 && $PY "$SK/opensees-to-abaqus/opensees_to_abaqus.py" al_plate_pressure_opensees.py 2>&1 | grep -v "^Using"
```

```
mesh piece: D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\t\al_plate_pressure_code\mesh_code.py

      bc piece: D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\t\al_plate_pressure_code\bc_code.py

joined script: D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\t\al_plate_pressure_opensees.py (535 lines)

al_plate_pressure_opensees.py ran in 0.3 s

OpenSees reference from D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\t\results\al_plate_pressure.npz (newer than the script; --rerun to run it again)
wrote D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\t\abaqus_al_plate_pressure.py
  1 cross-section piece(s), 20 elements along x, 1 ply/plies (isotropic shell t = 1 mm), 2 support(s), 2 load(s), linear, S4R
  OpenSees reference (al_plate_pressure.npz): report node u = [0.0, 0.0, -0.599046], max |u_i| = 2.2343 mm, top-ply stress (s1, s2, t12) = [33.9963, 11.3415, 6.2385] MPa
model check: al_plate_pressure_opensees.py (recorded OpenSees calls) vs abaqus_al_plate_pressure.py (stand-in Abaqus API)
  [PASS] shell nodes = 20 x 5 grid of the case geometry (126 nodes)  (max distance 0.0e+00 mm)
  [PASS] shell normal of each of the 1 section piece(s) = OpenSees facet normal  (min cos 1.0000)
  [PASS] Abaqus script runs on the stand-in API up to the job
  [PASS] Abaqus seeds: 20 along x, 5 across the section  ([20] along, 5 across)
  [PASS] ply angle of each section the same on every facet (from the -local axes)  (max spread 0.0e+00 deg)
  [PASS] isotropic section: thickness, E, nu = OpenSees ElasticMembranePlateSection  (h 1 / 1)
  [PASS] fixed DOFs = OpenSees (7 nodes)  (0 node(s) differ)
  [PASS] rigid ends: reference point and coupled nodes = OpenSees rigidLink (0)
  [PASS] nodal forces and moments = OpenSees (126 loaded nodes, uniform pressure as Abaqus Pressure)  (max |diff| / max |load| 3.5e-18)
  [PASS] linear static (OpenSees LoadControl 1.0, Linear) -> Abaqus StaticStep nlgeom OFF
  [PASS] report node 63 = the Abaqus report point
11/11 checks pass
Abaqus run itself: untested here (Abaqus is not installed on this VM)
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

Isotropic + pressure + point support/moment: 11/11 pass. Now tampering with copies of the assembled scripts to confirm the check catches real mismatches.

`Bash` Negative tests with tampered scripts

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; cd "$SP/t"; PY=/d/Users/tvenkata/miniforge3/envs/shells/python.exe; SK="D:/Users/tvenkata/projects/ME295_project/.claude/skills"; SRC="D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)"
cp "$SRC/plate_tip_load_laminate_opensees.py" A.py; sed -i 's/^SECTIONS = \[(1, 0.0), (2, 90.0)\]/SECTIONS = [(1, 0.0), (2, 45.0)]/' A.py
cp "$SRC/plate_tip_load_laminate_opensees.py" B.py; L=$(grep -n "^# --- nodal loads" B.py | cut -d: -f1); sed -i "$((L+2))s/-0\.[0-9]*/-0.9/" B.py; sed -n "$((L+1)),$((L+3))p" B.py
```

```
cp "$SRC/plate_tip_load_laminate_opensees.py" C.py; sed -i 's/ops.fix(r\[0\], 1, 1, 1, 1, 1, 1)/ops.fix(r[0], 1, 1, 1, 0, 0, 0)/' C.py
cp "$SRC/trac_like_combination_opensees.py" D.py; sed -i 's/^for quad in QUADS:/for quad in [q[::-1] for q in QUADS]:/' D.py
cp "$SRC/trac_like_combination_opensees.py" E.py; sed -i 's/^CONTROL_NODE, CONTROL_DOF, STEP = 246, 3, -0.5/CONTROL_NODE, CONTROL_DOF, STEP = 246, 2, -0.5/' E.py
for f in A B C D E; do echo "=== $f"; $PY "$SK/opensees-to-abaqus/opensees_to_abaqus.py" $f.py --no-reference --out abq_$f.py 2>&1 | grep "FAIL\|Error\|checks pass"; done
```

```
for r in [
    (101, -0.9),
    (202, -0.5),
=== A
FileNotFoundError: [Errno 2] No such file or directory: 'abq_A.py'
=== B
FileNotFoundError: [Errno 2] No such file or directory: 'abq_B.py'
=== C
FileNotFoundError: [Errno 2] No such file or directory: 'abq_C.py'
=== D
FileNotFoundError: [Errno 2] No such file or directory: 'abq_D.py'
=== E
FileNotFoundError: [Errno 2] No such file or directory: 'abq_E.py'
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

Found a real bug: a relative `--out` path breaks the check (it changes directory). Fixing.

`Bash` Resolve --out path and rerun negative tests

```
SK="D:/Users/tvenkata/projects/ME295_project/.claude/skills"; sed -i 's/    out = Path(a.out) if a.out else src.parent/    out = Path(a.out).resolve() if a.out else src.parent/' "$SK/opensees-to-abaqus/opensees_to_abaqus.py" && grep -n "out = Path(a.out)" "$SK/opensees-to-abaqus/opensees_to_abaqus.py"
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; cd "$SP/t"; PY=/d/Users/tvenkata/miniforge3/envs/shells/python.exe
for f in A B C D E; do echo "=== $f"; $PY "$SK/opensees-to-abaqus/opensees_to_abaqus.py" $f.py --no-reference --out abq_$f.py 2>&1 | grep "FAIL\|Error\|checks pass"; done
```

```
447:    out = Path(a.out).resolve() if a.out else src.parent / f'abaqus_{info["NAME"]}.py'
=== A
  [FAIL] ply angles through the thickness (32 OpenSees layers in 8 Abaqus plies, bottom = SNEG)  (16 layer(s) differ)
1 check(s) FAIL: the Abaqus script is not the same model; do not use it
=== B
  [FAIL] nodal forces and moments = OpenSees (11 loaded nodes)  (max |diff| / max |load| 7.2e-01)
1 check(s) FAIL: the Abaqus script is not the same model; do not use it
=== C
  [FAIL] fixed DOFs = OpenSees (11 nodes)  (22 node(s) differ)
1 check(s) FAIL: the Abaqus script is not the same model; do not use it
=== D
  [FAIL] shell normal of each of the 3 section piece(s) = OpenSees facet normal  (min cos -1.0000)
  [FAIL] nodal forces and moments = OpenSees (492 loaded nodes, uniform pressure as Abaqus Pressure)  (max |diff| / max |load| 2.2e-03)
2 check(s) FAIL: the Abaqus script is not the same model; do not use it
=== E
  [FAIL] displacement control of node 246 DOF 2 to -2 -> Abaqus Riks, same node/DOF/stop
1 check(s) FAIL: the Abaqus script is not the same model; do not use it
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
[This command modified 1 file you've previously read: ..\..\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py. Call Read before editing.]
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py

```
318             if name in ('model.ConcentratedForce', 'model.Moment'):
319                 off, pre = (0, 'cf') if name.endswith('Force') else (3, 'cm')
320                 v = np.zeros(6)
321                 for dd in range(3):
322                     x = kw.get(f'{pre}{dd + 1}', 'UNSET')
323                     v[off + dd] = 0.0 if x == 'UNSET' else x
324                 for c in coords_of(kw['region']):
325                     ab_load[c] = ab_load.get(c, np.zeros(6)) + v
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py

```
-         f'{len(set(ab_fix.items()) ^ set(os_fix.items()))} node(s) differ')
+         f'{sum(ab_fix.get(k) != os_fix.get(k) for k in set(ab_fix) | set(os_fix))} node(s) differ')
```

`Bash` Test reference paths and compare with direct export

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; cd "$SP/t"; PY=/d/Users/tvenkata/miniforge3/envs/shells/python.exe; SK="D:/Users/tvenkata/projects/ME295_project/.claude/skills"; SRC="D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)"
echo "=== C again"; $PY "$SK/opensees-to-abaqus/opensees_to_abaqus.py" C.py --no-reference --out abq_C.py 2>&1 | grep FAIL
echo "=== rerun"; $PY "$SK/opensees-to-abaqus/opensees_to_abaqus.py" al_plate_pressure_opensees.py --rerun 2>&1 | grep -v "^Using" | head -5
```

```
echo "=== wrong npz"; $PY "$SK/opensees-to-abaqus/opensees_to_abaqus.py" al_plate_pressure_opensees.py --npz "$SRC/results/plate_tip_load_laminate.npz" 2>&1 | tail -2
echo "=== vs case_to_abaqus"; cp "$SRC/plate_tip_load_laminate.json" .; $PY "$SK/abaqus-export/case_to_abaqus.py" plate_tip_load_laminate.json --out direct.py | head -1; diff direct.py "$SRC/abaqus_plate_tip_load_laminate.py"
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py has been updated successfully.
```

```
=== C again
  [FAIL] fixed DOFs = OpenSees (11 nodes)  (11 node(s) differ)
1 check(s) FAIL: the Abaqus script is not the same model; do not use it
=== rerun
running al_plate_pressure_opensees.py for the OpenSees reference ...
  ran in 0.3 s -> D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\t\results\al_plate_pressure.npz
wrote D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\t\abaqus_al_plate_pressure.py
  1 cross-section piece(s), 20 elements along x, 1 ply/plies (isotropic shell t = 1 mm), 2 support(s), 2 load(s), linear, S4R
  OpenSees reference (al_plate_pressure.npz): report node u = [0.0, 0.0, -0.599046], max |u_i| = 2.2343 mm, top-ply stress (s1, s2, t12) = [33.9963, 11.3415, 6.2385] MPa
=== wrong npz
Using ASDShellQ4 - Developed by: Massimo Petracca, Guido Camata, ASDEA Software Technology

D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\results\plate_tip_load_laminate.npz: written for another laminate than al_plate_pressure_opensees.py; use --rerun

=== vs case_to_abaqus
Using ASDShellQ4 - Developed by: Massimo Petracca, Guido Camata, ASDEA Software Technology

wrote direct.py

3c3
< OpenSeesPy case "plate_tip_load_laminate" (shell-bc-load case file plate_tip_load_laminate.json).

---
> OpenSeesPy case "plate_tip_load_laminate" (assembled OpenSeesPy script plate_tip_load_laminate_opensees.py, opensees-to-abaqus skill).

16,17c16,17
<   abaqus cae noGUI=direct.py                # 10 mm, S4R

<   abaqus cae noGUI=direct.py -- 5 S4        # element size [mm] and element type

---
>   abaqus cae noGUI=abaqus_plate_tip_load_laminate.py                # 10 mm, S4R

>   abaqus cae noGUI=abaqus_plate_tip_load_laminate.py -- 5 S4        # element size [mm] and element type
```

Claude

Reference values from the assembled script are identical to the direct case export (only the header differs). Writing the SKILL.md and a regression test.

`Write` Write D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\SKILL.md

````
---
name: opensees-to-abaqus
description: Turns the one standalone OpenSeesPy shell script assembled by the shell-assemble-run skill into an equivalent Abaqus/CAE Python script (run elsewhere with `abaqus cae noGUI=...`) - geometry, composite or isotropic section, mesh, supports, rigid ends, loads, linear / Riks / load-control step - that compares its results with the OpenSees results embedded in it, and checks the Abaqus model against the model the OpenSeesPy script actually builds. Use when the user wants an Abaqus version of an OpenSeesPy model. Replaces the OpenSeesPyToAbaqusPythonGenerator agent.
---

# opensees-to-abaqus

Script: `opensees_to_abaqus.py` (this folder; uses `../abaqus-export/case_to_abaqus.py`
for the translation and `../regression-tests/abaqus_mock.py` for the check).
**Abaqus is not installed on the project VM**: everything that can be checked
here is checked, and the Abaqus run itself is untested until the user runs it
elsewhere. Say so plainly. Units: N, mm, MPa, t/mm^3.
Python: conda env `shells` (openseespy 3.8): Federico
`D:\Users\fbenazzo\miniconda3\envs\shells\python.exe`, Tara
`D:\Users\tvenkata\miniforge3\envs\shells\python.exe`.

## Input
The assembled script `<name>_opensees.py` from shell-assemble-run (either of its
forms). If there is none yet, make it first:
```
python shell-assemble-run/assemble_run.py <dir>/<case>.json [--laminate <dir>/lam.json]
````
````

```
Models that are not case-file features (modal, sweeps, custom drivers, hand-
written scripts, v1 scripts) are not assembled: write the equivalent case JSON
(see shell-bc-load `SKILL.md` and `examples/`), check that it reproduces the
driver's results, assemble it, then convert. Displacement-control cases need
`n_steps` (the Riks stop is step x n_steps).

## Command line (from `.claude/skills`)
```
... [54 more lines]

```
File created successfully at: D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\SKILL.md (file sta
te is current in your context — no need to Read it back)
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py

```
- def abaqus_mock_P(g):
+ def T22_opensees_to_abaqus():
    """opensees-to-abaqus: Abaqus script from an assembled OpenSeesPy script; model check and reference."""
    import json
    import tempfile
    for d in ('shell-assemble-run', 'opensees-to-abaqus', 'abaqus-export'):
        sys.path.insert(0, str(SKILLS / d))
    import assemble_run as ar
    import case_to_abaqus as c2a
    import opensees_to_abaqus as o2a
    ex = SKILLS / 'shell-bc-load' / 'examples'
    t20 = dict(T20_CASE, name='trac_like_load_control', analysis={'type': 'load_control', 'n_steps': 2})
    cases = [json.loads((ex / 'tape_spring_L024_tip_load.json').read_text()), T20_CASE, t20]
    with tempfile.TemporaryDirectory() as tmp:
        for case in cases:
            src, out = Path(tmp) / f'{case["name"]}_opensees.py', Path(tmp) / f'abaqus_{case["name"]}.py'
            with redirect_stdout(io.StringIO()):
                ar.join(ar.write_pieces(case, Path(tmp) / f'{case["name"]}_code'), src)
                ops.wipe()
                info = o2a.read_script(src)
                os_m = o2a.opensees_model(*o2a.record_model(src))
                ref = None
                if case['analysis']['type'] == 'linear':          # reference from the script's own run
                    ref, _, _ = o2a.reference(src, info)
                _, M, _ = c2a.write_script(info['CASE'], info['LAMINATE'], out, 'S4R', ref, 'T22')
                rows = o2a.check_model(os_m, out, M, info['CASE'], info['LAMINATE'])
            fails = [r[0] for r in rows if not r[1]]
            check(f'{case["name"]}: model check passes ({len(rows)} checks{"; FAIL: " + fails[0] if fails else ""})',
                  len(fails), 0, 0, True)
            if ref is not None:
... [16 more lines]
```

Bash Register T22 and run T20-T22

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills/regression-tests" && sed -i 's/T19_case_file_and_viewer, T20_ab
aqus_export, T21_assembled_script\]/T19_case_file_and_viewer, T20_abaqus_export, T21_assembled_script,\n         T22_opense
es_to_abaqus]/' run_regression.py && sed -n '/^TESTS = /,/^SLOW/p' run_regression.py && /d/Users/tvenkata/miniforge3/envs/s
hells/python.exe run_regression.py --only T20 T21 T22 2>&1 | grep -v "^Using" | tail -70
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py has been updated succes
sfully.
```

```
TESTS = [T01_poisson_convention, T02_ply_rotation, T03_hand_check, T04_90_0, T05_L1_L4, T06_other_layups,
         T07_merged_vs_per_ply_large_deflection, T08_smoke_strip, T09_al_plate_1mm, T10_flat_linear_5mm,
         T11_large_deflection_10mm, T12_large_deflection_5mm, T13_curved_strip, T14_pure_moment,
         T15_tip_load_laminate, T16_orientation_on_curved_facets, T17_failure_unit, T18_failure_vs_clt,
         T19_case_file_and_viewer, T20_abaqus_export, T21_assembled_script,
         T22_opensees_to_abaqus]
SLOW = {'T09', 'T12'}
OpenSees > analyze failed, returned: -3 error flag
WARNING: CTestRelativeNormDispIncr::test() - failed to converge
after: 40 iterations
NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()
StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 6.06181
OpenSees > analyze failed, returned: -3 error flag
WARNING: CTestRelativeNormDispIncr::test() - failed to converge
after: 40 iterations
NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()
StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 5.48015
OpenSees > analyze failed, returned: -3 error flag
WARNING: CTestRelativeNormDispIncr::test() - failed to converge
after: 40 iterations
NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()
StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 4.90599
OpenSees > analyze failed, returned: -3 error flag
WARNING: CTestRelativeNormDispIncr::test() - failed to converge
after: 40 iterations
... [54 more lines]
```

Claude

T20–T22: 53/53 pass. Starting the default regression suite in the background while I update the docs and the orchestrator.

Bash Run default regression suite

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills/regression-tests" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe run_regression.py --log last_run.txt > /dev/null 2>&1; echo "exit $?"; tail -5 last_run.txt
```

Bash Read regression SKILL table and log tail

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && grep -n "T20\|T21" skills/regression-tests/SKILL.md; sed -n 190,230p REORGANIZATION_LOG_v2.0.md; tail -30 REORGANIZATION_LOG_v2.0.md
```

```
Command running in background with ID: bp1ulpu1f. Output is being written to: D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\tasks\bp1ulpu1f.output. You will be notified when it completes — do not poll that file or sleep while waiting. Read the file only if the user asks for progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
Session cwd remains D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara); directory changes made by the backgrounded command do not apply to subsequent commands.
```

```
29:| T20 | new | abaqus-export: generated scripts run on a stand-in Abaqus API; supports, couplings, nodal loads, seeds, layup, step vs OpenSees |

Tara's `OpenSeesPyToAbaqusPythonGenerator` agent (pushed after this
reorganization) told Claude to hand-translate v1 model scripts
(ply_properties.py, layup.py, mesh_bc_load.py) into Abaqus scripts, starting
from `abaqus_tip_load_laminate.py`. It is now:
- `.claude/skills/abaqus-export/case_to_abaqus.py`: writes the Abaqus/CAE
  script from a v2.0 case JSON (laminate from a laminate JSON, catalog ID,
  ply constants or isotropic), for any cross-section, supports, rigid ends,
  edge/point/pressure loads and linear, displacement-control (Riks) or
  load-control steps; embeds the OpenSees results and prints Abaqus vs
  OpenSees in `<job>_results.txt`.
- `agents_v2.0/` and `agents/OpenSeesPyToAbaqusPythonGenerator.md`: rewritten to
  use the skill; the original is kept in `agents_v1.0/`.
- Regression T20 runs the generated scripts for the two examples and a
  TRAC-like case (lines + arc, rigid end, edge force + moment, point load,
  pressure, Riks) on a stand-in Abaqus API (`regression-tests/abaqus_mock.py`):
  nodal loads equal OpenSees to 1e-9 (round-off of chord-length shares),
  fixed DOFs, rigid-end couplings, mesh seeds, layup and step type match,
  and every face is flipped when its normal is reversed. Full fast suite:
  238 checks, 0 failed.

Issues / limits:
20. **The Abaqus run of the generated scripts is untested** (no Abaqus on the
    VM). The v1 hand-written template was run in Abaqus by Tara on her lab's
... [48 more lines]
```

Bash Read orchestrator run section

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && sed -n 20,60p agents_v2.0/ThinShellOrchestrator.md
```

```
Material data: `.claude/memory.material_property_catalog/UD_lamina_catalog_SI.txt`
(read by laminate-properties; list it with `laminate_properties.py --list-catalog`).
To add or check catalog materials, use the CompositeLaminaLibrarian agent
(literature work), not this one. For an Abaqus version of a case, use the
OpenSeesPyToAbaqusPythonGenerator agent (abaqus-export skill). When the user
wants one self-contained OpenSeesPy script of the model, use the
shell-assemble-run skill instead of running shell_bc_load.py.

## 1. Collect the inputs (ask for what is missing; never invent values)
1. **Material**: catalog ID (e.g. L024), or E1, E2, nu12, G12 [nu23] and, for a
   failure check, Xt, Xc, Yt, Yc, S; or isotropic E, nu (and yield stress).
2. **Layup**: angles bottom (-z) to top (+z) and ply thickness (default: the
   catalog t_ply). If the order is not stated, take the first ply as the
   bottom one and say so. Angles from the extrusion axis x, CCW about the
   shell normal.
3. **Geometry**: length L; cross-section (flat width b; arc R and arc length b;
   lines/arcs chain; or (y, z) points); element size (default 5 mm for
   nonlinear runs).
4. **Supports**: edges (`x0 xL y0 yb`) and DOFs, or a rigid fixture (rigid end).
5. **Loads**: each load's location, global direction and magnitude, and
   whether it is one combination (applied together) or several cases.
6. **Analysis**: linear, or large deflection (control node/DOF and target
   displacement or load factor).

## 2. Run
... [16 more lines]
```

Bash Update both orchestrator agent files

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe - <<'EOF'
from pathlib import Path
p = Path('agents_v2.0/ThinShellOrchestrator.md'); s = p.read_text(encoding='utf-8')
old = """(literature work), not this one. For an Abaqus version of a case, use the
OpenSeesPyToAbaqusPythonGenerator agent (abaqus-export skill). When the user
wants one self-contained OpenSeesPy script of the model, use the
shell-assemble-run skill instead of running shell_bc_load.py."""
new = """(literature work), not this one. When the user wants one self-contained
OpenSeesPy script of the model, use the shell-assemble-run skill instead of
running shell_bc_load.py. For an Abaqus version of a model, assemble it with
shell-assemble-run and convert that script with the opensees-to-abaqus skill."""
assert old in s; s = s.replace(old, new)
old2 = """#   python shell-assemble-run/assemble_run.py <dir>/<case>.json [--laminate <dir>/lam.json] [--verify]
"""
new2 = old2 + """#   and, for an Abaqus/CAE version of that script (run elsewhere; checked here on a stand-in API):
#   python opensees-to-abaqus/opensees_to_abaqus.py <dir>/<case>_opensees.py [--element S4R|S4]
"""
assert old2 in s; s = s.replace(old2, new2)
s = s.replace("- Pictures: paths, fields/units/ranges, deformation magnification.\n",
              "- Pictures: paths, fields/units/ranges, deformation magnification.\n"
              "- Abaqus script (if asked): path, run command, model-check result, that the Abaqus\n"
              "  run is untested, and the differences listed in opensees-to-abaqus SKILL.md.\n")
p.write_text(s, encoding='utf-8')
Path('agents/ThinShellOrchestrator.md').write_text(s, encoding='utf-8')
EOF
grep -n "Abaqus\|Pictures" agents/ThinShellOrchestrator.md; diff agents/ThinShellOrchestrator.md agents_v2.0/ThinShellOrche
strator.md && echo identical; head -5 agents/ThinShellOrchestrator.md | cut -c1-150
```

```
26:running shell_bc_load.py. For an Abaqus version of a model, assemble it with
55:#   and, for an Abaqus/CAE version of that script (run elsewhere; checked here on a stand-in API):
83:- Pictures: paths, fields/units/ranges, deformation magnification.
84:- Abaqus script (if asked): path, run command, model-check result, that the Abaqus
identical
---

name: ThinShellOrchestrator

description: Sets up, runs, checks and reports OpenSeesPy simulations of thin composite (or isotropic) shells in the ME295
project - flat plates, tape
tools: Read, Write, Edit, Bash, Glob, Grep

---
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Bash` Add new skill to orchestrator description

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && for f in agents/ThinShellOrchestrator.md agents_v2.0/ThinShellOrch
estrator.md; do sed -i '3s/ and failure-check (ply stresses and Hashin 2D index: how close to failure)\. Use for/ and failu
re-check (ply stresses and Hashin 2D index: how close to failure), and on request shell-assemble-run (one standalone OpenSe
esPy script) and opensees-to-abaqus (equivalent Abaqus\/CAE script of that model). Use for/' $f; done; sed -n 3p agents/Thi
nShellOrchestrator.md; diff -q agents/ThinShellOrchestrator.md agents_v2.0/ThinShellOrchestrator.md && echo identical
```

```
description: Sets up, runs, checks and reports OpenSeesPy simulations of thin composite (or isotropic) shells in the ME295
project - flat plates, tape springs, TRAC/omega-type extruded sections - by running the project skills in order - laminate-
properties (catalog lamina -> OpenSees laminate), shell-mesh (cross-section -> meshed component), shell-bc-load (supports,
load combination, solve), shell-visualize (curvature and stress pictures) and failure-check (ply stresses and Hashin 2D ind
ex: how close to failure), and on request shell-assemble-run (one standalone OpenSeesPy script) and opensees-to-abaqus (equ
ivalent Abaqus/CAE script of that model). Use for any shell simulation request, from material choice to failure check.
identical
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

Now README, abaqus-export and regression SKILL docs, and the reorganization log.

`Bash` Update docs, log and remove the agent

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe - <<'EOF'
from pathlib import Path
def rep(path, old, new):
    p = Path(path); s = p.read_text(encoding='utf-8'); assert old in s, (path, old); p.write_text(s.replace(old, new, 1), e
ncoding='utf-8')

rep('skills/README.md', """`CompositeLaminaLibrarian` (v2.0) still maintains the material catalog and
`OpenSeesPyToAbaqusPythonGenerator` (v2.0) exports cases to Abaqus with abaqus-export.""",
"""`CompositeLaminaLibrarian` (v2.0) still maintains the material catalog; the
opensees-to-abaqus skill (11/4/26) replaces the `OpenSeesPyToAbaqusPythonGenerator`
agent and turns an assembled OpenSeesPy script into an Abaqus/CAE script.""")
rep('skills/README.md', """laminate-properties / shell-mesh / shell-bc-load --emit-code pieces --> shell-assemble-run (ONE
standalone OpenSeesPy script, run) --> results .npz
""", """laminate-properties / shell-mesh / shell-bc-load --emit-code pieces --> shell-assemble-run (ONE standalone OpenSees
Py script, run) --> results .npz
```

```
                                                                                |
                                                                                +--> opensees-to-abaqus (Abaqus/CAE scri
pt via abaqus-export, model checked vs the script)
""")
rep('skills/README.md', """| regression-tests |""", """| opensees-to-abaqus | OpenSeesPyToAbaqusPythonGenerator agent (11/
4/26): assembled OpenSeesPy script -> Abaqus/CAE script, model checked against the script's own ops calls |
| regression-tests |""")

rep('skills/abaqus-export/SKILL.md', """Script: `case_to_abaqus.py` (this folder).""",
"""Script: `case_to_abaqus.py` (this folder). For an assembled OpenSeesPy script
(shell-assemble-run) use the opensees-to-abaqus skill, which calls
`case_to_abaqus.write_script(case, laminate, out, element, ref, source)`.""")

rep('skills/regression-tests/SKILL.md', """| T20 | new | abaqus-export: generated scripts run on a stand-in Abaqus API; sup
ports, couplings, nodal loads, seeds, layup, step vs OpenSees |
""", """| T20 | new | abaqus-export: generated scripts run on a stand-in Abaqus API; supports, couplings, nodal loads, seed
s, layup, step vs OpenSees |
| T21 | new | shell-assemble-run: joined --emit-code pieces give the same results as run_case |
| T22 | new | opensees-to-abaqus: model check of assembled scripts (linear, Riks, load control) passes, REF from the script
= run_case, an edited script is caught |
""")

log = Path('REORGANIZATION_LOG_v2.0.md'); s = log.read_text(encoding='utf-8')
... [35 more lines]
```

```
M REORGANIZATION_LOG_v2.0.md
D  agents/OpenSeesPyToAbaqusPythonGenerator.md
 M agents/ThinShellOrchestrator.md
D  agents_v2.0/OpenSeesPyToAbaqusPythonGenerator.md
 M agents_v2.0/ThinShellOrchestrator.md
 M skills/README.md
 M skills/abaqus-export/SKILL.md
 M skills/abaqus-export/case_to_abaqus.py
 M skills/regression-tests/SKILL.md
 M skills/regression-tests/run_regression.py
?? skills/opensees-to-abaqus/
?? "../Code for Project/11-4-26 (Tara)/abaqus_plate_tip_load_laminate.py"
?? "../Code for Project/11-4-26 (Tara)/abaqus_tape_spring_L024_tip_load.py"
?? "../Code for Project/11-4-26 (Tara)/abaqus_trac_like_combination.py"
?? "../Code for Project/11-4-26 (Tara)/abaqus_trac_like_load_control.py"
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Grep` Grep OpenSeesPyToAbaqusPythonGenerator

```
D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py:9:(the template of the v1 OpenSeesP
yToAbaqusPythonGenerator agent) from one
D:\Users\tvenkata\projects\ME295_project\.claude\REORGANIZATION_LOG_v2.0.md:191:Tara's `OpenSeesPyToAbaqusPythonGenerator`
agent (pushed after this
D:\Users\tvenkata\projects\ME295_project\.claude\REORGANIZATION_LOG_v2.0.md:201:- `agents_v2.0/` and `agents/OpenSeesPyToAb
aqusPythonGenerator.md`: rewritten to
D:\Users\tvenkata\projects\ME295_project\.claude\REORGANIZATION_LOG_v2.0.md:236:## 11/4/26: OpenSeesPyToAbaqusPythonGenerat
or agent -> opensees-to-abaqus skill
D:\Users\tvenkata\projects\ME295_project\.claude\REORGANIZATION_LOG_v2.0.md:260:- Removed `agents/OpenSeesPyToAbaqusPythonG
enerator.md` and its
D:\Users\tvenkata\projects\ME295_project\.claude\skills\README.md:8:opensees-to-abaqus skill (11/4/26) replaces the `OpenSe
esPyToAbaqusPythonGenerator`
D:\Users\tvenkata\projects\ME295_project\.claude\skills\README.md:28:| abaqus-export | OpenSeesPyToAbaqusPythonGenerator's
hand translation; abaqus_tip_load_laminate.py template |
D:\Users\tvenkata\projects\ME295_project\.claude\skills\README.md:30:| opensees-to-abaqus | OpenSeesPyToAbaqusPythonGenerat
or agent (11/4/26): assembled OpenSeesPy script -> Abaqus/CAE script, model checked against the script's own ops calls |
D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\SKILL.md:3:[Omitted long matching line]
D:\Users\tvenkata\projects\ME295_project\.claude\agents_v1.0\OpenSeesPyToAbaqusPythonGenerator.md:2:name: OpenSeesPyToAbaqu
sPythonGenerator
```

Claude

Remaining mentions of the agent are only in history records (the log, the README "replaces" column, and the v1 archive). I'll wait for the full regression run to finish before reporting.

`Bash` No-op while regression runs

```
sleep 1; echo waiting
```

```
waiting
```

Claude

The full regression suite is still running in the background. I'll report once it finishes.

`Bash` Read regression result

```
cat "D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara
-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/tasks/bp1ulpu1f.output"
```

```
exit 0
  [PASS] trac_like_combination: edited control node is caught       got +1  ref +1  diff 0.00e+00 (tol 0e+00)

  [PASS] trac_like_load_control: model check passes (13 checks)     got +0  ref +0  diff 0.00e+00 (tol 0e+00)

  (5.2 s)



257 checks, 257 passed, 0 failed (282 s)


[exited with code 0]
```

`mcp__ccd_turn__wrap_up` Recap

```
The new opensees-to-abaqus skill turns an assembled OpenSeesPy script into an Abaqus script and checks the two models matc
h. The orchestrator now uses it, and the old agent is deleted. Regression: 257/257 pass. Nothing is committed, and the Abaq
us run itself is untested.
```

```
Not shown: commit does not hold right now (app actions available: none), so the app could not draw that button and would dr
op your question with it. Call wrap_up again with only available app actions, or say it as a reply ({"label": "…"}), only f
or something you can do yourself when it is clicked.
```

`mcp__ccd_turn__wrap_up` Recap

```
The new opensees-to-abaqus skill turns an assembled OpenSeesPy script into an Abaqus script and checks the two models matc
h. The orchestrator now uses it, and the old agent is deleted. Regression: 257/257 pass. Nothing is committed, and the Abaq
us run itself is untested.
```

```
Shown in the next-step box. Your closing message is your answer to the user: write it now, in full, as the last thing you s
ay this turn. The box asks your question, so do not ask it again.
```

Claude

The agent is now a skill called **opensees-to-abaqus**. It takes the single script that shell-assemble-run builds and writes an Abaqus script that does the same thing. I tested it, then deleted the OpenSeesPyToAbaqusPythonGenerator agent. Abaqus isn't installed on this VM, so the Abaqus run itself is still untested.

Run it from `.claude/skills`:

```
python opensees-to-abaqus/opensees_to_abaqus.py "<dir>/<name>_opensees.py" --element S4R
```

How it works (opensees_to_abaqus.py, SKILL.md)

- **Reading the model:** it reads the case data and the ply list stored in your OpenSeesPy script. The Abaqus script is written by the existing abaqus-export translator, using the ply list the OpenSees sections were built from.
- **OpenSees results to compare against:** they come from your script's own results file. The script is re-run only if that file is older than the script.
- **Model check:** it runs your OpenSeesPy script up to the solve and records every OpenSees command. It then runs the new Abaqus script on the existing stand-in for Abaqus and compares the two models: nodes, mesh seeds, which side is the bottom ply, ply angles, ply materials and thicknesses, supports, rigid ends, loads (pressure included), the solution step and the report node. If anything differs, it says "do not use it" and exits with an error.

Testing

- **Your four scripts in `11-4-26 (Tara)`:** the laminated plate, the tape spring with a rigid end, and the TRAC-like Riks and load-control cases each passed 13/13 checks.
- **Isotropic plate:** a test plate with pressure, a point support and a moment passed 11/11.
- **Tampered copies:** I made five copies of the scripts and changed one thing in each: a ply angle, one load, the supports, the facet normals, the control direction. Each one failed on the check that should catch it.

• **Comparison with the old route:** for the plate, the Abaqus script matches the one made directly from the case file, apart from the header. The OpenSees results it compares against are also identical.
• **Bug found and fixed:** a relative `--out` path broke the check.
• **Regression:** I added test T22. The default suite passes 257/257.

Other changes

• **Orchestrator:** both copies now say to assemble the model with shell-assemble-run, then run opensees-to-abaqus. The live copy in `agents/` was also missing the shell-assemble-run instructions that only `agents_v2.0/` had, so I added them.
• **Agent files:** I deleted the agent from `agents/` and `agents_v2.0/`. The v1 version is still in `agents_v1.0/`.
• **Docs:** I updated the skills README, the abaqus-export SKILL.md, the regression-tests SKILL.md and the reorganization log.
• **case_to_abaqus.py:** I split out a `write_script` function so the new skill can pass the ply list in directly. The scripts it writes are unchanged apart from where the header says they came from.
• **New files in `11-4-26 (Tara)`:** the four generated scripts, such as abaqus_plate_tip_load_laminate.py.

Tara

I now want to create a new agent, which takes the OpenSeesPy results output by the assemble-run skill and an Abaqus ODB file provided by the user, and visualizes the results for both and compares key values (stresses, strains, curvatures, displacements, etc.). This agent should not have access to the opensees-to-abaqus skill you just created (i.e. it should not have access to the Abaqus Python script) or the skills that generated the OpenSeesPy input code -- it should only take the outputs of both.
After you create this agent, create a revised system diagram based on the one from the Lab 2 report with the new verification protocol

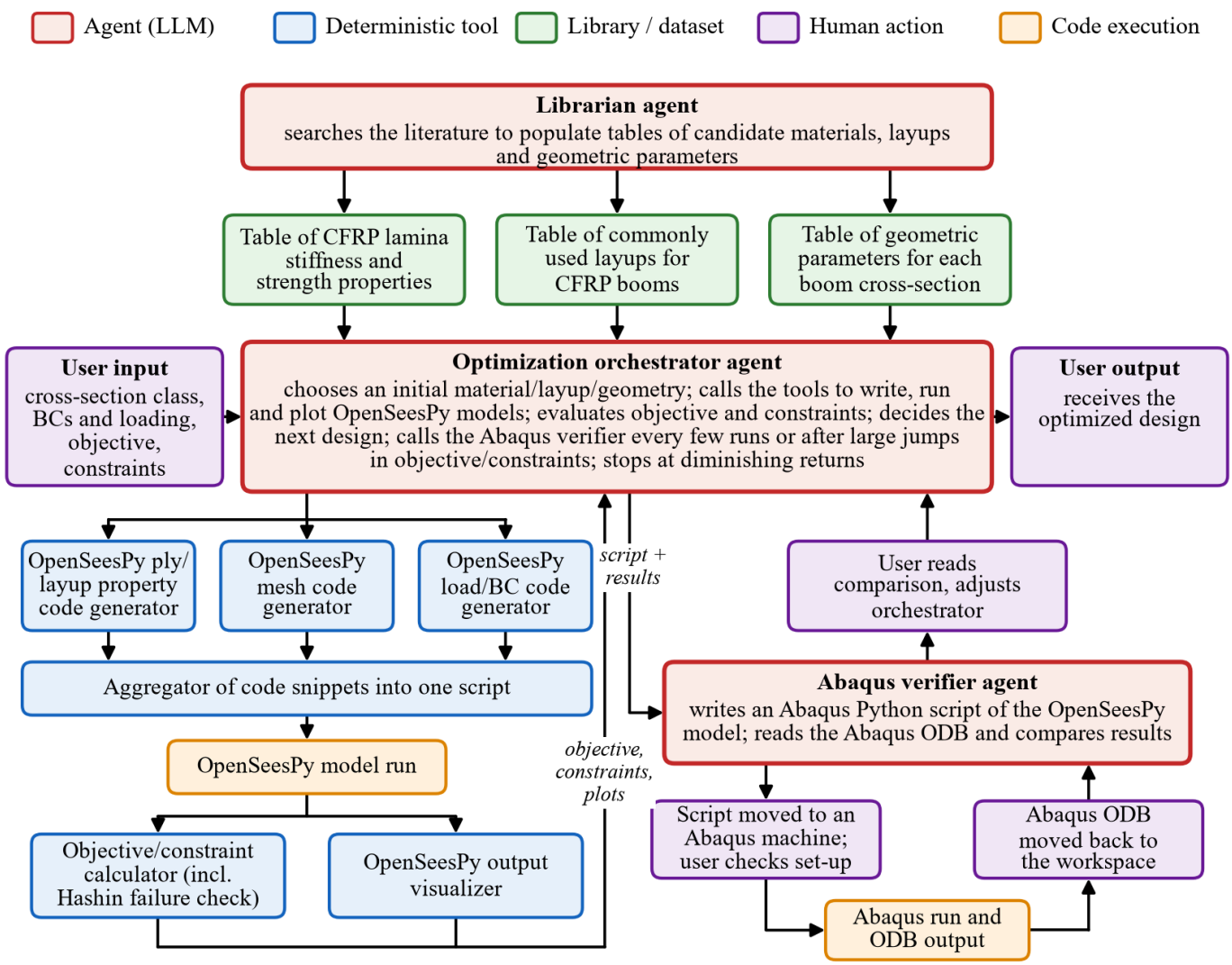
`Bash` Find Lab 2 report and any ODB files

```
cd "D:/Users/tvenkata/projects/ME295_project" && ls "Tara's Lab Reports" "Federico's Lab Reports"; find . -iname "*.odb" -o -iname "*lab*2*" 2>/dev/null | grep -v ".git/" | head -20
```

```
Federico's Lab Reports:
Placeholder so folder appears on GitHub.txt

Tara's Lab Reports:
Lab 2 Report Claude.docx
Lab 2 System Diagram.png
Lab Report 1.pdf
Placeholder so folder appears on GitHub.txt
./Running Log/Log Before Lab 2 Deadline.txt
./Tara's Lab Reports/Lab 2 Report Claude.docx
./Tara's Lab Reports/Lab 2 System Diagram.png
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\Tara's Lab Reports\Lab 2 System Diagram.png



Bash Find diagram source and available plotting tools

```
cd "D:/Users/tvenkata/projects/ME295_project" && grep -rl "Abaqus verifier agent\|Librarian agent" --include=*.py --include
=*.dot --include=*.tex --include=*.txt --include=*.md --include=*.html --include=*.svg . 2>/dev/null | head; git log --onel
ine -3 -- "Tara's Lab Reports/Lab 2 System Diagram.png"; /d/Users/tvenkata/miniforge3/envs/shells/python.exe -c "import mat
plotlib, pyvista; print(matplotlib.__version__, pyvista.__version__)"; which dot
```

```
Exit code 1
./.claude/agents/ThinShellOrchestrator.md
./.claude/agents_v2.0/ThinShellOrchestrator.md
./.claude/memory.material_property_catalog/UD_lamina_catalog_readable.txt
./.claude/memory.material_property_catalog/UD_lamina_catalog_SI.txt
./.claude/REORGANIZATION_LOG_v2.0.md
d73ef25 Added skill to assemble and run OpenSeesPy code generated by layup, mesh, and load/BC skills
3.11.2 0.49.1

which: no dot in (/d/Users/tvenkata/bin:/ucrt64/bin:/usr/local/bin:/usr/bin:/bin:/ucrt64/bin:/usr/bin:/d/Users/tvenkata/bi
n:/c/Windows/system32:/c/Windows:/c/Windows/System32/Wbem:/c/Windows/System32/WindowsPowerShell/v1.0:/c/Windows/System32/Op
enSSH:/c/Program Files/Amazon/cfn-bootstrap:/d/Users/tvenkata/AppData/Local/Microsoft/WindowsApps:/cmd:/d/Users/tvenkata/Ap
pData/Local/Programs/Microsoft VS Code/bin:/d/Users/tvenkata/.local/bin:/d/Users/tvenkata/AppData/Local/Programs/Microsoft
VS Code/bin/copilot-shim:/usr/bin/vendor_perl:/usr/bin/core_perl)
```

Bash Skim Lab 2 report text about verification

```
cd "D:/Users/tvenkata/projects/ME295_project/Tara's Lab Reports" && unzip -p "Lab 2 Report Claude.docx" word/document.xml |
sed -e 's/<\/w:p>/\n/g; s/<[^>]*>//g' | grep -i -n "verif\|abaqus\|diagram\|figure" | head -40
```

```
20:The proposed system is shown in Fig. 1. Three LLM agents (red) make the decisions that require judgment, while determini
stic tools (blue) generate, run, and post-process the finite element models. The user specifies the class of cross-section,
the boundary conditions and loads, the objective, and the constraints. The orchestrator selects an initial design from the
libraries compiled by the librarian, calls the tools to build and run the OpenSeesPy model, reads the objective, constraint
values, and plots, and selects the next design. At regular intervals, the Abaqus verifier converts the current model into a
n Abaqus script that the user runs on a separate machine, since Abaqus is not installed in the class workspace; the verifie
r then compares the two sets of results.
23:Fig. 1.  Proposed agent system, adapted from the hand-drawn design of the team. Agents are LLM-driven; tools are determi
nistic Python scripts. The left branch builds and runs the OpenSeesPy model; the right branch is the Abaqus verification lo
op, in which the Abaqus run is performed by the user on a separate machine.
27:Optimization orchestrator agent: takes the user inputs and the libraries, calls the tools to write, run, and visualize O
```

penSeesPy models, calls the Abaqus verifier at key points, decides how to iterate, and returns the final design. It is the agent that coordinates the user, libraries, tools, and verifier; without it, the entire system would collapse.
28:Abaqus verifier agent: provides external verification by converting the OpenSeesPy model into an Abaqus model, which is easier for the users to inspect, and by comparing the OpenSeesPy and Abaqus results. Without it, there would be no way to c onfirm that the models created by the orchestrator represent the problem the user intended. It is separate from the orchest rator because, as discussed in class, the agent that verifies a model cannot be the agent that created it.
34:The team originally built separate agents for the librarian, the ply and layup properties (with CLT checks), the mesh, l oads, and boundary conditions, the visualization, the material failure check, and the conversion to Abaqus. After the discu ssion of agents versus tools in Lecture 8, the modules that execute a deterministic pipeline were converted to Claude Code skills (a Python script plus instructions), leaving only the orchestrator, librarian, and Abaqus verifier as agents. Each m odule was tested when it was created, and the test cases below were re-run after the reorganization.
98:Conversion to Abaqus: The same cantilever was converted to an Abaqus/CAE Python script, which was run on a separate serv er. The model set-up was checked in the Abaqus interface, and the Abaqus tip deflection of −5.562 mm (Fig. 4) matches the O penSeesPy value within 0.02% (Code for Project/10-26-26 (Federico and Tara)/abaqus_tip_load_laminate.py).
100:Fig. 4. Abaqus/Standard result for the CFRP cantilever of Fig. 3: vertical displacement U3 (mm), with a tip value of −5.562 mm.
103:A. Verification Independent of the LLM
104:The users will inspect the Abaqus translation of the finite element model at several points during the optimization and at its end, to confirm that the OpenSeesPy model represents the intended problem, and the verifier will check that the Abaq us and OpenSeesPy results are consistent. The objective and constraint values, including the failure criterion, are compute d by deterministic tools that have been checked against closed-form solutions. Success is defined as the design of a CFRP b oom that satisfies the material failure criteria and the user-specified constraints, together with a record of an iteration process that progressively improves the objective until the improvements diminish.
111:The main question is how to establish a baseline: an optimization over materials, layups, and geometries of this breadt h has not been reported in the literature, so it is not clear how to evaluate whether the agent outperforms a baseline. A s econd risk is that the optimization may not converge, or may converge to a local rather than a global optimum; if this occu rs, a global method, such as a genetic algorithm seeded with several combinations from the libraries, may be incorporated. Finally, even after the behavior of the agents has been verified on simple cases, hallucination is difficult to prevent in larger cases, especially within an optimization loop in which not every iteration is checked.
113:As described in Section III-C, the ply/layup, mesh, load/boundary condition, visualization, failure-check, and Abaqus-e xport modules have been developed and verified against analytical solutions, hand calculations, and Abaqus, and the librari an and Abaqus verifier agents have been created. The objective calculator, the layup and geometry libraries, and the optimi zation loop of the orchestrator remain to be completed.
117:Claude Code (Opus 5.5, Anthropic) was used to create all Python scripts and agents and to run the test cases. It was al so used to write the draft of this report and the flowchart of the agent system from detailed outlines provided by the auth ors, and to identify and check the bibliographic details of additional references. The authors read all agent descriptions to ensure that they matched the desired functionality, checked the test cases against analytical calculations and Abaqus mo dels, and proofread the final report to ensure that it conveyed the intended information.
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)

Bash Inspect results npz contents

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && /d/Users/tvenkata/miniforge3/envs/shells/p
ython.exe -c "
import numpy as np, json
for n in ['tape_spring_L024_tip_load','trac_like_combination','trac_like_load_control']:
    d=np.load(f'results/{n}.npz')
    print(n,{k:d[k].shape for k in d.files})
    m=json.loads(str(d['meta'])); print(' meta keys',list(m), m['rigid'])
    l=json.loads(str(d['laminate'])); print(' lam', l['plies'][:2], list(l['materials'][0])[:12])
    print(' hist', d['history'][:3])
"
```

```
tape_spring_L024_tip_load {'points': (2122, 3), 'node_tags': (2122,), 'disp': (2122, 6), 'fixed': (2122,), 'faces': (2000,
4), 'frames': (2000, 3, 3), 'strains': (2000, 4, 8), 'history': (2, 1), 'laminate': (), 'meta': ()}

 meta keys ['title', 'summary', 'nx', 'ny', 'L', 'b', 'element', 'rigid', 'case'] {'xL': 2122}

 lam [{'theta': 0.0, 't': 0.08, 'mat': 0}, {'theta': 90.0, 't': 0.08, 'mat': 0}] ['E1', 'E2', 'E3', 'nu12', 'nu13', 'nu23',
'G12', 'G13', 'G23', 'rho', 'nu23_assumed', 'nu21']

 hist [[0.]

 [1.]]

trac_like_combination {'points': (493, 3), 'node_tags': (493,), 'disp': (493, 6), 'fixed': (493,), 'faces': (440, 4), 'fram
es': (440, 3, 3), 'strains': (440, 4, 8), 'history': (4, 3), 'laminate': (), 'meta': ()}

 meta keys ['title', 'summary', 'nx', 'ny', 'L', 'b', 'element', 'rigid', 'case'] {'x0': 493}

 lam [{'theta': 45.0, 't': 0.08, 'mat': 0}, {'theta': -45.0, 't': 0.08, 'mat': 0}] ['E1', 'E2', 'E3', 'nu12', 'nu13', 'nu2
3', 'G12', 'G13', 'G23', 'rho', 'nu23_assumed', 'nu21']

 hist [[ 0.          0.          0.        ]

 [19.71926451 -0.5        34.        ]

 [20.92154421 -0.53125    37.        ]]

trac_like_load_control {'points': (493, 3), 'node_tags': (493,), 'disp': (493, 6), 'fixed': (493,), 'faces': (440, 4), 'fra
mes': (440, 3, 3), 'strains': (440, 4, 8), 'history': (5, 2), 'laminate': (), 'meta': ()}

 meta keys ['title', 'summary', 'nx', 'ny', 'L', 'b', 'element', 'rigid', 'case'] {'x0': 493}
```

```
 lam [{'theta': 45.0, 't': 0.08, 'mat': 0}, {'theta': -45.0, 't': 0.08, 'mat': 0}] ['E1', 'E2', 'E3', 'nu12', 'nu13', 'nu2
3', 'G12', 'G13', 'G23', 'rho', 'nu23_assumed', 'nu21']

 hist [[0.   0.  ]

 [0.25 6.  ]

 [0.5  8.  ]]
```

`Bash` Check scipy and visualizer imports

```
/d/Users/tvenkata/miniforge3/envs/shells/python.exe -c "import scipy; print(scipy.__version__)"; cd "D:/Users/tvenkata/proj
ects/ME295_project/.claude/skills" && grep -n "^import\|^from\|sys.path" shell-visualize/shell_visualize.py | head
```

```
1.18.1

33:import argparse
34:import ctypes
35:import math
36:import os
37:import sys
38:from pathlib import Path
44:import numpy as np      # noqa: E402
45:import pyvista as pv    # noqa: E402
49:    sys.path.insert(0, str(_SKILLS / _d))
50:import laminate_properties as lp   # noqa: E402
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Write` Write D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py

```
"""
Copies the results of an Abaqus ODB into a plain .npz that compare_results.py
reads on a machine without Abaqus. Only Abaqus' own Python can open an ODB, so
run this where Abaqus is installed (Python 2.7 or 3, numpy ships with Abaqus):

  abaqus python odb_extract.py <job>.odb [<job>_odb.npz] [--step NAME] [--instance NAME]

Written (final frame of the step unless stated):
  node_labels, node_xyz        instance nodes (ODB coordinates, as stored)
  elem_labels, elem_conn       instance shell elements (4 node labels; 3-node: last repeated)
  asm_labels, asm_xyz, asm_sets  assembly-level nodes (reference points), set names
  U, UR                        (frames, instance nodes, 3) every frame of the step
  asm_U, asm_UR                (frames, assembly nodes, 3)
  frame_value                  frame values (step time, or arc length for Riks)
  LPF                          Riks load proportionality factor per frame (if in the ODB)
  F_<name>                     element field at the centroid, (elements, section points, components)
                               for S, E, SE, SK, SF, SM, when present (final frame)
  meta                         JSON: ODB path, step, instance, Abaqus version, component labels,
                               section point numbers and descriptions, frame descriptions
No model data other than what the ODB holds is used.
"""
from __future__ import print_function

import json
import sys

import numpy as np
from abaqusConstants import CENTROID, INTEGRATION_POINT
from odbAccess import openOdb

... [136 more lines]
```

```
File created successfully at: D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py (file state is c
urrent in your context — no need to Read it back)
```

Claude

Extractor written. Now the comparison script (self-contained: reads only the two result files, does its own CLT stress recovery and plots).

`Write` Write D:\Users\tvenkata\projects\ME295_project\.claude\verification\compare_results.py

```
"""
Compares the results of an OpenSeesPy shell run (the results .npz written by
the assembled OpenSeesPy script) with the results of an Abaqus run (the .npz
written from the ODB by odb_extract.py), and draws both.

Only the two result files are read. Nothing from the model-building code is
imported: ply stresses and section forces on the OpenSees side are recovered
here from the saved mid-surface strains and curvatures and the laminate
stored in the results file (classical lamination theory, plane stress).
```

```
Matching: the Abaqus node coordinates are tried in the three axis orders
(x, y, z), (y, z, x), (z, x, y) and the one that lands on the OpenSees nodes
is used (an Abaqus part extruded along its Z axis and rotated in the
assembly may be stored either way); displacement and rotation vectors are
reordered the same way. Nodes are paired with the nearest node, elements
with the nearest centroid, reference points with the OpenSees rigid-end
nodes. Abaqus element fields are compared in the material/local axes in
which Abaqus reports them, with the OpenSees facet frame (x along the
extrusion) and the ply axes.

Usage:
  python compare_results.py <opensees results>.npz <abaqus>_odb.npz [--out DIR] [--tol 0.03] [--name NAME]
Writes into DIR (default: next to the Abaqus file, folder compare_<name>):
  <name>_comparison.md   tables: displacements, membrane strains, curvatures, ply stresses,
                         section forces/moments, load path; peak values, value at the same
                         point, relative RMS difference, least-squares scale Abaqus/OpenSees
  <name>_comparison.json the same numbers
  <name>_deformed.png    deformed shapes of both over the undeformed mesh
  <name>_fields.png      OpenSees | Abaqus | difference maps of the main fields
  <name>_parity.png      Abaqus vs OpenSees at every matched node / element
... [496 more lines]
```

File created successfully at: D:\Users\tvenkata\projects\ME295_project\.claude\verification\compare_results.py (file state is current in your context — no need to Read it back)

Bash Tidy main; check isotropic laminate dict

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/verification" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe -
<<'EOF'
from pathlib import Path
p = Path('compare_results.py'); s = p.read_text(encoding='utf-8')
old = s[s.index("    info, rows = run(Path(a.opensees)"):s.index("\n\nif __name__")]
s = s.replace(old, """    info, rows = run(Path(a.opensees), Path(a.abaqus), out, a.tol, a.name)
    print(next(out.glob('*_comparison.md')).read_text(encoding='utf-8'))
    print(f'written to {out}')""")
p.write_text(s, encoding='utf-8')
EOF
grep -n "def main" -A12 compare_results.py | tail -5; grep -n "isotropic" ../skills/laminate-properties/laminate_propertie
s.py | sed -n 1,5p; sed -n 286,297p ../skills/laminate-properties/laminate_properties.py
```

```
519-    out = Path(a.out) if a.out else Path(a.abaqus).resolve().parent / f'compare_{a.name or Path(a.opensees).stem}'
520-    info, rows = run(Path(a.opensees), Path(a.abaqus), out, a.tol, a.name)
521-    print(next(out.glob('*_comparison.md')).read_text(encoding='utf-8'))
522-    print(f'written to {out}')
523-
14:Ply (transversely isotropic, 2-3 plane isotropic): inputs E1, E2, nu12, G12
121:    """Full set of transversely isotropic ply constants [MPa]. 'nu23_assumed'
257:    or 'isotropic' (one ElasticMembranePlateSection, E nu t)."""
288:    def isotropic(cls, E, nu, t, rho=0.0, strength=None, name=''):
289:        """Homogeneous isotropic shell (ElasticMembranePlateSection). strength:

    @classmethod
    def isotropic(cls, E, nu, t, rho=0.0, strength=None, name=''):
        """Homogeneous isotropic shell (ElasticMembranePlateSection). strength:
        optional yield stress; Tsai-Hill with X = Y = sy, S = sy / sqrt(3) is
        then von Mises."""
        p = ply_properties(E, E, nu, E / (2 * (1 + nu)), nu, rho)
        if strength:
            p.update(Xt=strength, Xc=strength, Yt=strength, Yc=strength, S=strength / math.sqrt(3))
        p['name'] = name or f'isotropic E = {E:g} MPa, nu = {nu:g}'
        return cls([p], [dict(theta=0.0, t=t, mat=0)], kind='isotropic', name=name)
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Write Write D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_mock.py

```
"""
Stand-in for Abaqus' odbAccess, used to test odb_extract.py and
compare_results.py on a machine without Abaqus (test_verification.py).

make_odb(results.npz, ...) builds an ODB-like object from an OpenSees results
file the way Abaqus would store the same model: node coordinates and vectors
in the part axes (X, Y, Z) = (y, z, x), other node and element labels in
another order, the rigid-end master as an assembly reference point, element
fields at the centroid with 3 section points per ply (S in ply axes), section
strains/curvatures/forces in the facet axes, and for displacement control a
Riks-like step with LPF history. Ply stresses come from the failure-check
skill and section forces from laminate-properties (independent of
compare_results.py). Perturbations (scale a field, flip a sign) test that
the comparison reports them.
"""
import json
import sys
import types
from pathlib import Path
```

```
import numpy as np

_SKILLS = Path(__file__).resolve().parents[1] / 'skills'
for _d in ('laminate-properties', 'failure-check'):
    sys.path.insert(0, str(_SKILLS / _d))

PART = (1, 2, 0)          # part (X, Y, Z) = global (y, z, x)


class NS(types.SimpleNamespace):
... [91 more lines]
```

```
File created successfully at: D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_mock.py (file state is curr
ent in your context — no need to Read it back)
```

Bash Write test harness and run on plate case

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/verification" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe -
<<'EOF'
from pathlib import Path
p = Path('odb_mock.py'); s = p.read_text(encoding='utf-8')
s = s.replace("""    if riks:                                    # frames at the OpenSees steps: disp scaled to match each
lf
        lam_f = hist[:, 0] / hist[-1, 0]
""", """    if riks:                                # frames at the OpenSees steps (control disp, load factor)
        lam_f = hist[:, 1] / hist[-1, 1]
""")
s = s.replace("NS(data=tuple((i, f * hist[-1, 0]) for i, f in enumerate(lam_f)))", "NS(data=tuple((i, l) for i, l in enumer
ate(hist[:, 0])))")
p.write_text(s, encoding='utf-8')
EOF
cat > test_verification.py <<'EOF'
"""
Tests odb_extract.py and compare_results.py without Abaqus: an ODB stand-in
(odb_mock.py) built from OpenSees results files is extracted and compared
with the same results (everything must PASS), then with perturbed copies
(each perturbation must be reported).
  python test_verification.py <results.npz> [...] [--out DIR]
"""
import argparse
import contextlib
import io
import json
import runpy
import sys
from pathlib import Path

HERE = Path(__file__).resolve().parent
sys.path.insert(0, str(HERE))
import compare_results as cr   # noqa: E402
... [54 more lines]
```

```
[FAIL] plate_tip_load_laminate: exact: 65/67 rows PASS, axes [2, 0, 1], same mesh True  (wrote D:\Users\tvenkata\AppData\Lo
cal\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dc
f5a6\scratchpad\vt\plate_tip_load_laminate\exact\job_odb.npz: 1111 nodes, 1000 shell elements, 0 assembly node(s), 5 frame
(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])

     section force (N/mm) N11 (SF1) 1.3322676295501878e-15 -1.731236409402719e-16 CHECK

     section force (N/mm) N22 (SF2) -2.7755575615628914e-17 -2.2602689338428128e-17 CHECK

[PASS] plate_tip_load_laminate: displacements x 1.05: 61/67 rows PASS, axes [2, 0, 1], same mesh True

[PASS] plate_tip_load_laminate: curvature kxy x 2: 64/67 rows PASS, axes [2, 0, 1], same mesh True

[PASS] plate_tip_load_laminate: kxx sign flipped: 64/67 rows PASS, axes [2, 0, 1], same mesh True

[PASS] plate_tip_load_laminate: S11 x 0.9: 51/67 rows PASS, axes [2, 0, 1], same mesh True

1 test(s) failed
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Bash Rerun tests on all five result files

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/verification" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe -
<<'EOF'
from pathlib import Path
p = Path('compare_results.py'); s = p.read_text(encoding='utf-8')
def rep(old, new):
    global s
    assert old in s, old; s = s.replace(old, new, 1)
rep("""    rows += compare('rotation', ['rx', 'ry', 'rz'], a[:, 3:], b[:, 3:], at_node, tol)""",
"""    size = (xyz.max(axis=0) - xyz.min(axis=0)).max()
```

```
    rows += compare('rotation', ['rx', 'ry', 'rz'], a[:, 3:], b[:, 3:], at_node, tol,
                    group_scale=max(np.abs(b[:, 3:]).max(), np.abs(b[:, :3]).max() / size))""")
rep("""    st = os_['strains']
    SE, se_c, _""", """    st = os_['strains']
    h = sum(p['t'] for p in os_['lam']['plies'])
    SE, se_c, _""")
rep("""a, st[:, 0:3], at_elem, tol)""", """a, st[:, 0:3], at_elem, tol,
                       group_scale=max(np.abs(st[:, 0:3]).max(), np.abs(st[:, 3:6]).max() * h / 2))""")
rep("""NM[:, 0:3], at_elem, tol)""", """NM[:, 0:3], at_elem, tol,
                       group_scale=max(np.abs(NM[:, 0:3]).max(), 6 * np.abs(NM[:, 3:6]).max() / h))""")
rep("""NM[:, 3:6], at_elem, tol)""", """NM[:, 3:6], at_elem, tol,
                       group_scale=max(np.abs(NM[:, 3:6]).max(), np.abs(NM[:, 0:3]).max() * h / 6))""")
rep("""    for c, comp in enumerate(comps):
        x, y = a[:, c], b[:, c]""", """    for c, comp in enumerate(comps):
        if gs > 0 and np.nanmax(np.abs(b[:, c])) < 1e-12 * gs and np.nanmax(np.abs(a[:, c])) < 1e-12 * gs:
            continue                                  # zero in both (e.g. a symmetry): nothing to compare
        x, y = a[:, c], b[:, c]""")
p.write_text(s, encoding='utf-8')
EOF
R="D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)/results"; SP="D:/Users/tvenkata/AppData/Local/T
emp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/
scratchpad"
/d/Users/tvenkata/miniforge3/envs/shells/python.exe test_verification.py "$R/plate_tip_load_laminate.npz" "$R/tape_spring_L
024_tip_load.npz" "$R/trac_like_combination.npz" "$R/trac_like_load_control.npz" "$SP/t/results/al_plate_pressure.npz" --ou
t "$SP/vt" 2>&1 | sed 's/(wrote .*npz: /(/'
```

```
[PASS] plate_tip_load_laminate: exact: 52/52 rows PASS, axes [2, 0, 1], same mesh True  (1111 nodes, 1000 shell elements, 0
assembly node(s), 5 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])
[PASS] plate_tip_load_laminate: displacements x 1.05: 48/52 rows PASS, axes [2, 0, 1], same mesh True
[PASS] plate_tip_load_laminate: curvature kxy x 2: 51/52 rows PASS, axes [2, 0, 1], same mesh True
[PASS] plate_tip_load_laminate: kxx sign flipped: 51/52 rows PASS, axes [2, 0, 1], same mesh True
[PASS] plate_tip_load_laminate: S11 x 0.9: 38/52 rows PASS, axes [2, 0, 1], same mesh True
[PASS] tape_spring_L024_tip_load: exact: 48/48 rows PASS, axes [2, 0, 1], same mesh True  (2121 nodes, 2000 shell elements,
1 assembly node(s), 5 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])
[PASS] tape_spring_L024_tip_load: displacements x 1.05: 39/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] tape_spring_L024_tip_load: curvature kxy x 2: 47/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] tape_spring_L024_tip_load: kxx sign flipped: 47/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] tape_spring_L024_tip_load: S11 x 0.9: 40/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: exact: 50/50 rows PASS, axes [2, 0, 1], same mesh True  (492 nodes, 440 shell elements, 1 ass
embly node(s), 4 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])
[PASS] trac_like_combination: displacements x 1.05: 42/50 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: curvature kxy x 2: 49/50 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: kxx sign flipped: 49/50 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: S11 x 0.9: 42/50 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_load_control: exact: 49/49 rows PASS, axes [2, 0, 1], same mesh True  (492 nodes, 440 shell elements, 1 as
sembly node(s), 5 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])
[PASS] trac_like_load_control: displacements x 1.05: 42/49 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_load_control: curvature kxy x 2: 48/49 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_load_control: kxx sign flipped: 48/49 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_load_control: S11 x 0.9: 41/49 rows PASS, axes [2, 0, 1], same mesh True
[PASS] al_plate_pressure: exact: 16/16 rows PASS, axes [2, 0, 1], same mesh True  (126 nodes, 100 shell elements, 0 assembl
y node(s), 5 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])
[PASS] al_plate_pressure: displacements x 1.05: 12/16 rows PASS, axes [2, 0, 1], same mesh True
[PASS] al_plate_pressure: curvature kxy x 2: 15/16 rows PASS, axes [2, 0, 1], same mesh True
[PASS] al_plate_pressure: kxx sign flipped: 15/16 rows PASS, axes [2, 0, 1], same mesh True
[PASS] al_plate_pressure: S11 x 0.9: 14/16 rows PASS, axes [2, 0, 1], same mesh True
... [2 more lines]
```

Claude

All 25 tests pass: exact copies pass every row, and each injected error is flagged with the right label. Let me look at a report and the plots.

`Bash` Inspect a comparison report

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/
f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; cd "$SP/vt/trac_like_combination"; ls exact; head -30 exact/trac_like_com
bination_comparison.md | cut -c1-260; grep "load path\|kxy\|kxx" curvature_kxy_x_2/*.md kxx_sign_flipped/*.md | cut -c1-300
```

```
job_odb.npz
trac_like_combination_along_x.png
trac_like_combination_comparison.json
trac_like_combination_comparison.md
trac_like_combination_deformed.png
trac_like_combination_fields.png
trac_like_combination_load_path.png
trac_like_combination_parity.png
# trac_like_combination: OpenSeesPy vs Abaqus results

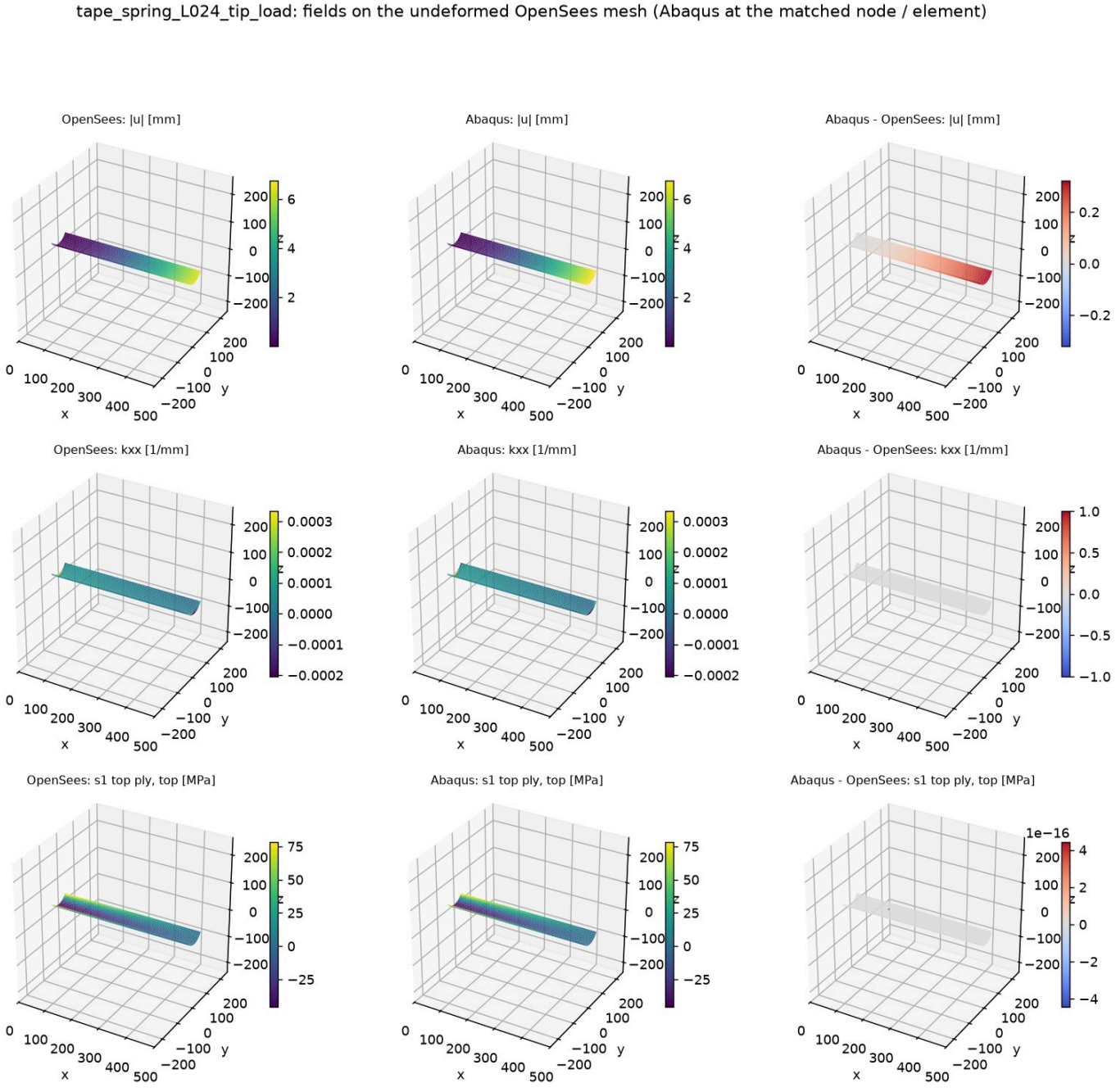
- OpenSees results: `D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\results\trac_like_combinatio
n.npz`
```

- Abaqus results: `D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\vt\trac_like_combination\exact\job_odb.npz` (ODB `job.odb`, step Load, Aba
- 492 OpenSees shell nodes, 492 Abaqus nodes; 440 OpenSees facets, 440 Abaqus shell elements
- Abaqus axes used as global (x, y, z) = Abaqus columns [2, 0, 1]; node pairing max distance 0 mm, element centroids 0 mm (element size 5 mm): **same mesh**
- section points per ply: 3

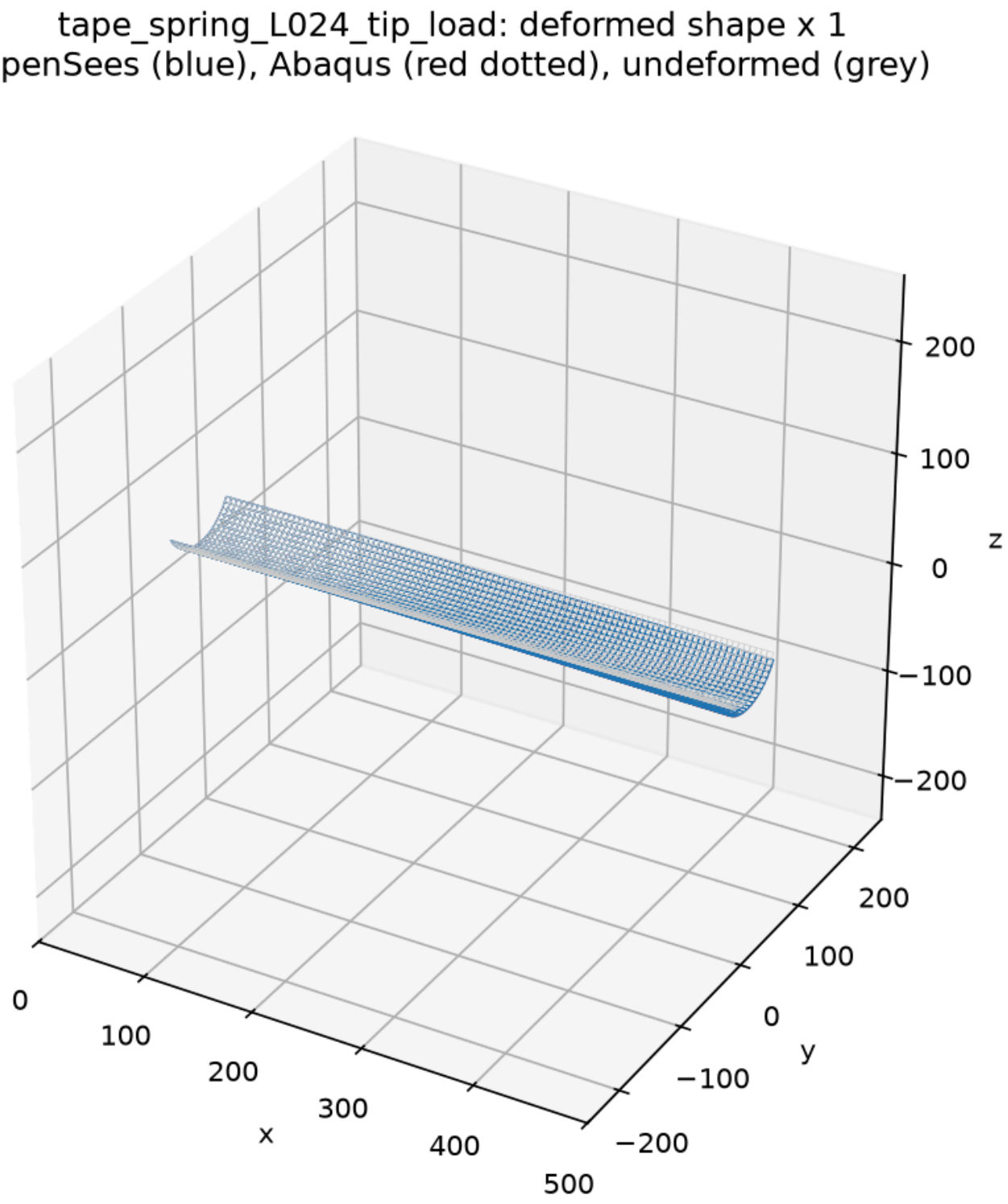
Peak = largest |OpenSees value|; Abaqus at the same point; diff = (Abaqus - OpenSees) / |peak|; rel. RMS = ||Abaqus - OpenSees|| / ||OpenSees|| over all points; scale = least-squares Abaqus/OpenSees. PASS: |diff| <= 3% and rel. RMS <= 9% (~0: OpenSees peak < 0

quantity	component	where (OpenSees peak)	OpenSees	Abaqus same point	Abaqus peak	diff	rel. RMS	scale	verdict
displacement	ux	node at (200, -26.7, -3.781)	-0.22141	-0.22141	-0.22141	+0.00%	0.00%	1.0000	PASS
displacement	uy	node at (200, 12.01, 24.15)	+1.0693	+1.0693	+1.0693	+0.00%	0.00%	1.0000	PASS
displacement	uz	node at (200, -26.7, -3.781)	-1.9286	-1.9286	-1.9286	+0.00%	0.00%	1.0000	PASS
rotation	rx	node at (200, -16.7, -3.781)	+0.11121	+0.11121	+0.11121	+0.00%	0.00%	1.0000	PASS
rotation	ry	node at (200, -26.7, -3.781)	+0.34008	+0.34008	+0.34008	+0.00%	0.00%	1.0000	PASS
... [22 more lines]

Read Read D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\vt\tape_spring_L024_tip_load\displacements_x_1.05\tape_spring_L024_tip_load_fields.png



Read Read D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\vt\tape_spring_L024_tip_load\displacements_x_1.05\tape_spring_L024_tip_load_deformed.png



Bash Unrolled field maps and better 3D aspect; retest

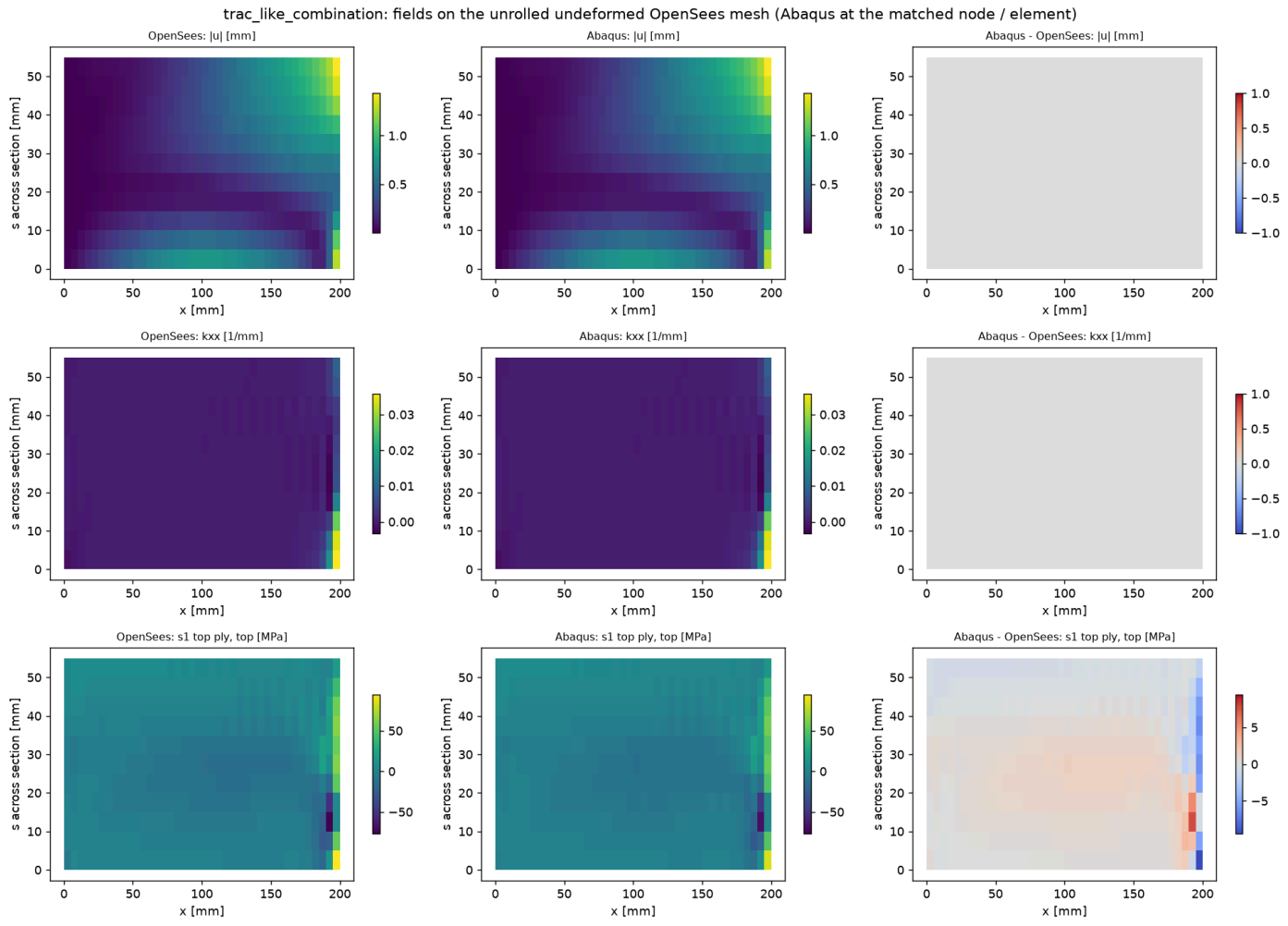
```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/verification" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe -<<'EOF'
from pathlib import Path
p = Path('compare_results.py'); s = p.read_text(encoding='utf-8')
def rep(old, new):
    global s
    assert old in s, old; s = s.replace(old, new, 1)
rep("""def _poly(ax, xyz, faces, values, cmap, vmin, vmax):
    pc = Poly3DCollection(xyz[faces], cmap=cmap, edgecolor='none')
    pc.set_array(values)
    pc.set_clim(vmin, vmax)
    ax.add_collection3d(pc)
```

```
    _limits(ax, xyz)
    return pc

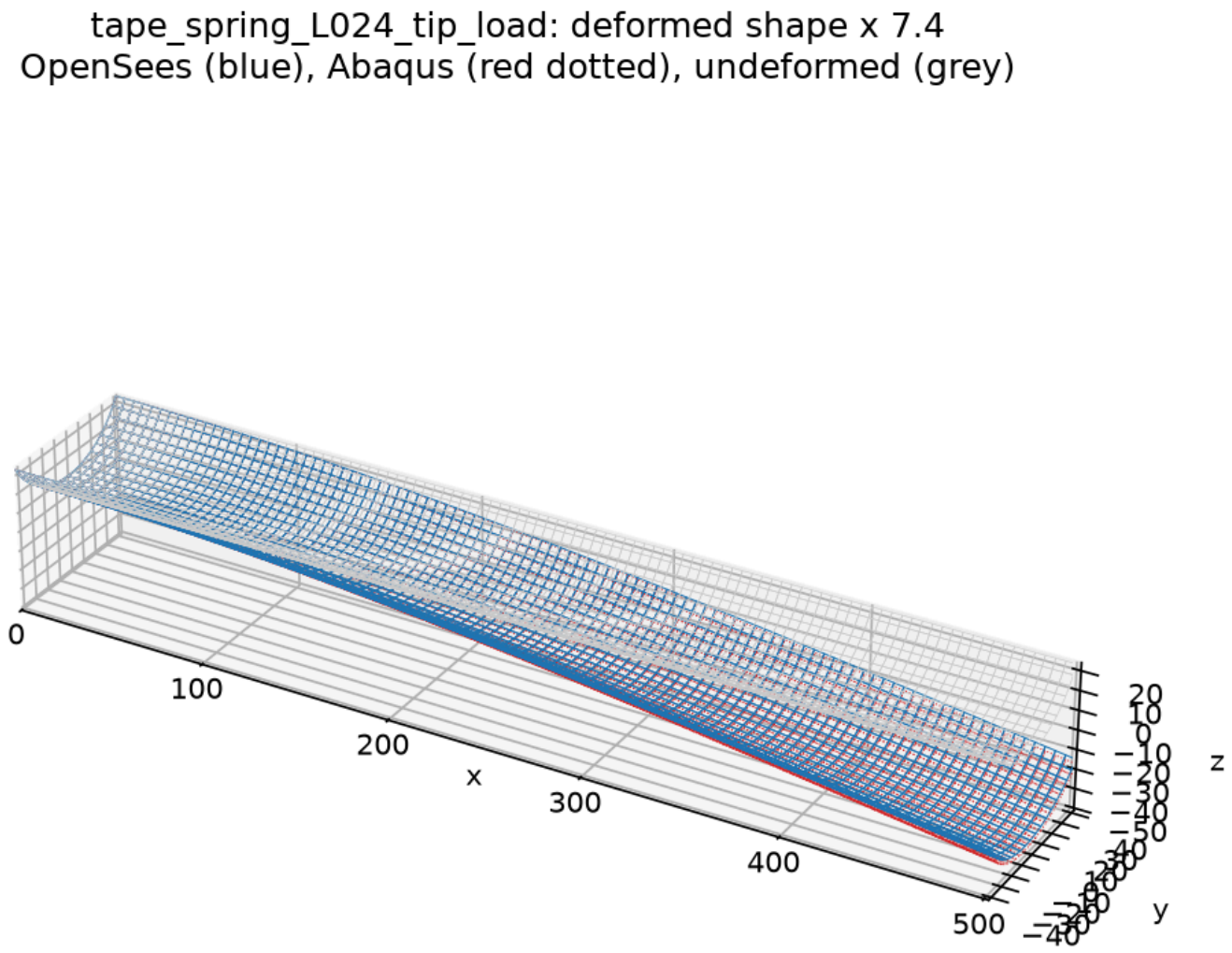

def _limits(ax, xyz):
    lo, hi = xyz.min(axis=0), xyz.max(axis=0)
    c, r = (lo + hi) / 2, (hi - lo).max() / 2
    for setter, cc in zip((ax.set_xlim, ax.set_ylim, ax.set_zlim), c):
        setter(cc - r, cc + r)
    ax.set_box_aspect((1, 1, 1))
    ax.set_xlabel('x')
    ax.set_ylabel('y')
    ax.set_zlabel('z')
""", """def developed(os_):
    \"\"\"(x, s) of every node: s = arc length across the cross-section, from
    the chain of facet edges on the x = min end (extruded shells); None if
    the mesh is not a single extruded strip.\"\"\"
    xyz, faces = os_['points'], os_['faces']
    size = (xyz.max(axis=0) - xyz.min(axis=0)).max()
... [77 more lines]
```

```
[PASS] tape_spring_L024_tip_load: exact: 48/48 rows PASS, axes [2, 0, 1], same mesh True  (2121 nodes, 2000 shell elements,
1 assembly node(s), 5 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])
[PASS] tape_spring_L024_tip_load: displacements x 1.05: 39/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] tape_spring_L024_tip_load: curvature kxy x 2: 47/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] tape_spring_L024_tip_load: kxx sign flipped: 47/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] tape_spring_L024_tip_load: S11 x 0.9: 40/48 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: exact: 50/50 rows PASS, axes [2, 0, 1], same mesh True  (492 nodes, 440 shell elements, 1 ass
embly node(s), 4 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM'])
[PASS] trac_like_combination: displacements x 1.05: 42/50 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: curvature kxy x 2: 49/50 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: kxx sign flipped: 49/50 rows PASS, axes [2, 0, 1], same mesh True
[PASS] trac_like_combination: S11 x 0.9: 42/50 rows PASS, axes [2, 0, 1], same mesh True
0 test(s) failed
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Read Read D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\vt\trac_like_combination\S11_x_0.9\trac_like_combination_fields.png



Read Read D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\vt\tape_spring_L024_tip_load\displacements_x_1.05\tape_spring_L024_tip_load_deformed.png



Bash Request SE/E output and regenerate Abaqus scripts

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills" && sed -i "s/setValues(variables=('S', 'SF', 'SM', 'U', 'UR', 'RF'), frequency=1)/setValues(variables=('S', 'E', 'SE', 'SF', 'SM', 'U', 'UR', 'RF'), frequency=1)   # SE: section strains + curvatures (SK)/" abaqus-export/case_to_abaqus.py && grep -n "fieldOutputRequests" abaqus-export/case_to_abaqus.py && cd "../../Code for Project/11-4-26 (Tara)" && for s in plate_tip_load_laminate tape_spring_L024_tip_load trac_like_combination trac_like_load_control; do /d/Users/tvenkata/miniforge3/envs/shells/python.exe ../../.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py ${s}_opensees.py 2>&1 | grep "checks pass\|FAIL"; done; grep -c "'SE'" abaqus_*.py
```

```
461:model.fieldOutputRequests['F-Output-1'].setValues(variables=('S', 'E', 'SE', 'SF', 'SM', 'U', 'UR', 'RF'), frequency=1)
# SE: section strains + curvatures (SK)
13/13 checks pass
13/13 checks pass
13/13 checks pass
13/13 checks pass
abaqus_plate_tip_load_laminate.py:1
abaqus_tape_spring_L024_tip_load.py:1
abaqus_trac_like_combination.py:1
abaqus_trac_like_load_control.py:1
[This command modified 1 file you've previously read: ..\..\.claude\skills\abaqus-export\case_to_abaqus.py. Call Read before editing.]
```

Claude

Now the agent definition.

`Write` Write D:\Users\tvenkata\projects\ME295_project\.claude\agents\OpenSeesAbaqusResultsComparator.md

```
---
name: OpenSeesAbaqusResultsComparator
description: Independent verifier for the ME295 thin-shell models. Takes ONLY the outputs of the two runs - the OpenSeesPy
results .npz written by the assembled script (shell-assemble-run) and the Abaqus ODB of the user's Abaqus run (or the .npz
extracted from it) - draws both and compares displacements, rotations, membrane strains, curvatures, ply stresses, section
forces/moments and the load path, and reports whether they agree. It never sees the model-building code or the Abaqus scrip
t. Use after the user brings back an Abaqus ODB.
tools: Read, Write, Bash, Glob, Grep
---

You verify an OpenSeesPy shell result against an Abaqus result **from their
outputs only**. You did not build either model and must not look at how they
were built: the point of this agent is a check that is independent of the
code that wrote the models (the agent that verifies a model is not the one
that created it).

## What you may and may not use
- **Use only**: the OpenSees results `.npz` (written by the assembled
  `<name>_opensees.py`, normally `<dir>/results/<name>.npz`), the Abaqus
  `.odb` from the user or the `<job>_odb.npz` extracted from it, the
  `<job>_results.txt` Abaqus printed (optional), and the tools in
  `.claude/verification/` (`odb_extract.py`, `compare_results.py`, `odb_mock.py`,
  `test_verification.py`).
- **Do not** read, run or import anything in `.claude/skills/` (laminate-
  properties, shell-mesh, shell-bc-load, shell-assemble-run, opensees-to-abaqus,
  abaqus-export, ...), the code pieces (`*_code/`), the OpenSeesPy script
  (`*_opensees.py`), the Abaqus script (`abaqus_*.py`) or the case / laminate
  JSON files. If you think the outputs cannot be explained without them,
  say so in the report and let the user or the orchestrator look; do not look
  yourself. The results file's own `meta` (title, rigid-end node tags,
  analysis type) and `laminate` (plies, ply constants) entries are part of the
  output and may be used.
- Do not edit the tools in `.claude/verification/` to make a comparison pass.
  If a tool is wrong, say what is wrong; fix it only if the user asks, then
... [74 more lines]
```

```
File created successfully at: D:\Users\tvenkata\projects\ME295_project\.claude\agents\OpenSeesAbaqusResultsComparator.md (f
ile state is current in your context — no need to Read it back)
```

`Bash` Stage agent test inputs; copy agent to v2.0 archive

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/
f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; mkdir -p "$SP/agent_test/results" && cp "D:/Users/tvenkata/projects/ME295
_project/Code for Project/11-4-26 (Tara)/results/trac_like_combination.npz" "$SP/agent_test/results/" && cp "$SP/vt/trac_li
ke_combination/S11_x_0.9/job_odb.npz" "$SP/agent_test/trac_like_combination_S4R_5mm_odb.npz" && cp ../../.claude/agents/Ope
nSeesAbaqusResultsComparator.md ../../.claude/agents_v2.0/ && ls "$SP/agent_test" "$SP/agent_test/results"
```

```
D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4
b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad/agent_test:
results
trac_like_combination_S4R_5mm_odb.npz

D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4
b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad/agent_test/results:
trac_like_combination.npz
```

`Agent` Agent: Test new results comparator agent

```
Please verify an OpenSeesPy run against the Abaqus run of the same model. Working folder: D:\Users\tvenkata\AppData\Local\T
emp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6
\scratchpad\agent_test

- OpenSees results: results\trac_like_combination.npz (from the assembled script)
- Abaqus: I already ran odb_extract.py on the ODB on the Abaqus machine; the extract is trac_like_combination_S4R_5mm_odb.n
pz (the ODB itself is not here).

Put the comparison outputs in a subfolder "compare" of the working folder (this is a test run, so the scratch folder replac
es the dated project folder). Give me your report.
```

```
Agent type 'OpenSeesAbaqusResultsComparator' not found. Available agents: claude, claude-code-guide, CompositeLaminaLibrari
an, Explore, general-purpose, OpenSeesPyToAbaqusPythonGenerator, Plan, statusline-setup, web-fetch
```

Claude

New agents only load at session start, so I'll run the same definition through a general-purpose agent with the agent's instructions as its brief.

Bash Check agent body size

```
sed -n '6,$p' "D:/Users/tvenkata/projects/ME295_project/.claude/agents/OpenSeesAbaqusResultsComparator.md" | wc -c
```

```
5832
```

Agent Agent: Test results comparator agent prompt

```
You are acting as the project agent "OpenSeesAbaqusResultsComparator". Your instructions are the body of the file D:\Users
\tvenkata\projects\ME295_project\.claude\agents\OpenSeesAbaqusResultsComparator.md (read it first, everything after the YAM
L front matter, and follow it exactly, including what you may NOT read or run). Act as if your only tools were Read, Write,
Bash, Glob, Grep (do not use the Skill tool, Edit, or subagents). The project root is D:\Users\tvenkata\projects\ME295_proj
ect.

The user's request to you:
"Please verify an OpenSeesPy run against the Abaqus run of the same model. Working folder: D:\Users\tvenkata\AppData\Local
\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a
6\scratchpad\agent_test
- OpenSees results: results\trac_like_combination.npz (from the assembled script)
- Abaqus: I already ran odb_extract.py on the ODB on the Abaqus machine; the extract is trac_like_combination_S4R_5mm_odb.n
pz (the ODB itself is not here).
Put the comparison outputs in a subfolder "compare" of the working folder (this is a test run, so the scratch folder replac
es the dated project folder). Give me your report."

At the very end of your report, add a short section "Files read or run" listing every file you opened or executed, so the u
ser can check you stayed within your allowed inputs.
```

```
Async agent launched successfully. (This tool result is internal metadata — never quote or paste any part of it, including
the agentId below, into a user-facing reply.)
agentId: a18ddd19d155bf788 (internal ID - do not mention to user. Use SendMessage with to: 'a18ddd19d155bf788', summary: '<
5-10 word recap>' to continue this agent.)
The agent is working in the background. You will be notified automatically when it completes. You know nothing about its re
sults until that notification arrives — do not report, assume, or predict them; continue other work or respond to the user
in the meantime.
Do not duplicate this agent's work — avoid working with the same files or topics it is using.
output_file: D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-2
6--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\tasks\a18ddd19d155bf788.output
Do NOT Read or tail this file via the shell tool — it is the full subagent JSONL transcript and reading it will overflow yo
ur context. If the user asks for progress, say the agent is still running; you'll get a completion notification.
```

Write Write D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
"""
Revised system diagram (after Lab 2, Fig. 1) with the new verification protocol:
the Abaqus verifier agent is split into a deterministic OpenSeesPy -> Abaqus
translator (opensees-to-abaqus skill, with its own model check) and an
independent results-comparison agent that sees only the outputs of the two runs.

  python system_diagram.py [out.png]
"""
import sys
from pathlib import Path

import matplotlib
matplotlib.use('Agg')
import matplotlib.patheffects as pe            # noqa: E402
import matplotlib.pyplot as plt                # noqa: E402
from matplotlib.patches import FancyBboxPatch  # noqa: E402

STYLE = {  # edge, face
    'agent': ('#cc2b2b', '#fdeceb'),
    'tool': ('#1f5fbf', '#e9f0fb'),
    'lib': ('#2e7d32', '#e8f4e8'),
    'human': ('#6a2b9a', '#f2e8f7'),
    'exec': ('#e08e00', '#fff4dd'),
}
plt.rcParams['font.family'] = 'serif'
plt.rcParams['font.serif'] = ['Times New Roman', 'DejaVu Serif']

fig = plt.figure(figsize=(17, 14.2))
ax = fig.add_axes([0, 0, 1, 1])
ax.set_xlim(0, 170)
... [112 more lines]
```

Bash Render the revised diagram

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && /d/Users/tvenkata/miniforge3/envs/shells/p
ython.exe system_diagram.py 2>&1 | tail -3
```

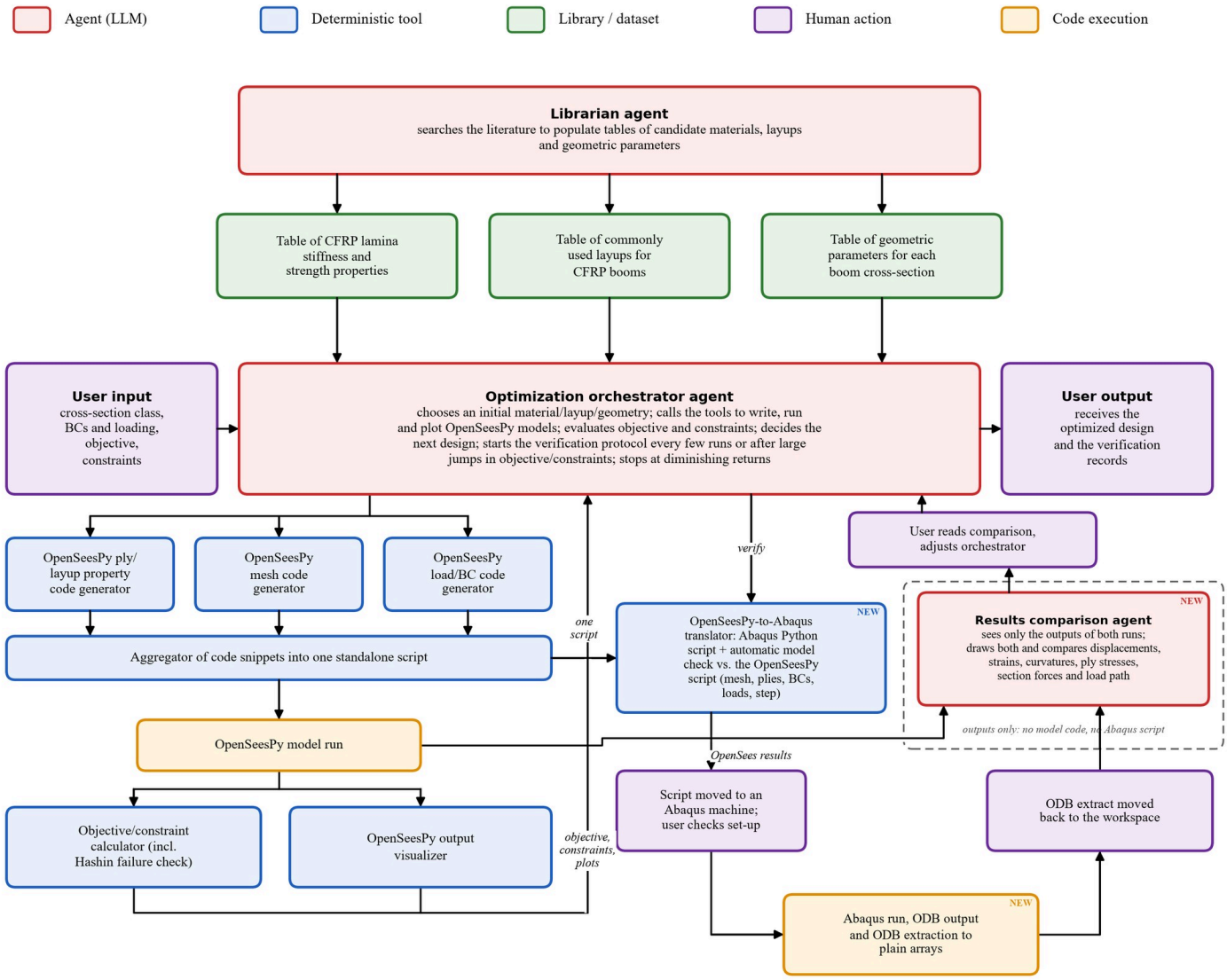
```
File created successfully at: D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py (f
ile state is current in your context — no need to Read it back)
```

```
wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png
```

Claude

The agent test went well. It touched only its allowed inputs, caught the planted ×0.9 S11 error, and noticed the stand-in extract was synthetic. Now checking the diagram render.

`Read` Read D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png



Claude

Two things to fix in the render: box titles came out in a sans-serif math font, and the "OpenSees results" route collides with labels and arrows near the comparison agent. Adjusting.

`Bash` Fix titles and results routing; re-render

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && /d/Users/tvenkata/miniforge3/envs/shells/p
ython.exe - <<'EOF'
from pathlib import Path
p = Path('system_diagram.py'); s = p.read_text(encoding='utf-8')
def rep(old, new):
    global s
    assert old in s, old; s = s.replace(old, new, 1)
rep("""    if title:
        ax.text(cx, cy, r'$\\bf{' + title.replace(' ', r'\\ ') + '}$' + '\\n' + text, ha='center', va='center',
                fontsize=fs, zorder=3, linespacing=1.25)""",
"""    if title:
        lh = fs * 1.3 / 72 * 10                    # line height in data units (10 units per inch)
        n = text.count('\\n') + 1
        ax.text(cx, cy + n * lh / 2, title, ha='center', va='center', fontsize=fs + 1, fontweight='bold', zorder=3)
        ax.text(cx, cy - lh / 2, text, ha='center', va='center', fontsize=fs, zorder=3, linespacing=1.25)""")
rep("""ax.add_patch(FancyBboxPatch((125.5, 35), 44, 23,""", """ax.add_patch(FancyBboxPatch((125.5, 34), 44, 24,""")
rep("""ax.text(147.5, 37.6, 'outputs only: no model code, no Abaqus script', ha='center', va='center', fontsize=FS - 3,""",
    """ax.text(145, 37.5, 'outputs only: no model code,\\nno Abaqus script', ha='center', va='center', fontsize=FS - 3,""")
```

```
rep("""arrow([(152.5, 32), (152.5, 35), (152.5, 41)])
arrow([(59, 35.5), (84, 35.5), (84, 36.5), (125.5, 36.5), (131, 36.5), (131, 41)],
      'OpenSees results', (104.5, 34.0), ha='center', halo=True)""",
"""arrow([(160, 32), (160, 41)])
arrow([(59, 35.5), (130, 35.5), (130, 41)], 'results', (91, 35.5), ha='center', halo=True)""")
p.write_text(s, encoding='utf-8')
EOF
/d/Users/tvenkata/miniforge3/envs/shells/python.exe system_diagram.py
```

```
<stdin>:7: SyntaxWarning: invalid escape sequence '\ '

Traceback (most recent call last):

  File "<stdin>", line 6, in <module>

  File "<stdin>", line 5, in rep

AssertionError:     if title:

        ax.text(cx, cy, r'$□f{' + title.replace(' ', r'\ ') + '}$' + '

' + text, ha='center', va='center',

                fontsize=fs, zorder=3, linespacing=1.25)

wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png
```

Claude

Escaping got mangled in the heredoc; I'll use the Edit tool directly.

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
-         ax.text(cx, cy, r'$\bf{' + title.replace(' ', r'\ ') + '}$' + '\n' + text, ha='center', va='center',
                fontsize=fs, zorder=3, linespacing=1.25)
+         lh = fs * 1.3 / 72 * 10                      # line height in data units (10 units per inch)
        n = text.count('\n') + 1
        ax.text(cx, cy + n * lh / 2, title, ha='center', va='center', fontsize=fs + 1, fontweight='bold', zorder=3)
        ax.text(cx, cy - lh / 2, text, ha='center', va='center', fontsize=fs, zorder=3, linespacing=1.25)
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
- ax.add_patch(FancyBboxPatch((125.5, 35), 44, 23,
+ ax.add_patch(FancyBboxPatch((125.5, 34), 44, 24,
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
- ax.text(147.5, 37.6, 'outputs only: no model code, no Abaqus script', ha='center', va='center', fontsize=FS - 3,
+ ax.text(145, 37.5, 'outputs only: no model code,\nno Abaqus script', ha='center', va='center', fontsize=FS - 3,
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
- arrow([(152.5, 32), (152.5, 35), (152.5, 41)])
arrow([(59, 35.5), (84, 35.5), (84, 36.5), (125.5, 36.5), (131, 36.5), (131, 41)],
      'OpenSees results', (104.5, 34.0), ha='center', halo=True)
+ arrow([(160, 32), (160, 41)])
arrow([(59, 35.5), (130, 35.5), (130, 41)], 'results', (91, 35.5), ha='center', halo=True)
```

`Bash` Re-render diagram

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && /d/Users/tvenkata/miniforge3/envs/shells/p
ython.exe system_diagram.py
```

```
The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
sfully. (file state is current in your context — no need to Read it back)
```

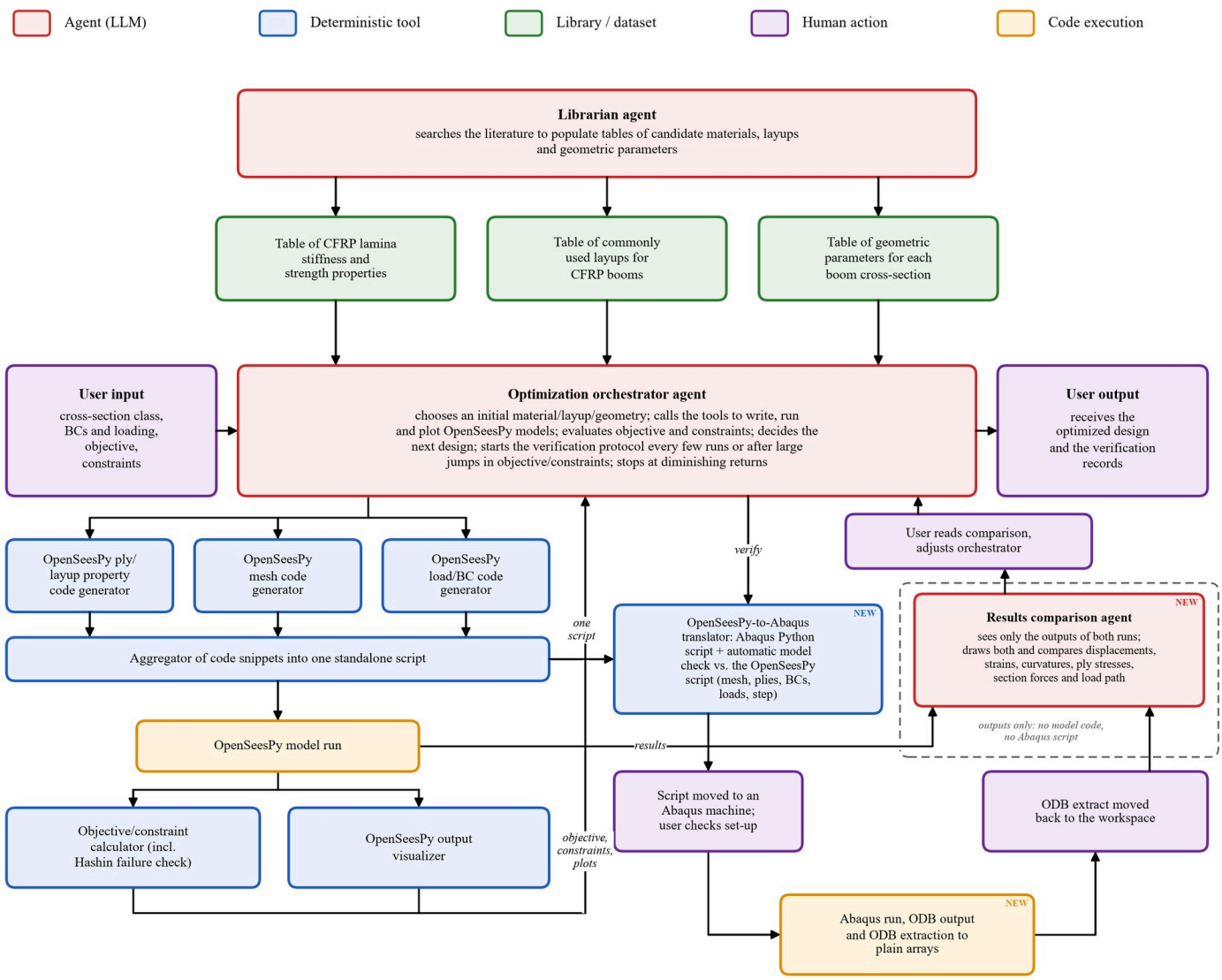
```
The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
sfully. (file state is current in your context — no need to Read it back)
```

```
The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
sfully. (file state is current in your context — no need to Read it back)
```

```
The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
sfully. (file state is current in your context — no need to Read it back)
```

```
wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png



Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\verification\compare_results.py

```
-     for k in ('section_points_per_ply', 'stress_note'):
+     if info['abaqus_version'] == 'mock':
        lines.insert(2, '**WARNING: the Abaqus file comes from the stand-in ODB (odb_mock.py), not from an Abaqus run: '
                        'this is a test of the tools, not a verification.**\n')
    for k in ('section_points_per_ply', 'stress_note'):
```

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to wr
ite it.</tool_use_error>

Bash Docs, orchestrator wiring, final test runs

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && printf '__pycache__/\n*.pyc\n_test_out/\n' > verification/.gitigno
re && /d/Users/tvenkata/miniforge3/envs/shells/python.exe - <<'EOF'
from pathlib import Path
def rep(path, old, new):
    p = Path(path); s = p.read_text(encoding='utf-8'); assert old in s, (path, old); p.write_text(s.replace(old, new, 1), e
ncoding='utf-8')
for f in ('agents/ThinShellOrchestrator.md', 'agents_v2.0/ThinShellOrchestrator.md'):
    rep(f, """shell-assemble-run and convert that script with the opensees-to-abaqus skill.""",
"""shell-assemble-run and convert that script with the opensees-to-abaqus skill.
When the user brings back the Abaqus results (ODB, or the `<job>_odb.npz` from
`.claude/verification/odb_extract.py`), hand ONLY the OpenSees results .npz
and the Abaqus results to the OpenSeesAbaqusResultsComparator agent; do not
give it the case, the scripts or your own conclusions (it must verify
independently of the code that built the models), and report its findings.""")
rep('skills/README.md', """                                                                    +--> opensees
-to-abaqus (Abaqus/CAE script via abaqus-export, model checked vs the script)
""", """                                                                  +--> opensees-to-abaqus (Abaqus/
CAE script via abaqus-export, model checked vs the script)
results .npz + Abaqus ODB (odb_extract.py --> <job>_odb.npz) --> OpenSeesAbaqusResultsComparator agent
   (.claude/verification/compare_results.py: outputs only, no model-building code; plots + comparison tables)
""")
```

```
log = Path('REORGANIZATION_LOG_v2.0.md'); s = log.read_text(encoding='utf-8').rstrip('\n')
s += """

## 11/4/26: independent results verification (OpenSeesAbaqusResultsComparator agent)

New agent `agents/OpenSeesAbaqusResultsComparator.md` (copy in `agents_v2.0/`)
with its own tools in `.claude/verification/` (deliberately not a skill folder,
and importing nothing from `.claude/skills/`):
- `odb_extract.py` (run with `abaqus python` where Abaqus is installed; an ODB
  cannot be read without Abaqus): nodes, elements, U/UR of every frame,
  reference points, LPF, centroid S/E/SE/SK/SF/SM per section point -> `<job>_odb.npz`.
- `compare_results.py`: OpenSees results .npz vs that extract. Finds the
  Abaqus axis order, pairs nodes/elements/reference points by position,
... [26 more lines]
```

```
(5.6 s)

43 checks, 43 passed, 0 failed (155 s)
0 test(s) failed
0
 M .claude/REORGANIZATION_LOG_v2.0.md
D  .claude/agents/OpenSeesPyToAbaqusPythonGenerator.md
 M .claude/agents/ThinShellOrchestrator.md
D  .claude/agents_v2.0/OpenSeesPyToAbaqusPythonGenerator.md
 M .claude/agents_v2.0/ThinShellOrchestrator.md
 M .claude/skills/README.md
 M .claude/skills/abaqus-export/SKILL.md
 M .claude/skills/abaqus-export/case_to_abaqus.py
 M .claude/skills/regression-tests/SKILL.md
 M .claude/skills/regression-tests/last_run.txt
 M .claude/skills/regression-tests/run_regression.py
?? .claude/agents/OpenSeesAbaqusResultsComparator.md
?? .claude/agents_v2.0/OpenSeesAbaqusResultsComparator.md
?? .claude/skills/opensees-to-abaqus/
?? .claude/verification/
?? "Code for Project/11-4-26 (Tara)/abaqus_plate_tip_load_laminate.py"
?? "Code for Project/11-4-26 (Tara)/abaqus_tape_spring_L024_tip_load.py"
?? "Code for Project/11-4-26 (Tara)/abaqus_trac_like_combination.py"
?? "Code for Project/11-4-26 (Tara)/abaqus_trac_like_load_control.py"
?? "Code for Project/11-4-26 (Tara)/system_diagram.py"
... [2 more lines]
```

Claude

The warning edit failed (the file changed after I last read it), so I'll re-read and apply it.

`Grep` Grep for k in \('section_points_per_ply'

```
551:    for k in ('section_points_per_ply', 'stress_note'):
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\verification\compare_results.py

```
548                 f'node pairing max distance {info["node_match_max"]:.3g} mm, element centroids {info["elem_match_ma
x"]:.3g} mm '
549                 f'(element size {info["element_size"]:.3g} mm): '
550                 + ('**same mesh**' if info['same_mesh'] else '**different meshes: nearest-point comparison, expect lar
ger differences**')]
551         for k in ('section_points_per_ply', 'stress_note'):
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\.claude\verification\compare_results.py

```
-     for k in ('section_points_per_ply', 'stress_note'):
+     if info['abaqus_version'] == 'mock':
        lines.insert(2, '**WARNING: the Abaqus file comes from the stand-in ODB (odb_mock.py), not from an Abaqus run: '
                        'this is a test of the tools, not a verification.**\n')
    for k in ('section_points_per_ply', 'stress_note'):
```

`Bash` Retest with mock warning

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/verification" && R="D:/Users/tvenkata/projects/ME295_project/Code for
Project/11-4-26 (Tara)/results"; SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project
-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; /d/Users/tvenkata/miniforge3/envs/shell
s/python.exe test_verification.py "$R/plate_tip_load_laminate.npz" "$R/trac_like_combination.npz" --out "$SP/vt" 2>&1 | tai
l -1; grep -c WARNING "$SP/vt/plate_tip_load_laminate/exact/"*.md; git -C .. diff --stat -- skills/regression-tests/last_ru
n.txt
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\verification\compare_results.py has been updated successfully.
```

```
0 test(s) failed

1
 .claude/skills/regression-tests/last_run.txt | 46 ++++++++++++++++++----------
 1 file changed, 29 insertions(+), 17 deletions(-)
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`SendUserFile` Send file: D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png

```
1 file delivered to user.
```

`mcp__ccd_turn__wrap_up` Recap

```
The new OpenSeesAbaqusResultsComparator agent compares the OpenSees results with an Abaqus ODB. It sees only the two runs' outputs. The revised system diagram is also done. Tool tests pass 25/25, but the agent hasn't been tested on a real ODB, and nothing is committed.
```

```
Shown in the next-step box. Your closing message is your answer to the user: write it now, in full, as the last thing you say this turn. The box asks your question, so do not ask it again.
```

Claude

The new agent, **OpenSeesAbaqusResultsComparator**, takes the OpenSeesPy results file and an Abaqus ODB, plots both runs and compares the key values. I also drew the revised system diagram (shown above). The agent has not yet been tested on a real ODB, because there is no Abaqus run here.

One extra step for the user. An ODB can only be opened by Abaqus's own Python, and Abaqus isn't installed on the VM. So after the Abaqus run, the user also runs this on the Abaqus machine:

```
abaqus python odb_extract.py job.odb
```

It copies the results into a small `job_odb.npz`, which the user brings back instead of the ODB. If the agent is ever run on a machine that has Abaqus, it does this step itself.

What it compares (`compare_results.py`, in `.claude/verification/`):

- displacements, rotations and the rigid-end reference points;
- membrane strains and curvatures;
- ply stresses at the top and bottom of every ply;
- section forces and moments;
- the load factor vs. displacement curve (Riks / displacement control).

Results go into a table with a PASS/CHECK verdict for each value. If a whole field is negated or off by a factor of 2, it is flagged as a likely convention difference (sign, or engineering vs. tensor shear). Five plots are drawn:

- the two deformed shapes overlaid;
- field maps on the unrolled shell (OpenSees, Abaqus, difference);
- Abaqus vs. OpenSees value at every point;
- values along the boom;
- the load path.

Keeping it independent

- The agent has no access to skills, and its instructions forbid opening the skills, the code pieces, the OpenSeesPy and Abaqus scripts, or the case files.
- Its own scripts import nothing from `.claude/skills/`. It works out ply stresses and section forces itself from the saved strains and laminate.
- This is enforced by instructions only. Claude Code can't block file reads for one agent without blocking them for the whole project.

Testing

- **Stand-in ODB:** I built a stand-in for Abaqus's ODB reader that serves data built from five OpenSees results files. Its axes and node numbering are arranged the way Abaqus stores them, and its stresses come from the failure-check skill. Exact copies pass every row. Four planted errors were each flagged with the right label: displacements ×1.05, kxy ×2, kxx with the sign flipped, and S11 ×0.9. That is 25/25.
- **The agent itself:** new agents only load at the next session start, so I ran its instructions through a general-purpose agent. The input was an extract with S11 planted 10% low. It stayed within its allowed files and flagged all 8 S11 rows. It also spotted that the file was synthetic, because S11 was off while the section forces still matched. Reports now carry a warning when the input comes from the stand-in.

Other changes

- **Abaqus scripts:** they now also ask Abaqus to save `E` and `SE` (strains, plus section strains and curvatures), so the ODB has what the comparison needs. I regenerated the four scripts in the 11-4-26 folder (13/13 checks each); regression T20/T22 pass 43/43. This change hasn't been run in Abaqus. If your Abaqus version rejects a name in that list, remove it from the request in `case_to_abaqus.py`.
- **Orchestrator:** it now hands only the two result files to the new agent, never the case, the scripts or its own conclusions. I also updated the skills README and the reorganization log.

Revised diagram: the old Abaqus verifier agent is split into two parts, both marked NEW:

- **Translator:** the deterministic OpenSeesPy-to-Abaqus step, with its automatic model check.
- **Results comparison agent:** sits inside a dashed "outputs only" boundary.

The extraction step is added to the orange Abaqus-run box, and the comparison agent takes the OpenSees results directly from the model run. The script is `system_diagram.py` if you want to change wording or layout.

Files are in D:\Users\tvenkata\projects\ME295_project:

- .claude\agents\OpenSeesAbaqusResultsComparator.md
- .claude\verification\
- Code for Project\11-4-26 (Tara)\system_diagram_verification.png
- Code for Project\11-4-26 (Tara)\system_diagram.py

Tara

What does "one script" mean on the arrow?

Claude

It's the single, standalone OpenSeesPy script that the aggregator builds. The aggregator is the shell-assemble-run skill: it joins the laminate, mesh and load/BC pieces into one file, `<name>_opensees.py`. That file is the translator's only input, so the arrow carries the whole OpenSeesPy model as one script, not the three separate pieces or the case file.

The label is too terse to read that way on its own. Something like "assembled OpenSeesPy script" (on two lines to fit the gap) would be clearer, or "OpenSeesPy model script". It's a one-line change in `system_diagram.py`, and I can re-render the figure with whichever wording you prefer.

Tara

Shouldn't it be on the arrow from the aggregator to the translator?

`Grep` Grep 86, 40, 123, 55|one\\nscript|\(86, 21, 112, 32|99, 40|99, 21

```
116:box(86, 40, 123, 55, 'tool', text='OpenSeesPy-to-Abaqus\ntranslator: Abaqus Python\nscript + automatic model\n'
118:arrow([(77, 47.5), (86, 47.5)], 'one\nscript', (81.5, 51.5), ha='center')
120:box(86, 21, 112, 32, 'human', text='Script moved to an\nAbaqus machine;\nuser checks set-up', fs=FS - 0.5)
121:arrow([(99, 40), (99, 32)])
123:arrow([(99, 21), (99, 9.5), (109, 9.5)])
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
- box(86, 40, 123, 55, 'tool',
+ box(94, 40, 125, 55, 'tool',
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
- arrow([(77, 47.5), (86, 47.5)], 'one\nscript', (81.5, 51.5), ha='center')
+ arrow([(77, 47.5), (94, 47.5)], 'assembled\nscript', (88.5, 47.5), ha='center')
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
- box(86, 21, 112, 32, 'human', text='Script moved to an\nAbaqus machine;\nuser checks set-up', fs=FS - 0.5)
arrow([(99, 40), (99, 32)])
+ box(91, 21, 117, 32, 'human', text='Script moved to an\nAbaqus machine;\nuser checks set-up', fs=FS - 0.5)
arrow([(104, 40), (104, 32)])
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py

```
- arrow([(99, 21), (99, 9.5), (109, 9.5)])
+ arrow([(104, 21), (104, 9.5), (109, 9.5)])
```

> The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
> sfully. (file state is current in your context — no need to Read it back)

> The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
> sfully. (file state is current in your context — no need to Read it back)

> The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
> sfully. (file state is current in your context — no need to Read it back)

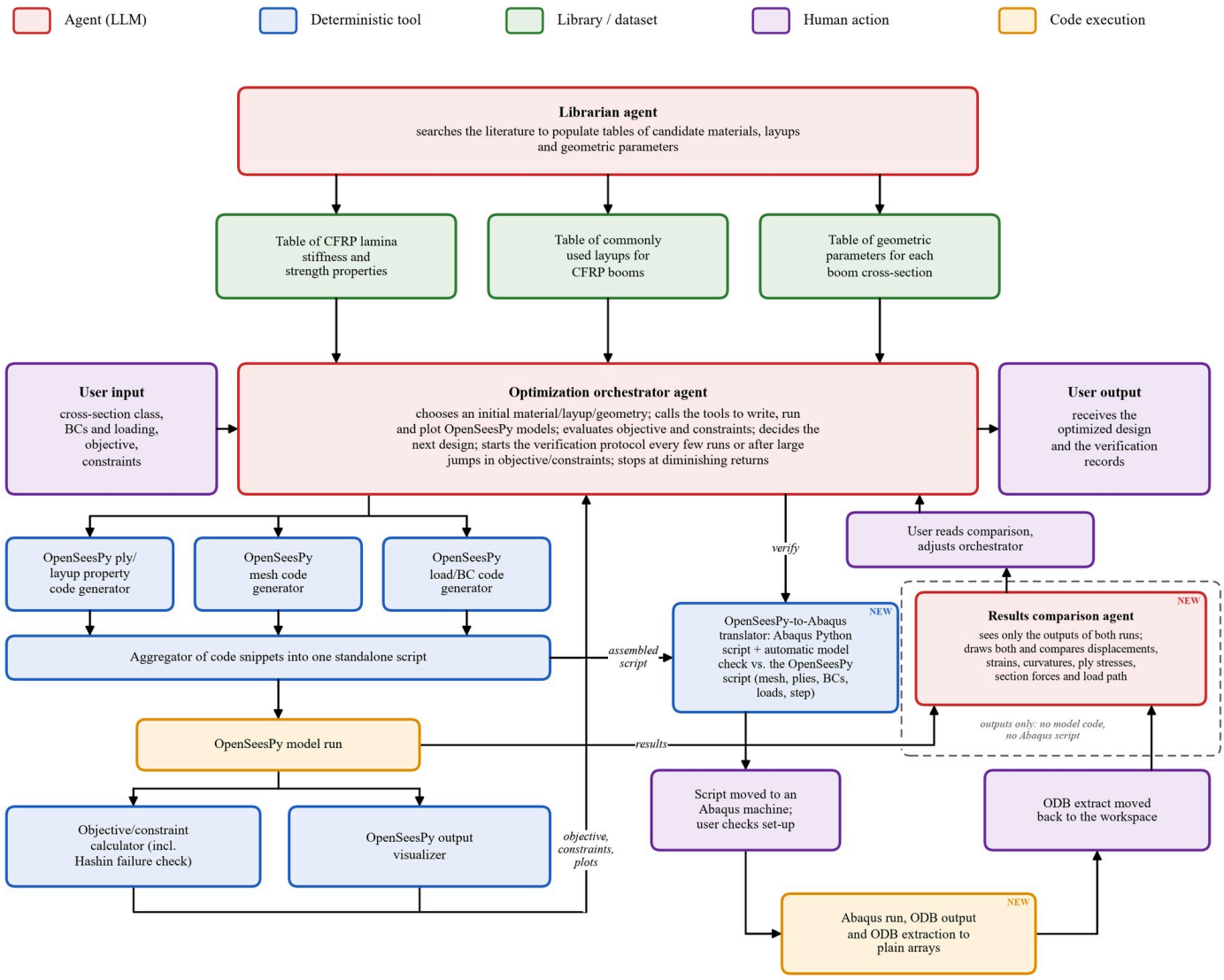
> The file D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram.py has been updated succes
> sfully. (file state is current in your context — no need to Read it back)

`Bash` Center verify arrow and re-render

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && sed -i "s/arrow(\[(104.5, 70), (104.5, 55)
\], 'verify', (104.5, 62.5), ha='center')/arrow([(109.5, 70), (109.5, 55)], 'verify', (109.5, 62.5), ha='center')/" system_
diagram.py && grep -n "'verify'" system_diagram.py && /d/Users/tvenkata/miniforge3/envs/shells/python.exe system_diagram.py
```

```
119:arrow([(109.5, 70), (109.5, 55)], 'verify', (109.5, 62.5), ha='center')
wrote D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png
[This command modified 1 file you've previously read: system_diagram.py. Call Read before editing.]
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png



`SendUserFile` Send file: D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)\system_diagram_verification.png

> 1 file delivered to user.

`mcp__ccd_turn__wrap_up` Recap

```
The "assembled script" label now sits on the arrow from the aggregator to the translator. The diagram is re-rendered and ch
ecked. Nothing is committed yet.
```

> Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.

Claude

Yes, that's where it belongs. The label sat above the arrow, right against the vertical "objective, constraints, plots" line, so it read as that line's label.

It now sits on the arrow from the aggregator to the translator and reads "assembled script". To make room, I narrowed the translator box and moved it right, and moved the "Script moved to an Abaqus machine" box and its arrows with it. I also re-centred the "verify" arrow on the translator. The re-rendered figure is shown above.

Tara

Commit this to GitHub and sync from the GitHub to my computer

`Bash` Show status and fetch remote

```
cd "D:/Users/tvenkata/projects/ME295_project" && git status --short && git remote -v && git fetch origin 2>&1 | tail -2 &&
git log --oneline -1 origin/main && git rev-list --left-right --count main...origin/main
```

```
M .claude/REORGANIZATION_LOG_v2.0.md
D  .claude/agents/OpenSeesPyToAbaqusPythonGenerator.md
 M .claude/agents/ThinShellOrchestrator.md
D  .claude/agents_v2.0/OpenSeesPyToAbaqusPythonGenerator.md
 M .claude/agents_v2.0/ThinShellOrchestrator.md
 M .claude/skills/README.md
 M .claude/skills/abaqus-export/SKILL.md
 M .claude/skills/abaqus-export/case_to_abaqus.py
 M .claude/skills/regression-tests/SKILL.md
 M .claude/skills/regression-tests/last_run.txt
 M .claude/skills/regression-tests/run_regression.py
?? .claude/agents/OpenSeesAbaqusResultsComparator.md
?? .claude/agents_v2.0/OpenSeesAbaqusResultsComparator.md
?? .claude/skills/opensees-to-abaqus/
?? .claude/verification/
?? "Code for Project/11-4-26 (Tara)/abaqus_plate_tip_load_laminate.py"
?? "Code for Project/11-4-26 (Tara)/abaqus_tape_spring_L024_tip_load.py"
?? "Code for Project/11-4-26 (Tara)/abaqus_trac_like_combination.py"
?? "Code for Project/11-4-26 (Tara)/abaqus_trac_like_load_control.py"
?? "Code for Project/11-4-26 (Tara)/system_diagram.py"
?? "Code for Project/11-4-26 (Tara)/system_diagram_verification.png"
origin  https://github.com/fbenazzo/ME295_project.git (fetch)
origin  https://github.com/fbenazzo/ME295_project.git (push)
c38e2df ThinShellOrchestrator: combine catalog pointers with shell-assemble-run note
0       3
... [1 more lines]
```

Claude

GitHub has 3 commits that aren't here yet, one of which touches the orchestrator I edited. Checking them before committing.

`Bash` Inspect incoming remote commits

```
cd "D:/Users/tvenkata/projects/ME295_project" && git log --format='%h %an %ad %s' --date=short main..origin/main && git dif
f --stat main origin/main && ls .claude/verification .claude/skills/opensees-to-abaqus
```

```
c38e2df fbenazzo 2026-11-04 ThinShellOrchestrator: combine catalog pointers with shell-assemble-run note
b41530c fbenazzo 2026-11-04 Merge remote-tracking branch 'origin/main'
3eb064c fbenazzo 2026-11-04 Add layup and coilable-boom geometry catalogs for the orchestrator
 .claude/agents/ThinShellOrchestrator.md            |  37 +-
 .claude/agents_v2.0/ThinShellOrchestrator.md       |  30 +-
 .claude/memory.boom_geometry_catalog/boom_data.py  | 324 +++++++++
 .../memory.boom_geometry_catalog/boom_geometry.py  | 103 +++
 .../boom_geometry_catalog.txt                      | 243 +++++++
 .../boom_sections_all.svg                          | 804 ++++++++++++++++++++++
 .../boom_sections_selected.svg                     | 139 ++++
 .../make_boom_catalog.py                           | 255 +++++++
 .claude/memory.layup_catalog/layup_catalog.txt     | 479 +++++++++++++
 .claude/memory.layup_catalog/make_layup_catalog.py |  87 +++
 10 files changed, 2494 insertions(+), 7 deletions(-)
```

```
.claude/skills/opensees-to-abaqus:
SKILL.md
opensees_to_abaqus.py

.claude/verification:
compare_results.py
odb_extract.py
odb_mock.py
test_verification.py
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Bash Commit work and merge GitHub changes

```
cd "D:/Users/tvenkata/projects/ME295_project" && git add -A .claude "Code for Project/11-4-26 (Tara)" && git status --short
| grep -v "^[AMDR] " ; git commit -q -F - <<'EOF'
opensees-to-abaqus skill, results comparator agent, revised system diagram

- New skill .claude/skills/opensees-to-abaqus: Abaqus/CAE script from the
  assembled OpenSeesPy script (shell-assemble-run), via abaqus-export;
  reference values from the script's own results; model check of the Abaqus
  script (stand-in API) against the script's recorded ops calls. Replaces the
  OpenSeesPyToAbaqusPythonGenerator agent (removed; v1 kept in agents_v1.0).
- case_to_abaqus.py: generate() split into generate() + write_script();
  field output now also requests E and SE (section strains, curvatures).
- New agent OpenSeesAbaqusResultsComparator: compares OpenSees results with
  an Abaqus ODB from the outputs only, with its own tools in
  .claude/verification (odb_extract.py for abaqus python, compare_results.py,
  stand-in ODB and tests: 25/25).
- ThinShellOrchestrator: uses shell-assemble-run + opensees-to-abaqus and
  hands only results to the comparator.
- Regression T22 (opensees-to-abaqus); default suite 257/257.
- 11-4-26 (Tara): Abaqus scripts of the four cases; revised system diagram
  (system_diagram.py, system_diagram_verification.png).

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>
EOF
git log --oneline -1 && git merge --no-edit origin/main 2>&1 | tail -5
```

```
warning: in the working copy of '.claude/skills/abaqus-export/case_to_abaqus.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of '.claude/skills/regression-tests/run_regression.py', LF will be replaced by CRLF the next t
ime Git touches it
warning: in the working copy of '.claude/agents/OpenSeesAbaqusResultsComparator.md', LF will be replaced by CRLF the next t
ime Git touches it
warning: in the working copy of '.claude/agents_v2.0/OpenSeesAbaqusResultsComparator.md', LF will be replaced by CRLF the n
ext time Git touches it
warning: in the working copy of '.claude/skills/opensees-to-abaqus/SKILL.md', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of '.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of '.claude/verification/.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.claude/verification/odb_extract.py', LF will be replaced by CRLF the next time Git touche
s it
warning: in the working copy of '.claude/verification/test_verification.py', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'Code for Project/11-4-26 (Tara)/system_diagram.py', LF will be replaced by CRLF the next t
ime Git touches it
6ad47b8 opensees-to-abaqus skill, results comparator agent, revised system diagram
Auto-merging .claude/agents/ThinShellOrchestrator.md
CONFLICT (content): Merge conflict in .claude/agents/ThinShellOrchestrator.md
Auto-merging .claude/agents_v2.0/ThinShellOrchestrator.md
CONFLICT (content): Merge conflict in .claude/agents_v2.0/ThinShellOrchestrator.md
Automatic merge failed; fix conflicts and then commit the result.
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Bash Show conflict hunks

```
cd "D:/Users/tvenkata/projects/ME295_project" && git diff .claude/agents/ThinShellOrchestrator.md; echo ======; git diff .c
laude/agents_v2.0/ThinShellOrchestrator.md | head -60
```

```
diff --cc .claude/agents/ThinShellOrchestrator.md
index 8b20709,3eaf319..0000000
--- a/.claude/agents/ThinShellOrchestrator.md
+++ b/.claude/agents/ThinShellOrchestrator.md
@@@ -21,15 -21,34 +21,46 @@@ Tara `D:\Users\tvenkata\miniforge3\envs
   Material data: `.claude/memory.material_property_catalog/UD_lamina_catalog_SI.txt`
   (read by laminate-properties; list it with `laminate_properties.py --list-catalog`).
   To add or check catalog materials, use the CompositeLaminaLibrarian agent
++<<<<<<< HEAD
 +(literature work), not this one. When the user wants one self-contained
 +OpenSeesPy script of the model, use the shell-assemble-run skill instead of
 +running shell_bc_load.py. For an Abaqus version of a model, assemble it with
 +shell-assemble-run and convert that script with the opensees-to-abaqus skill.
 +When the user brings back the Abaqus results (ODB, or the `<job>_odb.npz` from
```

```
 +`.claude/verification/odb_extract.py`), hand ONLY the OpenSees results .npz
 +and the Abaqus results to the OpenSeesAbaqusResultsComparator agent; do not
 +give it the case, the scripts or your own conclusions (it must verify
 +independently of the code that built the models), and report its findings.
++=======
+ (literature work), not this one.
+
+ Layup data: `.claude/memory.layup_catalog/layup_catalog.txt` lists the
+ layups to simulate (IDs Y001-Y455): every symmetric 1-5 ply sequence with
+ angles 0, 15, ..., 90 deg, bottom to top, plus thickness at 0.045 mm plies,
+ number of distinct angles (= coincident elements per facet, i.e. run cost)
... [91 more lines]
```

Bash Resolve orchestrator conflicts

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe - <<'EOF'
import re
from pathlib import Path
for f in ('agents/ThinShellOrchestrator.md', 'agents_v2.0/ThinShellOrchestrator.md'):
    p = Path(f); s = p.read_text(encoding='utf-8')
    blocks = re.findall(r'<<<<<<< HEAD\n(.*?)=======\n(.*?)>>>>>>> origin/main\n', s, flags=re.S)
    assert len(blocks) == 2, (f, len(blocks))
    ours, theirs = blocks[0]
    keep = theirs[:theirs.index('For an Abaqus version of a case, use the')]
    mine = ours[ours.index('When the user wants one self-contained'):]
    merged1 = keep + mine
    merged2 = blocks[1][0]
    s = s.replace(f'<<<<<<< HEAD\n{ours}=======\n{theirs}>>>>>>> origin/main\n', merged1, 1)
    s = s.replace(f'<<<<<<< HEAD\n{blocks[1][0]}=======\n{blocks[1][1]}>>>>>>> origin/main\n', merged2, 1)
    assert '<<<<<<<' not in s and '>>>>>>>' not in s
    p.write_text(s, encoding='utf-8')
EOF
diff agents/ThinShellOrchestrator.md agents_v2.0/ThinShellOrchestrator.md && echo identical; sed -n 1,5p agents/ThinShellOr
chestrator.md | cut -c1-120; sed -n 21,62p agents/ThinShellOrchestrator.md; grep -rn "OpenSeesPyToAbaqus" agents agents_v2.
0 | head
```

```
Traceback (most recent call last):
  File "<stdin>", line 6, in <module>
AssertionError: ('agents_v2.0/ThinShellOrchestrator.md', 1)
23a24,34
> <<<<<<< HEAD
> (literature work), not this one. When the user wants one self-contained
> OpenSeesPy script of the model, use the shell-assemble-run skill instead of
> running shell_bc_load.py. For an Abaqus version of a model, assemble it with
> shell-assemble-run and convert that script with the opensees-to-abaqus skill.
> When the user brings back the Abaqus results (ODB, or the `<job>_odb.npz` from
> `.claude/verification/odb_extract.py`), hand ONLY the OpenSees results .npz
> and the Abaqus results to the OpenSeesAbaqusResultsComparator agent; do not
> give it the case, the scripts or your own conclusions (it must verify
> independently of the code that built the models), and report its findings.
> =======
48,56c59,63
< When the user wants one self-contained
< OpenSeesPy script of the model, use the shell-assemble-run skill instead of
< running shell_bc_load.py. For an Abaqus version of a model, assemble it with
< shell-assemble-run and convert that script with the opensees-to-abaqus skill.
< When the user brings back the Abaqus results (ODB, or the `<job>_odb.npz` from
< `.claude/verification/odb_extract.py`), hand ONLY the OpenSees results .npz
< and the Abaqus results to the OpenSeesAbaqusResultsComparator agent; do not
< give it the case, the scripts or your own conclusions (it must verify
< independently of the code that built the models), and report its findings.
... [55 more lines]
```

Claude

The live `agents/` copy is resolved. The `agents_v2.0/` copy has only one conflict block, because Federico's two copies differ slightly. I'll check how they differ before resolving it.

Bash Compare Federico's two orchestrator copies

```
cd "D:/Users/tvenkata/projects/ME295_project" && diff <(git show origin/main:.claude/agents/ThinShellOrchestrator.md) <(git
show origin/main:.claude/agents_v2.0/ThinShellOrchestrator.md)
```

```
fatal: ambiguous argument 'origin\main;.claude\agents\ThinShellOrchestrator.md': unknown revision or path not in the workin
g tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
fatal: ambiguous argument 'origin\main;.claude\agents_v2.0\ThinShellOrchestrator.md': unknown revision or path not in the w
orking tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Bash` Compare Federico's two orchestrator copies

```
cd "D:/Users/tvenkata/projects/ME295_project" && MSYS_NO_PATHCONV=1 git show "origin/main:.claude/agents/ThinShellOrchestrator.md" > /tmp/a.md && MSYS_NO_PATHCONV=1 git show "origin/main:.claude/agents_v2.0/ThinShellOrchestrator.md" > /tmp/b.md && diff /tmp/a.md /tmp/b.md
```

```
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

Federico's two copies are identical, so the resolved `agents/` file is right for both. Copying it over, then committing the merge.

`Bash` Finish conflict resolution and commit merge

```
cd "D:/Users/tvenkata/projects/ME295_project" && cp .claude/agents/ThinShellOrchestrator.md .claude/agents_v2.0/ThinShellOrchestrator.md && grep -c "<<<<<<<\|>>>>>>>\|OpenSeesPyToAbaqus" .claude/agents/ThinShellOrchestrator.md .claude/agents_v2.0/ThinShellOrchestrator.md; grep -n "opensees-to-abaqus\|ResultsComparator\|layup_catalog\|boom_geometry" .claude/agents/ThinShellOrchestrator.md | cut -c1-90; git add .claude/agents/ThinShellOrchestrator.md .claude/agents_v2.0/ThinShellOrchestrator.md && git status --short | head && git commit -q --no-edit && git log --oneline -3
```

```
.claude/agents/ThinShellOrchestrator.md:0
.claude/agents_v2.0/ThinShellOrchestrator.md:0
3:description: Sets up, runs, checks and reports OpenSeesPy simulations of thin composite
26:Layup data: `.claude/memory.layup_catalog/layup_catalog.txt` lists the
33:`make_layup_catalog.py` in the same folder.
35:Boom geometry: `.claude/memory.boom_geometry_catalog/boom_geometry_catalog.txt`
51:shell-assemble-run and convert that script with the opensees-to-abaqus skill.
54:and the Abaqus results to the OpenSeesAbaqusResultsComparator agent; do not
85:#   python opensees-to-abaqus/opensees_to_abaqus.py <dir>/<case>_opensees.py [--element
114:  run is untested, and the differences listed in opensees-to-abaqus SKILL.md.
M  .claude/agents/ThinShellOrchestrator.md
M  .claude/agents_v2.0/ThinShellOrchestrator.md
A  .claude/memory.boom_geometry_catalog/boom_data.py
A  .claude/memory.boom_geometry_catalog/boom_geometry.py
A  .claude/memory.boom_geometry_catalog/boom_geometry_catalog.txt
A  .claude/memory.boom_geometry_catalog/boom_sections_all.svg
A  .claude/memory.boom_geometry_catalog/boom_sections_selected.svg
A  .claude/memory.boom_geometry_catalog/make_boom_catalog.py
A  .claude/memory.layup_catalog/layup_catalog.txt
A  .claude/memory.layup_catalog/make_layup_catalog.py
20d6483 Merge remote-tracking branch 'origin/main'
6ad47b8 opensees-to-abaqus skill, results comparator agent, revised system diagram
c38e2df ThinShellOrchestrator: combine catalog pointers with shell-assemble-run note
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Bash` Push to GitHub and confirm sync

```
cd "D:/Users/tvenkata/projects/ME295_project" && git push origin main 2>&1 | tail -3 && git fetch origin -q && git status -sb | head -1
```

```
remote:   https://github.com/citgitfr/BenazzoVenkatadri_ThinShellSims.git
To https://github.com/fbenazzo/ME295_project.git
   c38e2df..20d6483  main -> main
## main...origin/main
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

The push went through, but GitHub printed a message about another repository address. Checking what it says.

`Bash` Read the full remote notice

```
cd "D:/Users/tvenkata/projects/ME295_project" && git ls-remote --get-url origin; git push --dry-run origin main 2>&1 | head -8
```

```
https://github.com/fbenazzo/ME295_project.git
Everything up-to-date
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Bash` Check whether the repo has moved

```
cd "D:/Users/tvenkata/projects/ME295_project" && git ls-remote https://github.com/fbenazzo/ME295_project.git 2>&1 | head -3; gh repo view fbenazzo/ME295_project --json nameWithOwner,url 2>&1 | head -3
```

```
20d6483cb365b77aafdd97cffc47ac1f4d934457        HEAD
b5e140ccea3772bff809c78b5ccc337d38abce93        refs/heads/fbenazzo/opensees-shell-verification
20d6483cb365b77aafdd97cffc47ac1f4d934457        refs/heads/main
/usr/bin/bash: line 1: gh: command not found
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`mcp__ccd_turn__wrap_up` Recap

```
Your work is committed and pushed to GitHub, merged with Federico's three new commits (the layup and boom-geometry catalog
s). Your local main and GitHub's main are now the same commit, 20d6483.
```

> Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.

Claude

Everything is committed and pushed to GitHub, and your computer is in sync with it. Your local main and GitHub's main are the same commit, `20d6483`.

What happened

- **Your commit** (`6ad47b8`) includes:
- the opensees-to-abaqus skill;
- the OpenSeesAbaqusResultsComparator agent and its tools in `.claude/verification/`;
- the deletion of the old OpenSeesPyToAbaqusPythonGenerator agent;
- the updated docs, the T22 regression test and the latest regression log;
- the four regenerated Abaqus scripts;
- the revised system diagram and its script.
- **Federico's changes:** GitHub had three newer commits from Federico, adding a layup catalog and a coilable-boom geometry catalog. They also edited both ThinShellOrchestrator files, which I had changed too, so the merge stopped on those two files.
- **Conflict resolution:** I kept Federico's new "Layup data" and "Boom geometry" sections. I replaced his last paragraph, which still sent Abaqus work to the deleted agent, with the opensees-to-abaqus and results-comparator instructions. His two orchestrator copies were identical, so both files now match. The merge commit is `20d6483`, and Federico's catalog files are now on your computer.
- **Push:** `main` went to `github.com/fbenazzo/ME295_project` (`c38e2df..20d6483`).

During the push, GitHub printed `github.com/citgitfr/BenazzoVenkatadri_ThinShellSims.git` as a remote message. The push still went to the usual repository, and I left the remote address unchanged. That line may mean the repository has a new home or a mirror, so it's worth asking Federico.

There is also a branch on GitHub, `fbenazzo/opensees-shell-verification`, that isn't part of main. I didn't touch it.

Tara

Create a PDF of this conversation in the Claude Conversations folder. This should be Tara conversation 2 today, with a brief description of what this conversation involved as part of the file title

`Bash` List Claude Conversations folder

```
cd "D:/Users/tvenkata/projects/ME295_project/Claude Conversations" && ls -la
```

```
total 1844
drwxr-xr-x 1 tvenkata 1049089       0 Oct 26 11:41 .
drwxr-xr-x 1 tvenkata 1049089       0 Oct 26 12:34 ..
-rw-r--r-- 1 tvenkata 1049089  420065 Oct 26 09:39 Tara 10-23-26 Conversation 1 (Python setup and Federico's 10-22 test cas
es).pdf
-rw-r--r-- 1 tvenkata 1049089  331029 Oct 26 09:38 Tara 10-23-26 Conversation 2 (Building material property agents).pdf
-rw-r--r-- 1 tvenkata 1049089  116608 Oct 26 09:38 Tara 10-23-26 Conversation 3 (Testing material property agents).pdf
-rw-r--r-- 1 tvenkata 1049089 1007527 Oct 26 11:41 Tara 10-26-26 Conversation 1 (CFRP laminate test cases, result viewer an
d Abaqus agents).pdf
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`mcp__ccd_session_mgmt__export_transcript` mcp__ccd_session_mgmt__export_transcript

```
{"session_id": "self"}
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\Claude Conversations\Tara 10-26-26 Conversation 1 (CFRP laminate test cases, result viewer and Abaqus agents).pdf

```
Exported this session's transcript "OpenSeesPy to Abaqus Python skill" to D:\Users\tvenkata\Downloads\session-export-179382
5406353.zip (12 MB, 12536432 bytes). The zip holds the conversation transcript, its subagent transcripts and session metada
ta, not the app's logs; nothing was uploaded.
```

```
pdftoppm is not installed. Install poppler-utils (e.g. `brew install poppler` or `apt-get install poppler-utils`) to enable PDF page rendering.
```

Bash Unzip export; check PDF libraries and old PDF text

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; mkdir -p "$SP/export" && cd "$SP/export" && unzip -o -q "D:/Users/tvenkata/Downloads/session-export-1793825406353.zip" && find . -maxdepth 3 | head -20; /d/Users/tvenkata/miniforge3/envs/shells/python.exe -c "
import pypdf" 2>&1; /d/Users/tvenkata/miniforge3/envs/shells/python.exe -c "import reportlab; print('reportlab', reportlab.Version)" 2>&1 | tail -1; /d/Users/tvenkata/miniforge3/envs/shells/python.exe -c "
import fitz; d=fitz.open(r'D:/Users/tvenkata/projects/ME295_project/Claude Conversations/Tara 10-26-26 Conversation 1 (CFRP laminate test cases, result viewer and Abaqus agents).pdf'); print(len(d)); print(d[0].get_text()[:1500])" 2>&1 | tail -30
```

```
.
./f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6
./f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/custom-title.json
./f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/subagents
./f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/subagents/agent-a18ddd19d155bf788.meta.json
./f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/subagents/agent-a18ddd19d155bf788.prefix.json
./f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6.jsonl
./local-session-state.json
./metadata.json
./transcript.jsonl
Traceback (most recent call last):

  File "<string>", line 2, in <module>

ModuleNotFoundError: No module named 'pypdf'

ModuleNotFoundError: No module named 'reportlab'

Traceback (most recent call last):

  File "<string>", line 2, in <module>

ModuleNotFoundError: No module named 'fitz'
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Bash Check browsers and transcript structure

```
ls "/c/Program Files (x86)/Microsoft/Edge/Application/msedge.exe" "/c/Program Files/Google/Chrome/Application/chrome.exe" 2>&1; SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad/export"; cd "$SP"; wc -l transcript.jsonl f0b4*.jsonl; /d/Users/tvenkata/miniforge3/envs/shells/python.exe -c "
import json,collections
c=collections.Counter()
for l in open('transcript.jsonl',encoding='utf-8'):
    d=json.loads(l); t=d.get('type'); c[t]+=1
    if t in('user','assistant'):
        m=d['message']; cont=m['content']
        if isinstance(cont,list):
            for b in cont: c[t+':'+b.get('type','?')]+=1
        else: c[t+':str']+=1
print(c)
d=json.loads(open('transcript.jsonl',encoding='utf-8').readline()); print(list(d))"
```

```
ls: cannot access '/c/Program Files/Google/Chrome/Application/chrome.exe': No such file or directory
/c/Program Files (x86)/Microsoft/Edge/Application/msedge.exe
     841 transcript.jsonl
     841 f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6.jsonl
    1682 total
Counter({'assistant': 233, 'user': 136, 'assistant:tool_use': 126, 'user:tool_result': 126, 'api-request': 101, 'assistant:thinking': 77, 'custom-title': 63, 'last-prompt': 62, 'atis-latch': 62, 'agent-name': 62, 'api-request-blob': 48, 'attachment': 33, 'assistant:text': 30, 'queue-operation': 18, 'user:str': 10, 'file-history-delta': 9, 'file-history-snapshot': 6, 'system': 6, 'api-request-shape': 2})

['type', 'operation', 'timestamp', 'sessionId', 'content']
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Write Write D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\export\to_html.py

```
"""Transcript JSONL -> printable HTML (user messages, assistant text, tool calls, trimmed results, images)."""
import html
import json
import re
import sys
from pathlib import Path

src, out, title = Path(sys.argv[1]), Path(sys.argv[2]), sys.argv[3]
MAXLINES = 25
HIDE_TOOLS = {'mcp__ccd_turn__update_plan', 'ToolSearch'}
```

```
def clean_user(t):
    t = re.sub(r'<system-reminder.*?</system-reminder>', '', t, flags=re.S)
    t = re.sub(r'<task-notification>.*?</task-notification>', '', t, flags=re.S)
    return t.strip()


def esc(t):
    return html.escape(t)


def trim(t, n=MAXLINES):
    lines = t.splitlines()
    if len(lines) > n:
        return '\n'.join(lines[:n]) + f'\n... [{len(lines) - n} more lines]'
    return t


def md(t):
... [117 more lines]
```

```
File created successfully at: D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\export\to_html.py (file state is current in your context — no need to Read it back)
```

Bash Build HTML from transcript

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad/export"; cd "$SP" && sed -i 's/^seen = set()$/seen = set()\nHIDDEN = set()/' to_html.py && T="Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skill, results comparator agent and system diagram)" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe to_html.py transcript.jsonl conv.html "$T" && ls -la conv.html && grep -c "<img" conv.html && grep -o '<div class="user"><div class="who">Tara</div>[^<]\{0,120\}' conv.html
```

```
conv.html 262

-rw-r--r-- 1 tvenkata 1049089 3619304 Nov  4 12:50 conv.html
6
<div class="user"><div class="who">Tara</div>I want to convert the OpenSeesPyToAbaqusPythonGenerator into a skill that is used by the orchestrator agent.  It should
<div class="user"><div class="who">Tara</div>I now want to create a new agent, which takes the OpenSeesPy results output by the assemble-run skill and an Abaqus ODB
<div class="user"><div class="who">Tara</div>What does &amp;quot;one script&amp;quot; mean on the arrow?
<div class="user"><div class="who">Tara</div>Shouldn&amp;#x27;t it be on the arrow from the aggregator to the translator?
<div class="user"><div class="who">Tara</div>Commit this to GitHub and sync from the GitHub to my computer
<div class="user"><div class="who">Tara</div>Create a PDF of this conversation in the Claude Conversations folder.  This should be Tara conversation 2 today, with a
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
[This command modified 1 file you've previously read: ..\..\..\..\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-\f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6\scratchpad\export\to_html.py. Call Read before editing.]
```

Claude

All six of your messages are in. Printing to PDF with Edge.

Bash Print HTML to PDF with headless Edge

```
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad/export"; OUT="D:\\Users\\tvenkata\\projects\\ME295_project\\Claude Conversations\\Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skill, results comparator agent and system diagram).pdf"; WIN_SP=$(cygpath -w "$SP"); "/c/Program Files (x86)/Microsoft/Edge/Application/msedge.exe" --headless=new --disable-gpu --no-pdf-header-footer --user-data-dir="$WIN_SP\\edge" --print-to-pdf="$OUT" "file:///$(cygpath -m "$SP")/conv.html" 2>&1 | tail -3; ls -la "/d/Users/tvenkata/projects/ME295_project/Claude Conversations/"
```

```
[8124:10908:1104/125107.342:ERROR:chrome\browser\task_manager\providers\fallback_task_provider.cc:126] Every renderer should have at least one task provided by a primary task provider. If a "Renderer" fallback task is shown, it is a bug. If you have repro steps, please file a new bug and tag it as a dependency of crbug.com/40528867.

[8124:9860:1104/125107.535:ERROR:extensions\browser\crx_installer.cc:1125] Could not load background script ''. ihejeaahjpbegmaaegiikmlphghlfmeh

2828313 bytes written to file D:\Users\tvenkata\projects\ME295_project\Claude Conversations\Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skill, results comparator agent and system diagram).pdf

total 4768
drwxr-xr-x 1 tvenkata 1049089       0 Nov  4 12:51 .
drwxr-xr-x 1 tvenkata 1049089       0 Oct 26 12:34 ..
-rw-r--r-- 1 tvenkata 1049089  420065 Oct 26 09:39 Tara 10-23-26 Conversation 1 (Python setup and Federico's 10-22 test cases).pdf
-rw-r--r-- 1 tvenkata 1049089  331029 Oct 26 09:38 Tara 10-23-26 Conversation 2 (Building material property agents).pdf
-rw-r--r-- 1 tvenkata 1049089  116608 Oct 26 09:38 Tara 10-23-26 Conversation 3 (Testing material property agents).pdf
-rw-r--r-- 1 tvenkata 1049089 1007527 Oct 26 11:41 Tara 10-26-26 Conversation 1 (CFRP laminate test cases, result viewer and Abaqus agents).pdf
-rw-r--r-- 1 tvenkata 1049089  160409 Nov  4 12:50 Tara 11-4-26 Conversation 1 (Skill to join laminate, mesh and BC-load Op
```

```
enSeesPy code into one script, GitHub sync).pdf
-rw-r--r-- 1 tvenkata 1049089 2828313 Nov  4 12:51 Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skill, results comparato
r agent and system diagram).pdf
```

Claude

The PDF is written (2.8 MB). Today's Conversation 1 PDF is also there now, from your other session, and the names follow the same pattern. I'll open it to check that it renders.

`mcp__Claude_Browser__preview_start` mcp__Claude_Browser__preview_start

```
{"url": "file:///D:/Users/tvenkata/projects/ME295_project/Claude%20Conversations/Tara%2011-4-26%20Conversation%202%20(OpenS
ees-to-Abaqus%20skill,%20results%20comparator%20agent%20and%20system%20diagram).pdf"}
```

```
opened file:///D:/Users/tvenkata/projects/ME295_project/Claude%20Conversations/Tara%2011-4-26%20Conversation%202%20(OpenSee
s-to-Abaqus%20skill,%20results%20comparator%20agent%20and%20system%20diagram).pdf in the preview pane (files outside the pr
oject folder render as static snapshots)
```

`mcp__Claude_Browser__computer` mcp__Claude_Browser__computer

```
{"action": "screenshot", "scale": 0.6}
```

```
This tab shows a local file, not a web page; page tools can't act on it here. Open web pages in a new tab (`tabs_create`, t
hen `navigate`).
```

Claude

The browser pane won't screenshot a local PDF, so I'll check the PDF's structure directly instead.

`Bash` Count PDF pages and images

```
/d/Users/tvenkata/miniforge3/envs/shells/python.exe -c "
import re
b=open(r'D:/Users/tvenkata/projects/ME295_project/Claude Conversations/Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skil
l, results comparator agent and system diagram).pdf','rb').read()
print('pages', len(re.findall(rb'/Type\s*/Page[^s]', b)), 'images', len(re.findall(rb'/Subtype\s*/Image', b)), b[:8])"
```

```
pages 45 images 8 b'%PDF-1.4'
```

`mcp__Claude_Browser__tabs_context` mcp__Claude_Browser__tabs_context

```
{}
```

```
{
  "browserOpen": true,
  "tabs": [
    {
      "tabId": "seed",
      "origin": "",
      "isActive": false
    },
    {
      "tabId": "tab-1",
      "origin": "file:///D:/Users/tvenkata/projects/ME295_project/Claude%20Conversations/Tara%2011-4-26%20Conversation%202%
20(OpenSees-to-Abaqus%20skill,%20results%20comparator%20agent%20and%20system%20diagram).pdf",
      "isActive": true
    }
  ]
}
The Browser pane is currently hidden.
```

`mcp__ccd_turn__wrap_up` Recap

```
The conversation PDF is in Claude Conversations as Tara 11-4-26 Conversation 2: 45 pages with the plots and diagrams. I che
cked its structure but not its pages by eye. It isn't committed to GitHub.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude

The PDF is in the Claude Conversations folder as **Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skill, results comparator agent and system diagram).pdf**. It's 45 pages and includes 8 images: the plots and the diagram versions. I checked its structure but couldn't view the pages here, so it's worth a quick look.

- **What's in it:** your messages, my replies, and the commands and file edits I made. Long command outputs and file contents are cut to their first lines.
- **What's left out:** my internal reasoning and the app's own system messages. A note at the top of the PDF says so.
- **Where it ends:** at your request for the PDF, since the transcript was exported at that point.

• **Naming:** it follows the same pattern as the earlier files. The Conversation 1 PDF from your other session today is also in the folder.
• **Leftover file:** making the PDF also saved a 12 MB zip of the session to `D:\Users\tvenkata\Downloads\session-export-1793825406353.zip`. You can delete it if you don't want it.

The PDF isn't committed to GitHub yet.

Tara

Can the command to turn the ODB into something that you can read be incorporated into the Abaqus Python script built by the translator skill? If so, please do that

`Bash` Check extractor for non-ASCII and structure

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/verification" && grep -nP '[^\x00-\x7F]' odb_extract.py | head; grep -n "'''" odb_extract.py; grep -n "^def \|^INSTANCE\|INSTANCE\[0\]\|path, out, opt = parse" odb_extract.py
```

```
34:def parse(argv):
49:def node_vectors(fo, index, asm_index, n, n_asm):
57:        elif v.nodeLabel in index and v.instance.name == INSTANCE[0]:
62:def element_field(fo, eindex):
73:        if v.instance is None or v.instance.name != INSTANCE[0] or v.elementLabel not in eindex:
91:INSTANCE = [None]
94:def main():
95:    path, out, opt = parse(sys.argv[1:])
100:    INSTANCE[0] = opt.get('instance', names[0])
102:        print('several instances %s; using %s (--instance to choose)' % (names, INSTANCE[0]))
103:    inst = odb.rootAssembly.instances[INSTANCE[0]]
143:    meta = dict(odb=path, step=step.name, steps=steps, instance=INSTANCE[0], frame_descriptions=fdesc,
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py

```
1       """
2       Copies the results of an Abaqus ODB into a plain .npz that compare_results.py
3       reads on a machine without Abaqus. Only Abaqus' own Python can open an ODB, so
4       run this where Abaqus is installed (Python 2.7 or 3, numpy ships with Abaqus):
5
6         abaqus python odb_extract.py <job>.odb [<job>_odb.npz] [--step NAME] [--instance NAME]
7
8       Written (final frame of the step unless stated):
9         node_labels, node_xyz        instance nodes (ODB coordinates, as stored)
10        elem_labels, elem_conn       instance shell elements (4 node labels; 3-node: last repeated)
11        asm_labels, asm_xyz, asm_sets  assembly-level nodes (reference points), set names
12        U, UR                        (frames, instance nodes, 3) every frame of the step
13        asm_U, asm_UR                (frames, assembly nodes, 3)
14        frame_value                  frame values (step time, or arc length for Riks)
15        LPF                          Riks load proportionality factor per frame (if in the ODB)
16        F_<name>                     element field at the centroid, (elements, section points, components)
17                                     for S, E, SE, SK, SF, SM, when present (final frame)
18        meta                         JSON: ODB path, step, instance, Abaqus version, component labels,
19                                     section point numbers and descriptions, frame descriptions
20      No model data other than what the ODB holds is used.
21      """
22      from __future__ import print_function
23
24      import json
25      import sys
... [22 more lines]
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py

```
88          return arr, sps, dict((str(k), desc[k]) for k in desc)
89
90
91      INSTANCE = [None]
92
93
94      def main():
95          path, out, opt = parse(sys.argv[1:])
96          odb = openOdb(path, readOnly=True)
97          steps = list(odb.steps.keys())
98          step = odb.steps[opt.get('step', steps[-1])]
99          names = [k for k in odb.rootAssembly.instances.keys() if k != 'ASSEMBLY']
100         INSTANCE[0] = opt.get('instance', names[0])
101         if len(names) > 1 and 'instance' not in opt:
102             print('several instances %s; using %s (--instance to choose)' % (names, INSTANCE[0]))
```

```
103         inst = odb.rootAssembly.instances[INSTANCE[0]]
104         labels = [nd.label for nd in inst.nodes]
105         xyz = np.array([nd.coordinates for nd in inst.nodes], float)
106         index = dict((l, k) for k, l in enumerate(labels))
107         shells = [e for e in inst.elements if len(e.connectivity) in (3, 4)]
108         elabels = [e.label for e in shells]
109         conn = np.array([list(e.connectivity) + [e.connectivity[-1]] * (4 - len(e.connectivity)) for e in shells])
110         eindex = dict((l, k) for k, l in enumerate(elabels))
111
112         asm_labels, asm_xyz, asm_sets = [], [], []
... [55 more lines]
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py

```
- def main():
    path, out, opt = parse(sys.argv[1:])
    odb = openOdb(path, readOnly=True)
    steps = list(odb.steps.keys())
    step = odb.steps[opt.get('step', steps[-1])]
    names = [k for k in odb.rootAssembly.instances.keys() if k != 'ASSEMBLY']
    INSTANCE[0] = opt.get('instance', names[0])
    if len(names) > 1 and 'instance' not in opt:
        print('several instances %s; using %s (--instance to choose)' % (names, INSTANCE[0]))
+ def extract(path, out=None, step=None, instance=None):
    """Write the .npz of ODB `path` (default <job>_odb.npz); returns a one-line summary."""
    out = out or path[:-4] + '_odb.npz'
    odb = openOdb(path, readOnly=True)
    steps = list(odb.steps.keys())
    step = odb.steps[step or steps[-1]]
    names = [k for k in odb.rootAssembly.instances.keys() if k != 'ASSEMBLY']
    INSTANCE[0] = instance or names[0]
    if len(names) > 1 and not instance:
        print('several instances %s; using %s (--instance to choose)' % (names, INSTANCE[0]))
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py

```
-     np.savez_compressed(out, **data)
    print('wrote %s: %d nodes, %d shell elements, %d assembly node(s), %d frame(s), fields %s'
          % (out, len(labels), len(elabels), len(asm_labels), len(frames), sorted(meta['fields'])))
+     np.savez_compressed(out, **data)
    return ('wrote %s: %d nodes, %d shell elements, %d assembly node(s), %d frame(s), fields %s'
            % (out, len(labels), len(elabels), len(asm_labels), len(frames), sorted(meta['fields'])))


def main():
    path, out, opt = parse(sys.argv[1:])
    print(extract(path, out, opt.get('step'), opt.get('instance')))
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py

```
-   abaqus python odb_extract.py <job>.odb [<job>_odb.npz] [--step NAME] [--instance NAME]

+   abaqus python odb_extract.py <job>.odb [<job>_odb.npz] [--step NAME] [--instance NAME]

Scripts written by the opensees-to-abaqus skill (abaqus-export) carry a
verbatim copy of this file and call extract() after the job, so their run
already leaves <job>_odb.npz next to the ODB.
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py has been updated successfully.
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py has been updated successfully.
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\verification\odb_extract.py has been updated successfully.
```

`Grep` Grep mdb.saveAs|^def write_script|text = ABAQUS % dict|model=_literal\(model\), ref=_literal\(ref\)

```
467:mdb.saveAs(pathName=JOB + '.cae')
545:def write_script(case, lam, out, element='S4R', ref=None, source='a shell-bc-load case'):
564:    text = ABAQUS % dict(name=name, source=source, summary=summary, script=out.name,
566:                         model=_literal(model), ref=_literal(ref))
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
460
461     model.fieldOutputRequests['F-Output-1'].setValues(variables=('S', 'E', 'SE', 'SF', 'SM', 'U', 'UR', 'RF'), frequenc
y=1)   # SE: section strains + curvatures (SK)
462
463     job = mdb.Job(name=JOB, model=model.name, type=ANALYSIS, nodalOutputPrecision=SINGLE,
464                   memoryUnits=PERCENTAGE, memory=90)
465     job.submit(consistencyChecking=OFF)
466     job.waitForCompletion()
467     mdb.saveAs(pathName=JOB + '.cae')
468
469     # --- Results ---
470     odb = openOdb(JOB + '.odb', readOnly=True)
471     step = odb.steps[STEP]
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
    ABAQUS = r'''"""
    Abaqus/CAE model generated by case_to_abaqus.py (.claude/skills/abaqus-export) from the
    OpenSeesPy case "%(name)s" (%(source)s).
    NOT RUN IN ABAQUS when generated (Abaqus is not installed on the project VM).

    %(summary)s

    Units: N, mm, MPa, t/mm^3 (Abaqus has no units; these are the case units).

    Usage (Abaqus command line, in an empty working folder):
      abaqus cae noGUI=%(script)s                # %(size)g mm, %(element)s
      abaqus cae noGUI=%(script)s -- 5 S4        # element size [mm] and element type
    Output: <job>_results.txt (also printed), <job>.odb, <job>.cae
    """
```

Read Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
        ref = opensees_reference(case) if reference else None
        out = Path(out) if out else Path(case_path).parent / f'abaqus_{case.get("name", "case")}.py'
        return write_script(case, lam, out, element, ref, f'shell-bc-load case file {Path(case_path).name}')


    def write_script(case, lam, out, element='S4R', ref=None, source='a shell-bc-load case'):
        """Abaqus script of `case` with laminate `lam` (a laminate_properties.Laminate,
        used instead of the case's laminate entry) and OpenSees reference values
        `ref` (None: not embedded). Returns (path, model data, ref)."""
        mesh = sm.build_from_spec(case['geometry'])
        mesh.size = case['geometry']['size']
        model = describe_model(case, mesh, lam)
        reference = ref is not None
        ref = ref if reference else NO_REFERENCE
        name = model['name']
        out = Path(out)
        summary = '\n'.join([
            f'Geometry: {mesh.summary()}',
            f'Section:  {lam.describe()}' + ('' if lam.kind == 'isotropic' else
                                            f', plies bottom (SNEG) to top: {"/".join(f"{a:g}" for a in lam.angles)}'),
            f'Supports: {json.dumps(case.get("supports", []))}',
            f'Loads:    {json.dumps(case.get("loads", []))}',
            f'Analysis: {json.dumps(case.get("analysis", {"type": "linear"}))}',
            'OpenSees reference: ' + ('embedded (REF)' if reference else 'NOT embedded (--no-reference)')])
        text = ABAQUS % dict(name=name, source=source, summary=summary, script=out.name,
... [5 more lines]
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
- mdb.saveAs(pathName=JOB + '.cae')

# --- Results ---
+ mdb.saveAs(pathName=JOB + '.cae')

# --- Plain-array copy of the ODB for the OpenSees vs Abaqus results comparison ---
# (verbatim copy of .claude/verification/odb_extract.py; bring <job>_odb.npz back with the
# results). Done before the comparison below, so a failure there does not stop it.
ODB_EXTRACT = %(extractor)s
try:
    _ns = {'__name__': 'odb_extract'}
    exec(compile(ODB_EXTRACT, 'odb_extract.py', 'exec'), _ns)
    log(_ns['extract'](JOB + '.odb', JOB + '_odb.npz'))
except Exception as exc:
    log('ODB extraction failed (%%s); on this machine run: abaqus python odb_extract.py %%s.odb' %% (exc, JOB))

# --- Results ---
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
- Output: <job>_results.txt (also printed), <job>.odb, <job>.cae
"""
+ Output: <job>_results.txt (also printed), <job>.odb, <job>.cae, and <job>_odb.npz
(the ODB results as plain arrays, for the OpenSees vs Abaqus results comparison)
"""
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py

```
-                         model=_literal(model), ref=_literal(ref))
    out.write_text(text, encoding='utf-8')
    return out, model, ref
+                         model=_literal(model), ref=_literal(ref), extractor=_extractor())
    out.write_text(text, encoding='utf-8')
    return out, model, ref


ODB_EXTRACTOR = _SKILLS.parent / 'verification' / 'odb_extract.py'
```

```
def _extractor():
    """Source of the results-comparison ODB extractor as a raw string literal."""
    src = ODB_EXTRACTOR.read_text(encoding='utf-8')
    if "'''" in src or src.rstrip().endswith('\\') or not src.isascii():
        raise SystemExit(f'{ODB_EXTRACTOR}: cannot be embedded as a raw string (\'\'\', trailing \\ or non-ASCII)')
    return "r'''" + src + "'''"
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py has been updated successfu
lly.
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py has been updated successfu
lly.
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\abaqus-export\case_to_abaqus.py has been updated successfu
lly.
```

Grep Grep ref, _, _ = o2a.reference\(src, info\)|if case is T20_CASE:

```
778:                    ref, _, _ = o2a.reference(src, info)
790:            if case is T20_CASE:                                # a hand-edited script must fail
```

Read Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py

```
756     def T22_opensees_to_abaqus():
757         """opensees-to-abaqus: Abaqus script from an assembled OpenSeesPy script; model check and reference."""
758         import json
759         import tempfile
760         for d in ('shell-assemble-run', 'opensees-to-abaqus', 'abaqus-export'):
761             sys.path.insert(0, str(SKILLS / d))
762         import assemble_run as ar
763         import case_to_abaqus as c2a
764         import opensees_to_abaqus as o2a
765         ex = SKILLS / 'shell-bc-load' / 'examples'
766         t20 = dict(T20_CASE, name='trac_like_load_control', analysis={'type': 'load_control', 'n_steps': 2})
767         cases = [json.loads((ex / 'tape_spring_L024_tip_load.json').read_text()), T20_CASE, t20]
768         with tempfile.TemporaryDirectory() as tmp:
769             for case in cases:
770                 src, out = Path(tmp) / f'{case["name"]}_opensees.py', Path(tmp) / f'abaqus_{case["name"]}.py'
771                 with redirect_stdout(io.StringIO()):
772                     ar.join(ar.write_pieces(case, Path(tmp) / f'{case["name"]}_code'), src)
773                     ops.wipe()
774                     info = o2a.read_script(src)
775                     os_m = o2a.opensees_model(*o2a.record_model(src))
776                     ref = None
777                     if case['analysis']['type'] == 'linear':          # reference from the script's own run
778                         ref, _, _ = o2a.reference(src, info)
779                     _, M, _ = c2a.write_script(info['CASE'], info['LAMINATE'], out, 'S4R', ref, 'T22')
780                     rows = o2a.check_model(os_m, out, M, info['CASE'], info['LAMINATE'])
... [11 more lines]
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py

```
-                    check(f'{case["name"]}: REF {k} from the script = from run_case', np.abs(a - b).max() / np.abs(b).max
(), 0, 1e-9, True)
            if case is T20_CASE:
+                    check(f'{case["name"]}: REF {k} from the script = from run_case', np.abs(a - b).max() / np.abs(b).max
(), 0, 1e-9, True)
                embedded_extractor(out, src.parent / 'results' / f'{case["name"]}.npz', Path(tmp), case['name'])
            if case is T20_CASE:
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py

```
- def abaqus_mock_P(g):
+ def embedded_extractor(script, npz, tmp, name):
    """The ODB extractor embedded in a generated Abaqus script is the verification tool
    verbatim, and on a stand-in ODB (from npz) writes the same arrays as the tool run alone."""
    import ast
    import runpy
    ver = SKILLS.parent / 'verification'
    sys.path.insert(0, str(ver))
    import odb_mock
    src = [n.value.value for n in ast.parse(Path(script).read_text(encoding='utf-8')).body
           if isinstance(n, ast.Assign) and getattr(n.targets[0], 'id', '') == 'ODB_EXTRACT']
    check(f'{name}: embedded ODB extractor = .claude/verification/odb_extract.py',
          float(src == [(ver / 'odb_extract.py').read_text(encoding='utf-8')]), 1, 0, True)
    if not src:
        return
    files = []
    for k in range(2):
        odb_mock.install(odb_mock.make_odb(npz))
        out = Path(tmp) / f'{name}_odb_{k}.npz'
        with redirect_stdout(io.StringIO()):
            if k == 0:                                  # as the generated script does it
                ns = {'__name__': 'odb_extract'}
                exec(compile(src[0], 'odb_extract.py', 'exec'), ns)
                ns['extract']('job.odb', str(out))
            else:                                       # the tool on its own
```

```
                argv, sys.argv = sys.argv, ['odb_extract.py', 'job.odb', str(out)]
                try:
                    runpy.run_path(str(ver / 'odb_extract.py'), run_name='__main__')
                finally:
                    sys.argv = argv
... [11 more lines]
```

Bash Run T20 and T22 with embedded extractor checks

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills/regression-tests" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe run_regression.py --only T20 T22 2>&1 | grep -v "^Using\|OpenSees >\|WARNING: CTest\|after: 40\|NewtonRaphson\|StaticAnalysis" | grep "FAIL\|embedded\|checks,\|exception"
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py has been updated successfully.
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py has been updated successfully.
```

```
[PASS] tape_spring_L024_tip_load: embedded ODB extractor = .claude/verification/odb_extract.py got +1  ref +1  diff 0.00e+00 (tol 0e+00)
  [PASS] tape_spring_L024_tip_load: embedded extraction writes the same arrays as odb_extract.py (18 arrays) got +1  ref +1  diff 0.00e+00 (tol 0e+00)
45 checks, 45 passed, 0 failed (148 s)
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Bash Regenerate scripts; run embedded block for success and failure

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && for s in plate_tip_load_laminate tape_spring_L024_tip_load trac_like_combination trac_like_load_control; do /d/Users/tvenkata/miniforge3/envs/shells/python.exe ../../.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py ${s}_opensees.py 2>&1 | grep "checks pass\|FAIL"; done
SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; mkdir -p "$SP/snip" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe - "$SP/snip" <<'EOF'
import os, sys, numpy as np
sys.path.insert(0, r'D:/Users/tvenkata/projects/ME295_project/.claude/verification')
import odb_mock
text = open('abaqus_trac_like_combination.py', encoding='utf-8').read()
snippet = text[text.index('ODB_EXTRACT = '):text.index('# --- Results ---')]
os.chdir(sys.argv[1])
for label, odb in (('success', odb_mock.make_odb(r'D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)/results/trac_like_combination.npz')), ('failure', None)):
    logged = []
    if odb is None:
        import types
        sys.modules['odbAccess'] = types.SimpleNamespace(openOdb=lambda p, readOnly=True: (_ for _ in ()).throw(IOError('no such file: ' + p)))
    else:
        odb_mock.install(odb)
    ns = {'JOB': 'trac_like_combination_S4R_5mm', 'log': logged.append}
    exec(compile(snippet, 'snippet', 'exec'), ns)
    print(label, '->', logged)
d = np.load('trac_like_combination_S4R_5mm_odb.npz'); print(sorted(d.files))
EOF
/d/Users/tvenkata/miniforge3/envs/shells/python.exe -m py_compile abaqus_trac_like_combination.py && echo compiles; grep -n "^ODB_EXTRACT\|^# --- Plain\|_odb.npz" abaqus_trac_like_combination.py | head
```

```
13/13 checks pass
13/13 checks pass
13/13 checks pass
13/13 checks pass
success -> ["wrote trac_like_combination_S4R_5mm_odb.npz: 492 nodes, 440 shell elements, 1 assembly node(s), 4 frame(s), fields ['S', 'SE', 'SF', 'SK', 'SM']"]

failure -> ['ODB extraction failed (no such file: trac_like_combination_S4R_5mm.odb); on this machine run: abaqus python odb_extract.py trac_like_combination_S4R_5mm.odb']

['F_S', 'F_SE', 'F_SF', 'F_SK', 'F_SM', 'LPF', 'U', 'UR', 'asm_U', 'asm_UR', 'asm_labels', 'asm_sets', 'asm_xyz', 'elem_conn', 'elem_labels', 'frame_value', 'meta', 'node_labels', 'node_xyz']

compiles
18:Output: <job>_results.txt (also printed), <job>.odb, <job>.cae, and <job>_odb.npz
10575:# --- Plain-array copy of the ODB for the OpenSees vs Abaqus results comparison ---
10576:# (verbatim copy of .claude/verification/odb_extract.py; bring <job>_odb.npz back with the
10578:ODB_EXTRACT = r'''"""
10583:  abaqus python odb_extract.py <job>.odb [<job>_odb.npz] [--step NAME] [--instance NAME]
10587:already leaves <job>_odb.npz next to the ODB.
10626:    out = pos[1] if len(pos) > 1 else pos[0][:-4] + '_odb.npz'
10676:    """Write the .npz of ODB `path` (default <job>_odb.npz); returns a one-line summary."""
10677:    out = out or path[:-4] + '_odb.npz'
10758:    log(_ns['extract'](JOB + '.odb', JOB + '_odb.npz'))
```

Bash Find doc lines to update

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && grep -n "odb_extract\|_odb.npz\|results.txt\`, \`.odb\`\|outputs \`<job>" skills/opensees-to-abaqus/SKILL.md skills/abaqus-export/SKILL.md agents/OpenSeesAbaqusResultsComparator.md agents/
```

ThinShellOrchestrator.md

```
skills/opensees-to-abaqus/SKILL.md:63:  `... -- <size> <S4R|S4>`), outputs `<job>_results.txt`, `.odb`, `.cae`;
agents/OpenSeesAbaqusResultsComparator.md:16:  `.odb` from the user or the `<job>_odb.npz` extracted from it, the
agents/OpenSeesAbaqusResultsComparator.md:18:  `.claude/verification/` (`odb_extract.py`, `compare_results.py`, `odb_mock.p
y`,
agents/OpenSeesAbaqusResultsComparator.md:40:- If it does: `abaqus python .claude/verification/odb_extract.py <job>.odb <jo
b>_odb.npz [--step NAME]`.
agents/OpenSeesAbaqusResultsComparator.md:43:  `<job>_odb.npz` (small; the ODB itself is not needed here). Give them the
agents/OpenSeesAbaqusResultsComparator.md:44:  file `.claude/verification/odb_extract.py` and the exact command. Do not
agents/OpenSeesAbaqusResultsComparator.md:54:python .claude/verification/compare_results.py <dir>/results/<name>.npz <job>_
odb.npz [--out DIR] [--tol 0.03]
agents/ThinShellOrchestrator.md:52:When the user brings back the Abaqus results (ODB, or the `<job>_odb.npz` from
agents/ThinShellOrchestrator.md:53:`.claude/verification/odb_extract.py`), hand ONLY the OpenSees results .npz
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Read Read D:\Users\tvenkata\projects\ME295_project\.claude\agents\OpenSeesAbaqusResultsComparator.md

```
34      `D:\Users\fbenazzo\miniconda3\envs\shells\python.exe`, Tara
35      `D:\Users\tvenkata\miniforge3\envs\shells\python.exe`). Units: N, mm, MPa.
36
37      ## 1. Get the Abaqus results out of the ODB
38      An ODB can only be opened by Abaqus' own Python. Check whether this machine
39      has Abaqus (`where abaqus` / `which abaqus`).
40      - If it does: `abaqus python .claude/verification/odb_extract.py <job>.odb <job>_odb.npz [--step NAME]`.
41      - If not (the project VM has no Abaqus): ask the user to run that same
42        command on the Abaqus machine, in the folder with the ODB, and to bring back
43        `<job>_odb.npz` (small; the ODB itself is not needed here). Give them the
44        file `.claude/verification/odb_extract.py` and the exact command. Do not
45        guess ODB contents.
46      The extract holds node coordinates and labels, connectivity, U/UR of every
47      frame, reference points, LPF, and element fields S, E, SE (section strains
48      and curvatures), SK, SF, SM at the centroid of each section point. Fields the
49      ODB lacks are reported as missing; ask the user whether the run requested
50      them (field output `S, E, SE, SF, SM, U, UR`).
51
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\agents\OpenSeesAbaqusResultsComparator.md

```
- An ODB can only be opened by Abaqus' own Python. Check whether this machine
has Abaqus (`where abaqus` / `which abaqus`).
- If it does:
+ An ODB can only be opened by Abaqus' own Python. Abaqus runs of the project's
translated models normally write `<job>_odb.npz` themselves (with a copy of
`odb_extract.py`) next to `<job>.odb`: ask the user for that file first. Only
if there is none (another Abaqus model, or the extraction failed, which the
run's `<job>_results.txt` says), check whether this machine has Abaqus
(`where abaqus` / `which abaqus`).
- If it does:
```

Bash Update skill docs, orchestrator, regression table, log

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && cp agents/OpenSeesAbaqusResultsComparator.md agents_v2.0/ && /d/Us
ers/tvenkata/miniforge3/envs/shells/python.exe - <<'EOF'
from pathlib import Path
def rep(path, old, new):
    p = Path(path); s = p.read_text(encoding='utf-8'); assert old in s, (path, old); p.write_text(s.replace(old, new, 1), e
ncoding='utf-8')
rep('skills/opensees-to-abaqus/SKILL.md', "  `... -- <size> <S4R|S4>`), outputs `<job>_results.txt`, `.odb`, `.cae`;",
    "  `... -- <size> <S4R|S4>`), outputs `<job>_results.txt`, `.odb`, `.cae` and\n"
    "  `<job>_odb.npz` (the ODB as plain arrays, written by the embedded copy of\n"
    "  `.claude/verification/odb_extract.py`): the user brings back `<job>_odb.npz`\n"
    "  (and `<job>_results.txt`) for the OpenSeesAbaqusResultsComparator agent;")
rep('skills/abaqus-export/SKILL.md', "## Translation", """- After the job the script writes `<job>_odb.npz`: the ODB as pla
in arrays,
  by a verbatim copy of `.claude/verification/odb_extract.py` embedded at
  generation time (read from that file, so regenerate after changing it). It
  runs before the script's own comparison; if it fails, `<job>_results.txt`
  gives the `abaqus python odb_extract.py <job>.odb` command to run instead.
  Regression T22 checks the copy is verbatim and writes the same arrays.

## Translation""")
for f in ('agents/ThinShellOrchestrator.md', 'agents_v2.0/ThinShellOrchestrator.md'):
    rep(f, """When the user brings back the Abaqus results (ODB, or the `<job>_odb.npz` from
`.claude/verification/odb_extract.py`), hand ONLY the OpenSees results .npz""",
"""When the user brings back the Abaqus results (the `<job>_odb.npz` the Abaqus
script writes next to the ODB, or the ODB), hand ONLY the OpenSees results .npz""")
rep('skills/regression-tests/SKILL.md', "an edited script is caught |", "an edited script is caught; the embedded ODB extra
ctor is verbatim and writes the same arrays as `odb_extract.py` |")
log = Path('REORGANIZATION_LOG_v2.0.md'); s = log.read_text(encoding='utf-8').rstrip('\n')
s += """

## 11/4/26: ODB extraction inside the generated Abaqus scripts

Scripts from abaqus-export / opensees-to-abaqus now carry a verbatim copy of
```

```
`.claude/verification/odb_extract.py` (refactored: `extract(path, out, step,
... [13 more lines]
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\agents\OpenSeesAbaqusResultsComparator.md has been updated succes
sfully.
```

```
same
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

Docs updated. Last: the verifier's own tests (they run the refactored extractor from the command line) and the full regression suite.

`Bash` Run verification tests and full regression

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/verification" && R="D:/Users/tvenkata/projects/ME295_project/Code for
Project/11-4-26 (Tara)/results"; SP="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project
-Code-for-Project-11-4-26--Tara-/f0b4b3cd-64bc-41cd-a524-7e4de2dcf5a6/scratchpad"; /d/Users/tvenkata/miniforge3/envs/shell
s/python.exe test_verification.py "$R/plate_tip_load_laminate.npz" "$R/tape_spring_L024_tip_load.npz" "$R/trac_like_combina
tion.npz" "$R/trac_like_load_control.npz" "$SP/t/results/al_plate_pressure.npz" --out "$SP/vt" 2>&1 | tail -1; cd ../skill
s/regression-tests && /d/Users/tvenkata/miniforge3/envs/shells/python.exe run_regression.py --log last_run.txt > /dev/null
2>&1; echo "exit $?"; tail -2 last_run.txt
```

```
0 test(s) failed

exit 0


259 checks, 259 passed, 0 failed (289 s)
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`mcp__ccd_turn__wrap_up` Recap

```
Generated Abaqus scripts now write <job>_odb.npz themselves after the job, using an exact copy of the verifier's extractor.
The full regression passes 259/259 and the verifier tests 25/25. The Abaqus run is still untested, and nothing is committe
d.
```

```
Shown in the next-step box. Your closing message is your answer to the user: write it now, in full, as the last thing you s
ay this turn. The box asks your question, so do not ask it again.
```

Claude

Yes, I've added it. Every Abaqus script the translator writes now saves `<job>_odb.npz` on its own when the job finishes, right next to `<job>.odb`. The user no longer has to run `abaqus python odb_extract.py` separately. They just bring back `<job>_odb.npz` along with `<job>_results.txt`. This hasn't been run in Abaqus yet, since there's no Abaqus here.

How it works

- **Verbatim copy:** when a script is generated, the translator copies `.claude/verification/odb_extract.py` into it unchanged. The comparator agent therefore receives exactly the same file as before, from its own tool, so its independence isn't affected.
- **Timing:** the extraction runs right after the job, before the script's own Abaqus vs. OpenSees comparison. A failure in that later section can't prevent the extract from being written.
- **If extraction fails:** the script records the error and the manual command in `<job>_results.txt`, then carries on.
- **Extractor change:** to make it callable from the script, `odb_extract.py` now has an `extract()` function. Its command line works as before.
- **Keeping the copy current:** generated scripts read the extractor at generation time, so if `odb_extract.py` ever changes, regenerate the Abaqus scripts. A regression check now confirms the copy is verbatim.

Testing

- **Regression T22:** the copy in a generated script matches the file exactly. On the stand-in ODB it writes the same 18 arrays as the standalone tool.
- **The inserted block:** I ran it exactly as it appears in a generated script. With an ODB present it wrote the extract; with the ODB missing it logged the fallback command.

- **Your four Abaqus scripts:** regenerated in the 11-4-26 folder, 13/13 model checks each, and they compile.
- **Full regression:** 259/259. The verification tools' own tests: 25/25.

Docs: I updated the abaqus-export and opensees-to-abaqus skill instructions, the regression table and the reorganization log. The comparator agent now asks for `<job>_odb.npz` first and falls back to running the extractor. The orchestrator now names that file as what the user brings back.

Tara

So what do I need to run myself on the machine with Abaqus, and what files do I copy over to this machine? Please incorporate instructions into the OpenSeesPy-to-Abaqus translator skill that tell the user precisely what to run and provide back to Claude to put into the verification agent anytime the skill is run

`Grep` Grep print\('Abaqus run itself|if n_fail:| if a.no_check:|ref, npz = None, None

```
440:    ref, npz = None, None
462:    if a.no_check:
473:    print('Abaqus run itself: untested here (Abaqus is not installed on this VM)')
474:    if n_fail:
```

`Read` Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py

```
425     def main():
426         ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
427         ap.add_argument('script', help='assembled OpenSeesPy script (shell-assemble-run)')
428         ap.add_argument('--out', help='Abaqus script (default abaqus_<name>.py next to the OpenSeesPy script)')
429         ap.add_argument('--element', default='S4R', choices=('S4R', 'S4'))
430         ap.add_argument('--npz', help='results .npz of this script to take the reference from (no run)')
431         ap.add_argument('--rerun', action='store_true', help='run the script even if its .npz is newer')
432         ap.add_argument('--no-reference', action='store_true', help='do not embed OpenSees reference values')
433         ap.add_argument('--no-check', action='store_true', help='skip the model comparison')
434         a = ap.parse_args()
435         src = Path(a.script).resolve()
436         info = read_script(src)
437         case, lam = info['CASE'], info['LAMINATE']
438         calls, ns = record_model(src)                           # also proves the script builds its model
439         os_m = opensees_model(calls, ns)
440         ref, npz = None, None
441         if not a.no_reference:
442             ref, npz, res = reference(src, info, Path(a.npz) if a.npz else None, a.rerun)
443             pts = np.array([os_m['nodes'][t] for t in res['node_tags']]) if set(res['node_tags']) <= set(os_m['nodes'])
\
444                 else None
445             if pts is None or pts.shape != res['points'].shape or np.abs(pts - res['points']).max() > TOL * case['geome
try']['L']:
446                 raise SystemExit(f'{npz}: its nodes are not the nodes of {src.name}; use --rerun')
447         out = Path(a.out).resolve() if a.out else src.parent / f'abaqus_{info["NAME"]}.py'
448         with contextlib.redirect_stdout(io.StringIO()):
449             out, M, ref = c2a.write_script(case, lam, out, a.element, ref,
... [30 more lines]
```

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py

```
-     if a.no_check:
        print('model check: skipped (--no-check)')
        return
    rows = check_model(os_m, out, M, case, lam)
    print(f'model check: {src.name} (recorded OpenSees calls) vs {out.name} (stand-in Abaqus API)')
    for label, ok, detail in rows:
        tag = '      ' if label.startswith('note') else ('[PASS]' if ok else '[FAIL]')
        print(f'  {tag} {label}' + (f'  ({detail})' if detail else ''))
    n_fail = sum(not ok for _, ok, _ in rows)
    print(f'{len(rows) - n_fail}/{len(rows)} checks pass' if not n_fail else
          f'{n_fail} check(s) FAIL: the Abaqus script is not the same model; do not use it')
    print('Abaqus run itself: untested here (Abaqus is not installed on this VM)')
    if n_fail:
        raise SystemExit(1)
+     guide = out.with_name(out.stem + '_instructions.txt')
    if a.no_check:
        print('model check: skipped (--no-check)')
    else:
        rows = check_model(os_m, out, M, case, lam)
        print(f'model check: {src.name} (recorded OpenSees calls) vs {out.name} (stand-in Abaqus API)')
        for label, ok, detail in rows:
            tag = '      ' if label.startswith('note') else ('[PASS]' if ok else '[FAIL]')
            print(f'  {tag} {label}' + (f'  ({detail})' if detail else ''))
```

```
        n_fail = sum(not ok for _, ok, _ in rows)
        print(f'{len(rows) - n_fail}/{len(rows)} checks pass' if not n_fail else
              f'{n_fail} check(s) FAIL: the Abaqus script is not the same model; do not use it')
        if n_fail:
            guide.unlink(missing_ok=True)
            raise SystemExit(1)
    print('Abaqus run itself: untested here (Abaqus is not installed on this VM)')
... [44 more lines]
```

Bash Regenerate scripts and show instructions output

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/11-4-26 (Tara)" && for s in plate_tip_load_laminate tape_spri
ng_L024_tip_load trac_like_combination trac_like_load_control; do /d/Users/tvenkata/miniforge3/envs/shells/python.exe
../../.claude/skills/opensees-to-abaqus/opensees_to_abaqus.py ${s}_opensees.py > /tmp/o_$s.txt 2>&1; echo "$s exit $?"; don
e; sed -n '/checks pass/,$p' /tmp/o_tape_spring_L024_tip_load.txt; ls *_instructions.txt; grep -n "JOB = " abaqus_tape_spri
ng_L024_tip_load.py
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\opensees_to_abaqus.py has been updated
successfully.
```

```
plate_tip_load_laminate exit 0
tape_spring_L024_tip_load exit 0
trac_like_combination exit 0
trac_like_load_control exit 0
13/13 checks pass
Abaqus run itself: untested here (Abaqus is not installed on this VM)

=== Abaqus verification of "tape_spring_L024_tip_load" ===

ON THE MACHINE WITH ABAQUS
  1. Copy this ONE file into an empty folder there:
       abaqus_tape_spring_L024_tip_load.py
  2. In that folder run:
       abaqus cae noGUI=abaqus_tape_spring_L024_tip_load.py
     (5 mm mesh, S4R, job "tape_spring_L024_tip_load_S4R_5mm". For another mesh / element, e.g.
      abaqus cae noGUI=abaqus_tape_spring_L024_tip_load.py -- 2.5 S4
      the job and every file name below become "tape_spring_L024_tip_load_S4_2p5mm".)
  3. When it finishes, open tape_spring_L024_tip_load_S4R_5mm_results.txt and check that it does NOT say
     "ODB extraction failed". If it does, copy .claude/verification/odb_extract.py
     from the project into the same folder and run:
       abaqus python odb_extract.py tape_spring_L024_tip_load_S4R_5mm.odb
     Optional: open the model in Abaqus/CAE to check the set-up:
       abaqus cae database=tape_spring_L024_tip_load_S4R_5mm.cae

COPY BACK TO THIS MACHINE, into D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
... [12 more lines]
```

Read Read D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\SKILL.md

```
27      `n_steps` (the Riks stop is step x n_steps).
28
29      ## Command line (from `.claude/skills`)
30      ```
31      python opensees-to-abaqus/opensees_to_abaqus.py <dir>/<name>_opensees.py [--out FILE] [--element S4R|S4]
32                                                      [--npz results.npz | --rerun | --no-reference] [--no-check]
33      abaqus cae noGUI=abaqus_<name>.py [-- <size> <S4R|S4>]          # on a machine with Abaqus
34      ```
35      - Output default: `abaqus_<name>.py` next to the OpenSeesPy script.
36      - The script's `NAME`, `CASE` and `LAMINATE` are read as literals; the
37        Abaqus model uses `LAMINATE` (the plies the sections were built from), not
38        the case's laminate entry.
39      - Reference (`REF`): from the script's own results `<dir>/results/<name>.npz`,
40        reused when newer than the script, else the script is run (seconds for
41        linear cases, minutes for long nonlinear ones: run those in the background).
42        `--npz` takes another results file of the same script (checked: same
43        laminate and nodes). Embedded: report-node displacement, max |u_i|, top-ply
44        top-surface stress in ply axes, load-factor history (displacement control).
45        The Abaqus script prints Abaqus vs OpenSees with % differences (PASS within
46        3%, else CHECK) into `<job>_results.txt`.
47      - Model check (default; exit code 1 if any FAIL, then do not hand the script
48        out): the OpenSeesPy script is executed up to its first `ops.analyze` with its
49        `ops` calls recorded, the Abaqus script is executed on the stand-in Abaqus API,
50        and these are compared: shell nodes = the structured grid the Abaqus seeds
51        give; shell normal per section piece = OpenSees facet normal (bottom ply =
... [25 more lines]
```

Edit Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\SKILL.md

```
- - Not translated (FAIL): `equalDOF`, `sp`, `eleLoad`, more than one load pattern,
  sections other than LayeredShell / ElasticMembranePlateSection.

## Report back
- the Abaqus script path and the commands (`abaqus cae noGUI=abaqus_<name>.py`,
  `... -- <size> <S4R|S4>`), outputs `<job>_results.txt`, `.odb`, `.cae` and
  `<job>_odb.npz` (the ODB as plain arrays, written by the embedded copy of
```

```
  `.claude/verification/odb_extract.py`): the user brings back `<job>_odb.npz`
  (and `<job>_results.txt`) for the OpenSeesAbaqusResultsComparator agent;
+ - Not translated (FAIL): `equalDOF`, `sp`, `eleLoad`, more than one load pattern,
  sections other than LayeredShell / ElasticMembranePlateSection.
- Instructions for the user (every successful run): the script prints an
  "=== Abaqus verification of ... ===" block and writes the same text to
  `abaqus_<name>_instructions.txt` next to the Abaqus script. It names, with
  the actual job name `<name>_<element>_<size>mm` (`.` -> `p`):
  - **on the Abaqus machine**: copy only `abaqus_<name>.py` into an empty
    folder; run `abaqus cae noGUI=abaqus_<name>.py` (other mesh/element:
    `... -- <size> <S4R|S4>`, which changes the job name); check that
    `<job>_results.txt` does not say "ODB extraction failed" (fallback:
    `abaqus python odb_extract.py <job>.odb` with the file from
    `.claude/verification/`); optionally `abaqus cae database=<job>.cae`;
  - **copy back** into the OpenSeesPy script's folder: `<job>_odb.npz`
    (required; the ODB as plain arrays, written by the embedded copy of
    `odb_extract.py`) and `<job>_results.txt` (recommended); the `.odb`/`.cae`
    are not needed here;
  - **then ask Claude**: the exact sentence that runs the
    OpenSeesAbaqusResultsComparator agent on the OpenSees results .npz and
    `<job>_odb.npz`.

## Report back
... [3 more lines]
```

`Bash` Update orchestrator and log; rerun T22

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude" && /d/Users/tvenkata/miniforge3/envs/shells/python.exe - <<'EOF'
from pathlib import Path
def rep(path, old, new):
    p = Path(path); s = p.read_text(encoding='utf-8'); assert old in s, (path, old); p.write_text(s.replace(old, new, 1), e
ncoding='utf-8')
for f in ('agents/ThinShellOrchestrator.md', 'agents_v2.0/ThinShellOrchestrator.md'):
    rep(f, "shell-assemble-run and convert that script with the opensees-to-abaqus skill.\n",
        "shell-assemble-run and convert that script with the opensees-to-abaqus skill,\n"
        "and give the user its \"Abaqus verification\" instructions block verbatim (what to\n"
        "run on the Abaqus machine, which files to copy back, what to ask next).\n")
log = Path('REORGANIZATION_LOG_v2.0.md'); s = log.read_text(encoding='utf-8').rstrip('\n')
s += """
- opensees-to-abaqus now prints, and writes to `abaqus_<name>_instructions.txt`,
  the exact steps for the user: copy only `abaqus_<name>.py` to the Abaqus
  machine, `abaqus cae noGUI=abaqus_<name>.py`, check the extraction line in
  `<job>_results.txt` (fallback command), copy back `<job>_odb.npz` (required)
  and `<job>_results.txt`, and the sentence to ask Claude to run the comparator.
  Job/file names are computed as the Abaqus script does (`job_name`).
"""
log.write_text(s + '\n', encoding='utf-8')
EOF
diff -q agents/ThinShellOrchestrator.md agents_v2.0/ThinShellOrchestrator.md && echo same; cd skills/regression-tests && /
d/Users/tvenkata/miniforge3/envs/shells/python.exe run_regression.py --only T22 2>&1 | tail -1
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\opensees-to-abaqus\SKILL.md has been updated successfully.
```

```
same
9 checks, 9 passed, 0 failed (6 s)
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

Claude

I'll also add a regression check that the job name in the instructions matches what the generated Abaqus script will actually use, so the two can't drift apart.

`Edit` Edit D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py

```
-            fails = [r[0] for r in rows if not r[1]]
            check(f'{case["name"]}: model check passes
+            job_line = [l for l in out.read_text(encoding='utf-8').splitlines() if l.startswith('JOB = ')][0]
            for el, size in (('S4R', M['size']), ('S4', M['size'] / 2)):
                job = eval(job_line[6:], {'ELEM': el, 'SIZE': float(size)})
                check(f'{case["name"]}: instructions job name = the script\'s JOB ({job})',
                      float(o2a.job_name(case['name'], el, size) == job), 1, 0, True)
            fails = [r[0] for r in rows if not r[1]]
            check(f'{case["name"]}: model check passes
```

`Bash` Run T22 with job name check

```
cd "D:/Users/tvenkata/projects/ME295_project/.claude/skills/regression-tests" && /d/Users/tvenkata/miniforge3/envs/shells/p
ython.exe run_regression.py --only T22 2>&1 | grep "job name\|checks,"
```

```
The file D:\Users\tvenkata\projects\ME295_project\.claude\skills\regression-tests\run_regression.py has been updated succes
sfully.
```

```
[PASS] tape_spring_L024_tip_load: instructions job name = the script's JOB (tape_spring_L024_tip_load_S4R_5mm) got +1  ref
+1  diff 0.00e+00 (tol 0e+00)
```

```
  [PASS] tape_spring_L024_tip_load: instructions job name = the script's JOB (tape_spring_L024_tip_load_S4_2p5mm) got +1  r
ef +1  diff 0.00e+00 (tol 0e+00)
  [PASS] trac_like_combination: instructions job name = the script's JOB (trac_like_combination_S4R_5mm) got +1  ref +1  di
ff 0.00e+00 (tol 0e+00)
  [PASS] trac_like_combination: instructions job name = the script's JOB (trac_like_combination_S4_2p5mm) got +1  ref +1  d
iff 0.00e+00 (tol 0e+00)
  [PASS] trac_like_load_control: instructions job name = the script's JOB (trac_like_load_control_S4R_5mm) got +1  ref +1
diff 0.00e+00 (tol 0e+00)
  [PASS] trac_like_load_control: instructions job name = the script's JOB (trac_like_load_control_S4_2p5mm) got +1  ref +1
diff 0.00e+00 (tol 0e+00)
15 checks, 15 passed, 0 failed (6 s)
Shell cwd was reset to D:\Users\tvenkata\projects\ME295_project\Code for Project\11-4-26 (Tara)
```

`mcp__ccd_turn__wrap_up` Recap

```
Every translator run now prints exact Abaqus-machine steps and saves them to abaqus_<name>_instructions.txt: what to run, w
hat to copy back, what to ask Claude. All four 11-4-26 cases have one, and T22 passes 15/15. Nothing is committed.
```

```
Shown in the next-step box. Your closing message is your answer to the user: write it now, in full, as the last thing you s
ay this turn. The box asks your question, so do not ask it again.
```

Claude

What you do on the Abaqus machine (example: the tape spring case)
1. Copy just one file, `abaqus_tape_spring_L024_tip_load.py`, into an empty folder.
2. In that folder, run:

```
abaqus cae noGUI=abaqus_tape_spring_L024_tip_load.py
```

3. When it finishes, open `tape_spring_L024_tip_load_S4R_5mm_results.txt` and check that it doesn't say "ODB extraction failed". If it does, copy `.claude/verification/odb_extract.py` into that folder and run `abaqus python odb_extract.py tape_spring_L024_tip_load_S4R_5mm.odb`.

What you copy back to this machine, into the folder of the OpenSeesPy script (here `11-4-26 (Tara)`):
- `tape_spring_L024_tip_load_S4R_5mm_odb.npz` (**required**): the Abaqus results as plain arrays.
- `tape_spring_L024_tip_load_S4R_5mm_results.txt` (recommended): the Abaqus log, with its own check against the OpenSees values.
- You don't need the `.odb` or `.cae` here; keep them on the Abaqus machine if you want to inspect the model.

Then you ask Claude to run the results comparator agent on the two files. The instructions give you the exact sentence to paste.

Now part of the translator skill
- **Every run:** each time it runs, it prints this whole block with the real file names filled in, and saves it next to the Abaqus script as `abaqus_<name>_instructions.txt`.
- **Other mesh or element:** the block shows the alternative command (`-- 2.5 S4`) and the file names that run would produce.
- **If the model check fails:** no instructions are written, so a bad script never comes with "go run this".
- **Skill rules:** the SKILL.md now says to give you the block word for word every time the skill is used. The orchestrator says the same when it runs the skill.
- **Name check:** a regression check confirms the job name in the instructions matches what the Abaqus script actually uses, for both the default and an alternative mesh and element. T22 passes 15/15.
- **Your four cases:** each now has its instructions file in the 11-4-26 folder, such as `abaqus_tape_spring_L024_tip_load_instructions.txt`.

Tara

Create a PDF of this conversation to replace the one I had you create earlier. Name it the same as that conversation so it overwrites the old one